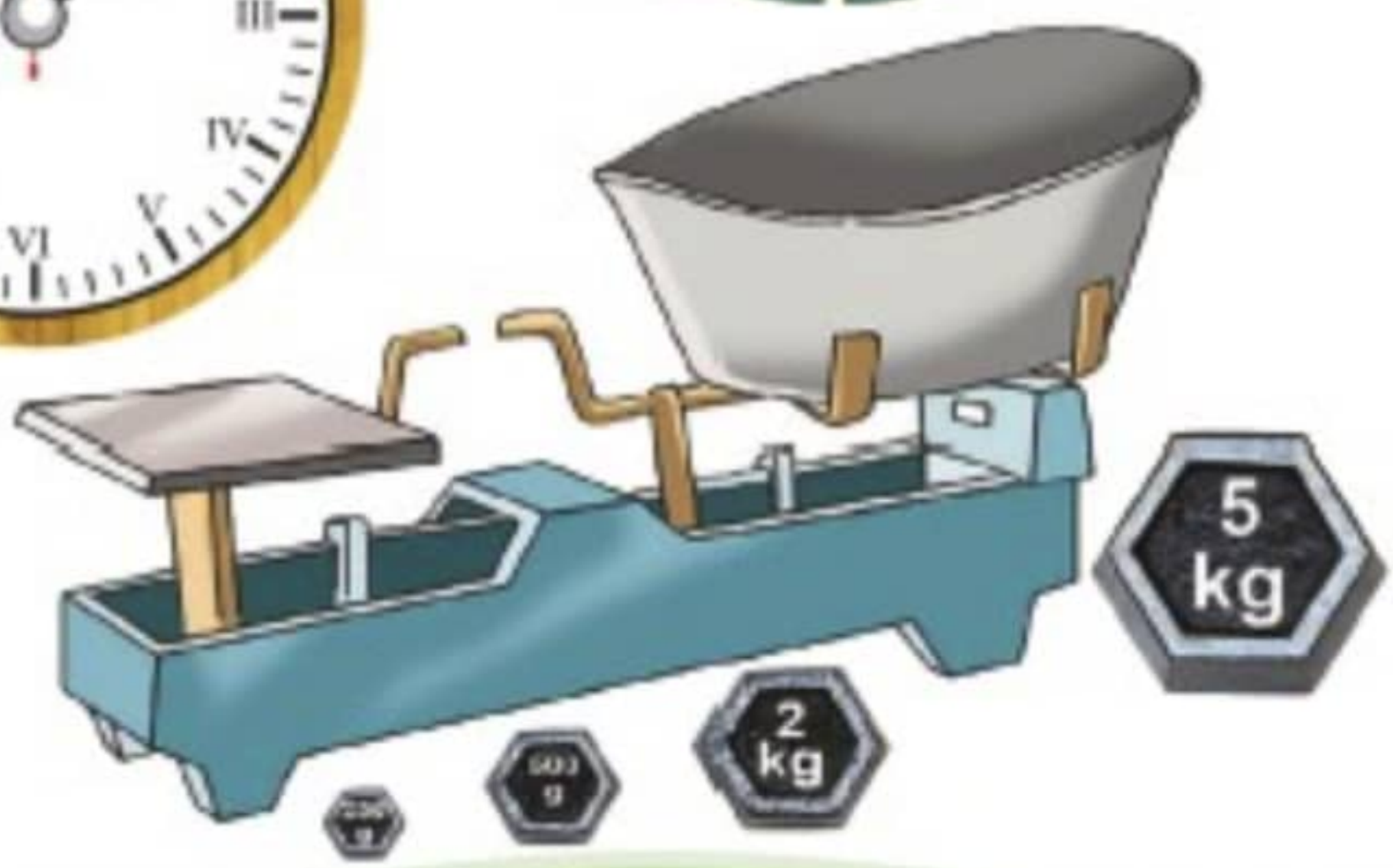
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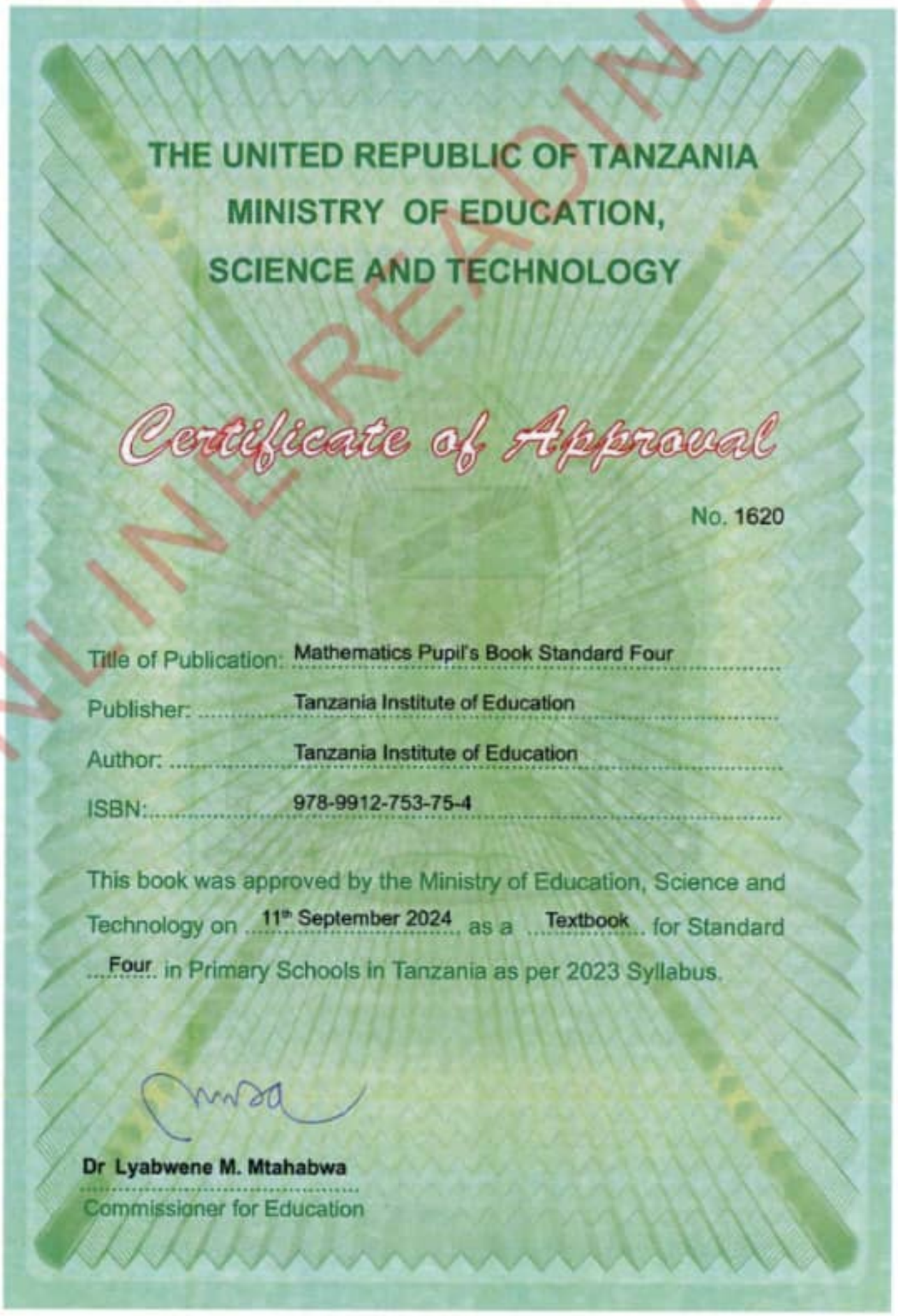
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Tanzania Institute of Education

Mathematics

Pupil's Book

Standard Four



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Introduction

This textbook, Mathematics Pupil's Book for Standard Four, is written in accordance with the 2023 Mathematics Syllabus for Primary Schools issued by the Ministry of Education, Science and Technology. This is the second edition after the improvement of the Standard Four Textbook published in 2019, based on 2016 syllabus.

The book consists of ten chapters, namely, Roman numbers; Multiplication of whole numbers, Division of whole numbers; Measurements of length, mass and volume; Perimeters of plane figures; Area of plane figures; Fractions; Decimals; Measurements of time and Tanzanian Currency. The competencies developed will enable you to acquire basic mathematical skills and their applications, logical thinking and solve real life problems.

You are advised to engage yourself and with friends in all activities and exercises in this book in order to develop the intended competencies.

Additional learning resources are available in the TIE-e-Library at *https://ol.tie.go.tz* or *ol.tie.go.tz*

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Chapter One

Roman numbers

Introduction

In this chapter, you will learn to read and write numbers from 1 to 1,000 in Roman numbers. These numbers range from I to M in Roman numbers. They are used in various contexts such as on the introductory pages of books and on clock faces. The competencies developed will enable you to use Roman numbers in different contexts.



Think

Life without application of numbers

Roman numbers from I to X

Whole numbers are composed of ten digits, which are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. Roman numbers are written by using some or combinations of the letters of the alphabet.

Read the following numbers:

Numerals	Roman numbers	Numbers in words
1	I	one
2	II	two
3	III	three
4	IV	four
5	V	five

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6	VI	six
7	VII	seven
8	VIII	eight
9	IX	nine
10	X	ten

Activity 1 **Identifying Roman numbers**

1. Use various sources, such as Khan Academy, to learn items that use Roman numbers.
2. Write down the names of the items obtained from Step 1.
3. Why is it important for you to do this activity?

Writing Roman numbers from I to X

The Roman number one (I) can be written by repeating it to form various numbers. For example,

Numerals	1	2	3
Roman numbers	I	II	III

This number (I) cannot be repeated more than three times in one number.

The Roman numbers I, V, and X are also used to create other Roman numbers. For example,

Numerals	4	5	6	7	8	9
Roman numbers	IV	V	VI	VII	VIII	IX

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When a smaller Roman number is written to the right of a Roman number which is equal or greater in its value, the resulting number is obtained by adding the values of those numbers.

Example 1

Write the following Roman numbers in numerals:

(a) II (b) VI (c) VII (d) VIII

Solution

(a) II = 1+1 = 2 Therefore, II is 2.	(b) VI = 5 + 1 = 6 Therefore, VI is 6.
(c) VII = 5 + 2 = 7 Therefore, VII is 7.	(d) VIII = 5 + 3 = 8 Therefore, VIII is 8.

A Roman number with a smaller value written to the left of a Roman number with a greater value is obtained by subtracting the value of the smaller number from the greater number.

Example 2

Write the following Roman numbers in numerals:

(a) IV (b) IX

Solution

(a) IV = 5 – 1 = 4 Therefore, IV is 4.	(b) IX = 10 – 1 = 9 Therefore, IX is 9.

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Exercise 1

1. Write IX, VI, X, IV, VII, V and VIII in ascending order.
2. Convert VII, IV, VI, V, VIII and IX into numerals.
3. Convert 3, 6, 7, 1, 8, 4, 10, 9 and 5 into Roman numbers.
4. Fill in the missing numbers in the following table.

Number in words	six	seven	nine	four	ten
Roman numbers				IV	

6. A parent bought shoes with sizes X, V, and VII for the kids. Write the shoes sizes in numerals.

Roman numbers from XI to XX

Read the numbers in the following table.

Roman numbers	Numerals	Numbers in words
XI	11	eleven
XII	12	twelve
XIII	13	thirteen
XIV	14	fourteen
XV	15	fifteen
XVI	16	sixteen
XVII	17	seventeen
XVIII	18	eighteen
XIX	19	nineteen
XX	20	twenty

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Example 1

Write the following numerals in Roman numbers:

(a) 14 (b) 17 (c) 19

Solution

(a) 14 = X + IV = XIV Therefore, 14 is XIV.	(b) 17 = X + VII = 17 Therefore, 17 is XVII.
(c) 19 = X + IX = XIX Therefore, 19 is XIX.	

Example 2

Convert the following into numerals:

(a) XII (b) XV (c) XVIII

Solution

(a) XII = 10 + 2 = 12 Therefore, XII is 12.	(b) XV = 10 + 5 = 15 Therefore, XV is 15.
(c) XVIII = 10 + 8 = 18 Therefore, XVIII is 18.	

Exercise 2

1. Write the Roman number with the highest value in the following list: XII, XIII, XV, XIX, XVI, XVII, XVIII and XIV.
2. Write the following Roman numbers in words:
   (a) XIV (b) XVI (c) XX (d) XI (e) XV (f) XIX

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3. Several cards were marked with the numbers 19, 17, 13, 16, 20, 14, 12, 18, 11, and 15. Write the numbers in Roman numbers, then arrange them in ascending order.
4. Convert XI, XVIII, XV, XVII, and XII into numerals.
5. Write the missing numbers in each of the following sequence:
   (a) X, ____, XIV, XVI, ____, XX
   (b) ____, XIV, ____, XII, ____, X
6. Read the clock face, then write the time in numerals.



Roman numbers from XXI to XXXIII

Read the numbers in the following table.

Numerals	21	22	23	24	25	26	27	28	29	30	31	32	33
Roman numbers	XXI	XXII	XXIII	XXIV	XXV	XXVI	XXVII	XXVIII	XXIX	XXX	XXXI	XXXII	XXXIII

Example 1

Write the following numerals in Roman numbers:

(a) 24 (b) 37 (c) 27

Solution

(a) 24 = XX + IV
= XXIV
Therefore, 24 is XXIV.

(b) 37 = XXX + VII
= XXXVII
Therefore, 37 is XXXVII.

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(c) 27 = XX + VII
= XXVII
Therefore, 27 is XXVII.

Example 2

Write the following Roman number in numerals:

(a) XXI (b) XXXIII

Solution

(a) XXI = 20 + 1
= 21
Therefore, XXI is 21.

(b) XXXIII = 30 + 3
= 33
Therefore, XXXIII is 33.

Exercise 3

1. Write the following numerals in Roman numbers:
   (a) 29 (b) 26 (c) 32 (d) 23
2. Write each of the following Roman number in numerals.
   (a) XXII (b) XXXI (c) XXX (d) XXV
3. Write the missing Roman numbers in each sequence:
   (a) XXI, ____, XXIII, ____, ____, XXVI
   (b) XXIV, ____, XXVI, ____, ____, XXIX, ____, ____, ____
   (c) XXXII, ____, XXX ____, XXVIII, ____, XXVI
   (d) ____, ____, XXV, ____, XXIII, XXII, ____

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4. Pupils were given number cards with the following Roman numbers:

Pupils	John	Asha	Ali	Rose
Cards	XXIX	XX	XXXII	XVII

(a) Who was given a card with high value?

(b) Write the card with high value in numeral.

Roman numbers from XXXIV to L

Read the following Roman numbers:

XXXIV, XXXV, XXXVI, XXXVII, XXXVIII, XXXIX, XL, XLI, XLII, XLIII, XLIV, XLV, XLVI, XLVII, XLVIII, XLIX and L.

Example 1

Write the following numbers in Roman numbers:

(a) 34 (b) 41 (c) 45

Solution

(a) 34= XXX + IV
= XXXIV
Therefore, 34 is XXXIV.

(b) 41 = XL + I
= XLI
Therefore, 41 is XLI.

(c) 45 = XL + V
= XLV
Therefore, 45 is XLV.

Example 2

Write the following Roman numbers in numerals:

(a) XXXVII (b) XLIII (c) XLIX

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Solution

(a) XXXVII = 30 + 7 = 37 Therefore, XXXVII is 37.	(b) XLIII = 40 + 3 = 43 Therefore, XLIII is 43.
(c) XLIX = 40 + 9 = 49 Therefore XLIX is 49.	

Exercise 4

1. Write the following numerals in Roman numbers:
   (a) 44 (b) 39 (c) 35 (d) 46
2. Write the following Roman numbers in numerals.
   (a) XXXVI (b) XLVIII (c) XLI (d) XXXIV
3. Write the missing Roman numbers in each sequence:
   (a) XXXIV, ____, XXXVI, ____, ____, XXXIX
   (b) XXXVI, XXXVIII, ____, XLII, XLIV, ____, ____, ____
   (c) L, ____, XLIV, ____, XXXVIII, XXXII
4. Fill in the missing numbers in the following table.

Numerals		39		46
Roman numbers	XXXVI		XLIII	

5. Fill in the missing numbers in each sequence:
   (a) XXXII, ____, XXXVI, ____, ____

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(b) X, ____, XX, XXV, ____, ____, XL

(c) L, ____, ____, ____, XXX, XXV

(d) XXXV, XXXVII, ____, XLI, ____, XLV

6. Arrange sticks in a shape which shows 48 in Roman numbers, then write the resulting Roman number.

7. Nuru saw a sign post marked with a Roman number XLIV. What number is that in numerals?

8. Write the following Roman numbers in words:

   (a) XLIX (b) XXXVI (c) XLV (d) XXXVII

9. Write the following in Roman numbers:

   (a) Forty-two (b) Thirty-seven

   (c) Thirty-nine (d) Forty-three

10. Write the following Roman numbers in words:

    (a) XLII (b) XXXVIII (c) XXXIX (d) XLVII

Roman numbers from L to C

Read the following numbers:

Numerals	50	60	70	80	90	100
Roman numbers	L	LX	LXX	LXXX	XC	C
Numbers in words	fifty	sixty	seventy	eighty	ninety	one hundred

The numbers I, X, and C are the only Roman numbers that, when written to the left of a larger number are subtracted, and when written to the right, are added. These numbers cannot be repeated when written to the left of a larger number. Also, these numbers

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cannot be repeated more than three times when written to the right of a larger number. In numerals, these numbers correspond to 1, 10 and 100.

The numbers IV and IX represent ones. Therefore, when these numbers are written after higher value, then, the numbers are added to the larger number.

Read the numbers in the following table:

Roman numbers	Numbers in words	Numerals
LV	fifty-five	55
LIX	fifty-nine	59
LXXXIX	eighty-nine	89
LXXII	seventy-two	72
XCIX	ninety-nine	99
LXIV	sixty-four	64
LVIII	fifty-eight	58
LXI	sixty-one	61
LXXIII	seventy-three	73
LXXXV	eight-five	85

Example 1

Write the following Roman numbers in words:

(a) LXV (b) LXXVI (c) XCIX (d) LIV

Solution

(a) $LXV = 60 + 5$

$= 65$

Therefore, the number LXV is sixty-five.

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(b) $LXXVI = 70 + 6$

$= 76$

Therefore, the number LXXVI is seventy-six.

(c) $XCIX = 90 + 9$

$= 99$

Therefore, the number XCIX is ninety-nine.

(d) $LIV = 50 + 4$

$= 54$

Therefore, the number LIV is fifty-four.

Example 2

Write the following numerals in Roman numbers:

(a) 53 (b) 67 (c) 57 (d) 88

Solution

(a) $53 = L + III$

$= LIII$

Therefore, 53 is LIII.

(b) $67 = LX + VII$

$= LXVII$

Therefore, 67 is LXVII.

(c) $57 = L + VII$

LVII

Therefore, 57 is LVII.

(d) $88 = LXXX + VIII$

$= LXXXVIII$

Therefore, 88 is LXXXVIII.

Exercise 5

1. Write the following Roman numbers in words:

   (a) LXVII (b) XC (c) LXXXII (d) XCVIII

2. Write the following numerals in Roman numbers:

   (a) 52 (b) 66 (c) 70 (d) 93

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3. Write the following Roman numbers from the smallest to the largest:

   C, I, X, L, V, LXV, LXXXV, XCV, LV, and LXXV.

4. Write the missing Roman numbers in the following sequence:

   LXXX, _____, LXXXII, _____, LXXXIV, _____.

Roman numbers from C to D

The Roman numbers C and D are the only representations of hundreds whereas C stands for 100, while D stands for 500. When C is written to the left of D, its value is subtracted from D; however, when C is written to the right of D, its value can only be added up to three times.

For example,

(i) CD (ii) DCC (iii) DCCC

Read the following numbers:

Numerals	100	200	300	400	500
Roman numbers	C	CC	CCC	CD	D
Number in words	one hundred	two hundred	three hundred	four hundred	five hundred

Read the following numbers:

Roman numbers	Number in words	Numerals
CIV	one hundred and four	104
CXIX	one hundred and nineteen	119
CLXXVI	one hundred and seventy-six	176
CCXXII	two hundred and twenty-two	222

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CCCXLIII	three hundred and forty-three	343
CCXXV	two hundred and twenty-five	225
CCCXXXVIII	three hundred and thirty-eight	338
CDXVII	four hundred and seventeen	417
CCLIV	two hundred and fifty-four	254
CDXXXI	four hundred and thirty-one	431

Example 1

Write the following numerals in Roman numbers:

(a) 205 (b) 362

Solution

(a) $205 = CC + V$
$= CCV$
Therefore, 205 is CCV.

(b) $362 = CCC + LXII$
$= CCCLXII$
Therefore, 362 is CCCLXII.

Example 2

Write the following Roman numbers in words:

(a) CXVII (b) CDXLVI

Solution

(a) CXVII - One hundred and seventeen.

(b) CDXLVI - Four hundred and forty-six.

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Exercise 6

1. Write the following Roman numbers in words:
   (a) CCCXXX (b) CCXIX (c) CDXLVII (d) CCXCII
2. Write the following in Roman numbers:
   (a) Four hundred and forty-four
   (b) One hundred and fifteen
   (c) Two hundred and nine
   (d) Four hundred and fifty-four
3. Convert the following numerals into Roman numbers.
   (a) 108 (b) 374 (c) 435 (d) 442
4. The distance between City A and City B is CDLXXVIII kilometres. Write these distance in numerals.

Roman numbers from D to M

The Roman numbers M is 1000 in numerals. The Roman numbers C can only be written to the left of M, allowing it to be subtracted from M.

Read the following Roman numbers:

Roman numbers	Number in words
D	five hundred
DC	six hundred
DCC	seven hundred
DCCC	eight hundred
CM	nine hundred
M	one thousand

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Read the following Roman numbers.

Number in words	Roman numbers
five hundred and eleven	DXI
seven hundred and ninety-nine	DCCXCIX
six hundred and seventy-four	DCLXXIV
nine hundred and eighty-four	CMLXXXIV
five hundred and fifty	DL
seven hundred and twenty-eight	DCCXXVIII

Example 1

Write the following Roman numbers in numerals:

(a) CMIX (b) DCCLXXIV

Solution

(a) $CMIX = 900 + 9$
$= 909$
Therefore, CMIX is 909.

(b) $DCCLXXIV = 700 + 74$
$= 774$
Therefore, DCCLXXIV is 774.

Example 2

Write the following numerals in Roman numbers:

(a) 576 (b) 688 (c) 842 (d) 964

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Solution

(a) $576 = D + LXXVI$

$= DLXXVI$

Therefore, 576 is DLXXVI.

(b) $688 = DC + LXXXVIII$

$= DCLXXXVIII$

Therefore, 688 is DCLXXXVIII.

(c) $842 = DCCC + XLII$

$= DCCCXLII$

Therefore, 842 is DCCCXLII.

(d) $964 = CM + LXIV$

$= CMLXIV$

Therefore, 964 is CMLXIV.

Activity 2 Learning Roman numbers

Description

Use various reliable internet sources such as Khan Academy to learn more about Roman numbers.

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Exercise 7

1. Write the following Roman numbers in words:
   (a) CMLXXIX (b) DLXXVIII
   (c) CMXCIV (d) DCCCVI

2. Write the following in Roman numbers:
   (a) Five hundred fifty-nine
   (b) Seven hundred sixty-four
   (c) Nine hundred eighty-two
   (d) Six hundred sixty-seven

3. Convert the following numerals into Roman numbers:
   (a) 572 (b) 698 (c) 755 (d) 916

4. Karol was given a card marked with the number DLVII. Anna was given a card marked with the number DXLVII.
   (a) Who was given a card with high value?
   (b) Write the number on each card in numerals.

Summary

1. Numerals are composed of ten digits which are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.
2. There is no Roman number representing 0.
3. The basic Roman numbers I, V, X, L, C, D, and M, are used to create other Roman numbers.
4. In Roman numbers, letters are not repeated more than three times consecutively in one number.

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5. The Roman numbers V, L, and D cannot be subtracted in other numbers.

6. The Roman numbers I, X, and C can be subtracted from other numbers. Therefore,
   (a) I can be subtracted from V and X only, for example IV and IX.
   (b) X can be subtracted from L and C only, for example XL and XC.
   (c) C can be subtracted from D and M only, for example CD and CM.

7. The number I, X, and C can be written by repeating up to three times consecutively. For example; III, XXX, and CCC.

Revision Exercise

1. Write the following Roman numbers in words:
   (a) CMXC (b) DLXIV (c) CMLXXI (d) DCXCVII

2. Write the following in Roman numbers.
   (a) Six hundred and twenty-one
   (b) Eight hundred and fifty-five
   (c) Eight hundred and eighteen
   (d) Nine hundred and fifty

3. Write the following Roman numbers in words:
   (a) XCI (b) CCXCII (c) DCCCLXXI (d) CMXCIX

4. The length of the road is CMXCVI kilometres. Write this length in numerals.

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5. Two cards were marked with the following numbers; DCVI and CMXI.
   (a) Which card had the higher value?
   (b) Write the number of the card with the smaller value in numerals.

6. Write the missing Roman numbers in each sequence:
   (a) CXV, CCXXX, _____, _____, DLXXV, _____, _____, _____.
   (b) DCXIX, DCXCIX, _____, _____, _____.

7. Arrange the following numbers from the smallest to the largest:
   XCV, XLVII, CCIX, CLI, LXXIII, CMV, CMXCII, III, XV, XCII.

8. The distance between two towns E and F is CMXLIX kilometres. Write this distance in numeral:

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Chapter Two

Multiplication of whole numbers

Introduction

In this chapter, you will learn about multiplication of whole numbers. The concept of multiplication is applied in everyday activities such as business, construction, statistics, sports, and financial transactions. The competencies developed will enable you to use multiplication in various contexts.



Think

Mathematics without the knowledge of multiplication

Activity 1 Developing the concept of multiplication

Steps

1. In your environment, prepare groups of similar objects according to the quantity you need.
2. Count the number of groups you have in Step 1.
3. Identify the number of groups and the total number of objects in each group to find the overall total.
4. Is there any relationship between the number of objects, the number of groups and the overall total?

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Multiplication of two-digit numbers by a one-digit multiplier

Example 1

$62 \times 4 =$

Steps

1. Multiply 2 ones by 4: $2 \times 4 = 8$. Write 8 in the ones place.
2. Multiply 6 tens by 4: $6 \times 4 = 24$. Write 4 in the tens place. Regroup 20 tens into 2 hundreds.

   Therefore, $62 \times 4 = 248$.

Example 2

$25 \times 6 =$

Steps

1. Multiply 5 ones by 6: $5 \times 6 = 30$. Write 0 in the ones place. Shift 3 tens to the tens place.
2. Multiply 2 tens by 6: $2 \times 6 = 12$. Add tens: $2 + 3 = 5$. Write 5 in the tens place. Regroup 10 tens into 1 hundreds. Write 1 in the hundreds place.

   Therefore, $25 \times 6 = 150$.

Exercise 1

1. $41 \times 9 =$
2. $64 \times 2 =$
3. $38 \times 7 =$
4. $56 \times 7 =$
5. $49 \times 4 =$
6. $72 \times 8 =$

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7. $\begin{array}{r} 88 \\ \times\ 8 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 59 \\ \times\ 3 \\ \hline \\ \hline \end{array}$

9. $\begin{array}{r} 78 \\ \times\ 5 \\ \hline \\ \hline \end{array}$

10. $\begin{array}{r} 27 \\ \times\ 9 \\ \hline \\ \hline \end{array}$

11. $\begin{array}{r} 32 \\ \times\ 2 \\ \hline \\ \hline \end{array}$

12. $\begin{array}{r} 90 \\ \times\ 8 \\ \hline \\ \hline \end{array}$

13. $\begin{array}{r} 79 \\ \times\ 6 \\ \hline \\ \hline \end{array}$

14. $\begin{array}{r} 98 \\ \times\ 7 \\ \hline \\ \hline \end{array}$

15. $\begin{array}{r} 65 \\ \times\ 4 \\ \hline \\ \hline \end{array}$

Multiplication of two-digit numbers by a two-digit multiplier

Example 1

$23 \times 21 =$

Write multiplication problem vertically aligning the digits by place values

Solution	Steps
$\begin{array}{r} 23 \\ \times\ 21 \\ \hline 23 \\ +\ 460 \\ \hline 483 \\ \hline \end{array}$	1. $1 \times 3 = 3$. Write 3 in the ones. 2. $1 \times 20 = 20$. Write 2 in the tens place. 3. $20 \times 3 = 60$. Write 6 in the tens place and 0 in the ones place. 4. $20 \times 20 = 400$. Write 4 in the hundreds place. 5. Add the results from Steps 1 to 4: $23 + 460 = 483$.

Therefore, $23 \times 21 = 483$.

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Example 2

$27 \times 12 =$

Solution	Steps
27 × 12 54 +270 324	1. $2 \times 7 = 14$. Write 4 in the ones place. Regroup 10 ones into 1 tens. Shift 1 tens to the tens place. 2. $2 \times 20 = 40$. Write 4 tens. Add tens: $4 + 1 = 5$. Write 5 in the tens place. 3. $10 \times 7 = 70$. Write 7 in the tens place and 0 in the ones place. 4. $10 \times 20 = 200$. Write 2 in the hundreds place. 5. Add the results from Steps 1 to 4: $54 + 270 = 324$.

Therefore, $27 \times 12 = 324$.

Example 3

$78 \times 69 =$

Solution	Steps
78 × 69 702 +4680 5382	1. $9 \times 8 = 72$. Write 2 in the ones place. Regroup 70 ones into 7 to tens. Shift 7 tens to the tens place 2. $9 \times 70 = 630$. Write 63 tens. Add tens: $63 + 7 = 70$. Write 0 in the tens place and 7 in the hundreds place.

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3. $60 \times 8 = 480$. Write 0 in the ones place and 8 in the tens place. Regroup 40 tens into 4 hundreds. Shift 4 hundreds to the hundreds place.
4. $60 \times 70 = 4200$. Write 42 hundreds. Add 4 hundreds: $42 + 4 = 46$. Write 6 in the hundreds place. Regroup 40 hundreds into 4 thousands. Write 4 in the thousands place.
5. Add the results from Steps 1 to 4: $702 + 4680 = 5382$.

Therefore, $78 \times 69 = 5{,}382$.

Exercise 2

1. $\begin{array}{r} 22 \\ \times\ 14 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 28 \\ \times\ 18 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 11 \\ \times\ 13 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 24 \\ \times\ 19 \\ \hline \\ \hline \end{array}$

5. $\begin{array}{r} 23 \\ \times\ 13 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 33 \\ \times\ 33 \\ \hline \\ \hline \end{array}$

7. $\begin{array}{r} 40 \\ \times\ 18 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 42 \\ \times\ 12 \\ \hline \\ \hline \end{array}$

9. $\begin{array}{r} 32 \\ \times\ 11 \\ \hline \\ \hline \end{array}$

10. $\begin{array}{r} 41 \\ \times\ 22 \\ \hline \\ \hline \end{array}$

11. $\begin{array}{r} 97 \\ \times\ 88 \\ \hline \\ \hline \end{array}$

12. $\begin{array}{r} 85 \\ \times\ 60 \\ \hline \\ \hline \end{array}$

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13. $\begin{array}{r} 47 \\ \times\ 18 \\ \hline \\ \hline \end{array}$

14. $\begin{array}{r} 53 \\ \times\ 34 \\ \hline \\ \hline \end{array}$

15. $\begin{array}{r} 65 \\ \times\ 24 \\ \hline \\ \hline \end{array}$

16. $25 \times 17 =$

17. $37 \times 31 =$

18. $62 \times 13 =$

19. $56 \times 29 =$

20. $74 \times 53 =$

Multiplication of three-digit numbers by a multiplier with up to two-digits

Example 1

$217 \times 4 =$

Solution	Steps
$\begin{array}{r} 217 \\ \times\ 4 \\ \hline 868 \\ \hline \end{array}$	1. $4 \times 7 = 28$. Write 8 in the ones place. Regroup 20 ones into 2 tens. Shift 2 tens to the tens place. 2. $4 \times 10 = 40$. Write 4 tens. Add tens: $4 + 2 = 6$. Write 6 in the tens place. 3. $200 \times 4 = 800$. Write 8 in the hundreds place. 4. Add the results from Step 1 to 3: $68 + 800 = 868$

Therefore, $217 \times 4 = 868$.

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Example 2

$278 \times 13 =$

Steps

The answer can be obtained through a horizontal arrangement as follows:

1. Multiply 278 by 3 ones: $278 \times 3 = 834$.
2. Multiply 278 by 1 tens 1: $278 \times 10 = 2780$.
3. Add the results from Steps 1 and 2: $834 + 2780 = 3614$.

Therefore, $278 \times 13 = 3614$.

Example 3

$624 \times 35 =$

Solution	Steps
624 × 35 3120 +18720 21840	1. Multiply 624 by 5 ones: $624 \times 5 = 3120$. 2. Multiply 624 by 3 tens: $624 \times 30 = 18720$. 3. Add the results from Steps 1 and 2: $3120 + 18720 = 21840$.

Therefore, $624 \times 35 = 21{,}840$.

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Exercise 3

1. $417 \times 11 =$
2. $530 \times 8 =$
3. $292 \times 37 =$
4. $709 \times 5 =$
5. $500 \times 40 =$
6. $810 \times 10 =$
7. $345 \times 8 =$
8. $359 \times 79 =$
9. $916 \times 58 =$
10. $999 \times 97 =$
11. $904 \times 9 =$
12. $675 \times 55 =$
13. $742 \times 79 =$
14. $379 \times 27 =$
15. $174 \times 71 =$
16. $486 \times 43 =$
17. $737 \times 35 =$
18. $837 \times 61 =$
19. $496 \times 63 =$
20. $515 \times 7 =$

Multiplication of numbers with up to six digits by a multiplier with up to three digits

Example 1

$$\begin{array}{r} 3125 \\ \times \quad 12 \\ \hline \\ \hline \end{array}$$

Solution

$$\begin{array}{r} 3125 \\ \times \quad 12 \\ \hline 6250 \\ +31250 \\ \hline 37500 \\ \hline \end{array}$$

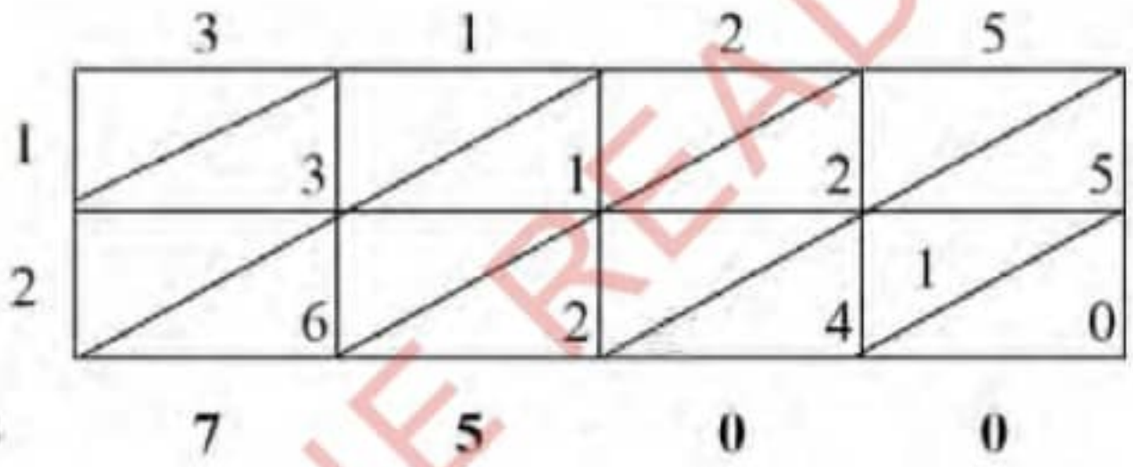
Therefore, the answer is 37,500.

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Alternative method

You can use the method of boxes by following the steps;

1. Draw boxes according to the digits given both horizontally and vertically.
2. Write the multiplier across the top horizontally and the multiplicand on the right and vertically.
3. Divide each box diagonally to create two triangles.
4. Multiply each digit of the multiplier by each digit of the multiplicand.
5. In each box write the tens in the top triangle and the ones in the bottom triangle.
6. Add the numbers diagonally.



Therefore, the answer is 37,500.

Example 2

$276 \times 123 =$

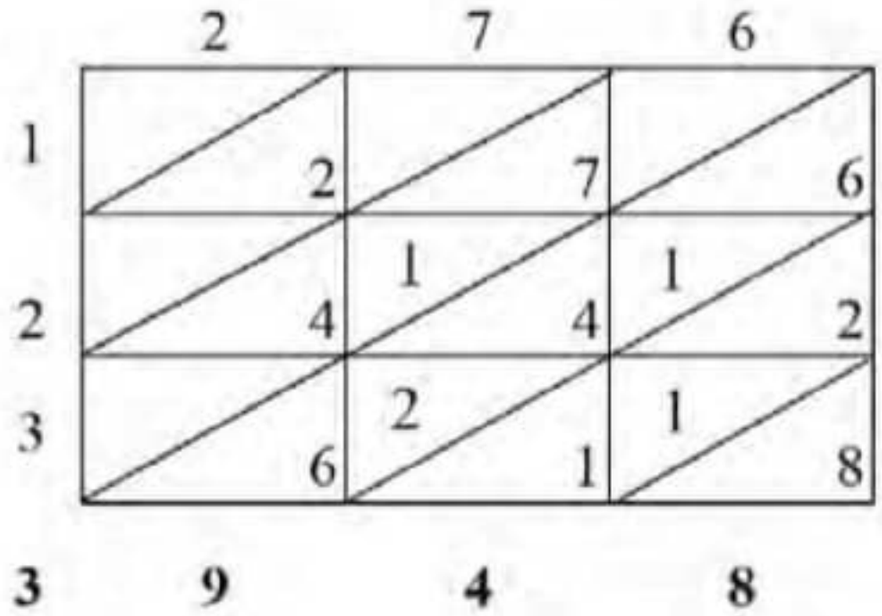
Solution

$$\begin{array}{r} 276 \\ \times\ 123 \\ \hline 828 \\ 5520 \\ +27600 \\ \hline 33948 \\ \hline \end{array}$$

Therefore, the answer is 33,948.

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Alternative method



Therefore, the answer is 33,948.

Example 3

$42538 \times 39 =$

The answer can be obtained through a horizontal arrangement method as follows:

1. Multiply 42538 by 9 ones: $42538 \times 9 = 382842$.
2. Multiply 42538 by 3 tens: $42538 \times 30 = 1276140$.
3. Add the results from Steps 1 and 2:
   $382842 + 1276140 = 1658982$.

Therefore, $42538 \times 39 = 1,658,982$.

Example 4

$$\begin{array}{r} 771481 \\ \times \quad 8 \\ \hline \\ \hline \end{array}$$

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$$\begin{array}{r} 771481 \\ \times \quad 8 \\ \hline 6171848 \\ \hline \end{array}$$

Therefore, the answer is 6,171,848.

Exercise 4

1. $109 \times 125 =$
2. $1592 \times 145 =$
3. $25578 \times 24 =$
4. $8202 \times 200 =$
5. $97227 \times 20 =$
6. $384567 \times 19 =$
7. $6088 \times 40 =$
8. $50009 \times 120 =$
9. $90000 \times 90 =$
10. $231 \times 213 =$
11. $4181 \times 54 =$
12. $3722 \times 946 =$
13. $\begin{array}{r} 10979 \\ \times \quad 241 \\ \hline \\ \hline \end{array}$
14. $\begin{array}{r} 95413 \\ \times \quad 42 \\ \hline \\ \hline \end{array}$
15. $\begin{array}{r} 521 \\ \times 132 \\ \hline \\ \hline \end{array}$
16. $\begin{array}{r} 239586 \\ \times \quad 24 \\ \hline \\ \hline \end{array}$
17. $\begin{array}{r} 3416 \\ \times \quad 244 \\ \hline \\ \hline \end{array}$
18. $\begin{array}{r} 42762 \\ \times \quad 52 \\ \hline \\ \hline \end{array}$
19. $\begin{array}{r} 105000 \\ \times \quad 102 \\ \hline \\ \hline \end{array}$
20. $\begin{array}{r} 81271 \\ \times \quad 708 \\ \hline \\ \hline \end{array}$

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Word problems involving multiplication of whole numbers

Example 1

A businessman bought 170 boxes of bar soap. Each box contains 20 bars of soap. How many bars of soap were bought in total?

Solution

1 box contains 20 bars of soap.

170 boxes contain $170 \times 20 = 3400$ bars.

Therefore, the businessman bought a total of 3,400 bars of soap.

Example 2

The school bought 5624 crates of soda. If each crate had 24 bottles of soda, how many bottles of soda were bought in total?

Solution

Number of crates 5,624

Each crate contained 24 bottles of soda.

$$\begin{array}{r} 5624 \\ \times \quad 24 \\ \hline 22496 \\ +112480 \\ \hline 134976 \\ \hline \end{array}$$

Therefore, the school bought a total of 134,976 bottles of soda.

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Exercise 5

1. A school bus carries 60 sacks of beans per trip. How many sacks of beans will the bus carry in 19 trips?
2. An average of 128 children are born each month in a certain hospital. How many children are expected to be born in 12 months?
3. Pupils from 309 schools sat for an examination. If each school had 208 pupils, how many pupils in total did the examination?
4. An airport serves 462015 passengers per month. How many passengers will the airport serve in five months if the number remains constant each month?
5. Villagers planted 3841 trees in each nursery. If they planted trees in 389 nurseries, how many trees did the villagers plant in total?
6. A warehouse can store 8985 sacks. How many sacks can 728 such warehouses store in total?
7. A factory prints 685390 books per day. How many books will the factory print in 317 days?
8. A farmer keeps 5998 laying hens. If each hen lays two eggs per day, how many eggs will be layed in nine weeks?

Activity 2 — Learning multiplication through online lessons

Description

Use reliable online lessons to learn more about the concept of multiplication of whole numbers.

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Summary

1. When multiplying numbers, arrange them in either vertical or horizontal order.
2. Multiply the digits starting from the ones, tens, hundreds, and so on.
3. The multiplier is the number used to multiply another number.

Revision Exercise

1. $86 \times 11 =$
2. $710 \times 12 =$
3. $8714 \times 126 =$
4. $2060 \times 73 =$
5. $650 \times 130 =$
6. $8475 \times 410 =$
7. $15001 \times 14 =$
8. $61721 \times 102 =$
9. $172 \times 13 =$
10. $102510 \times 103 =$
11. $30016 \times 12 =$
12. $22381 \times 111 =$
13. $511 \times 105 =$
14. $4151 \times 16 =$
15. $315233 \times 24 =$
16. $93115 \times 104 =$
17. $\begin{array}{r} 7492 \\ \times \quad 11 \\ \hline \\ \hline \end{array}$
18. $\begin{array}{r} 165 \\ \times \quad 21 \\ \hline \\ \hline \end{array}$
19. $\begin{array}{r} 414 \\ \times 232 \\ \hline \\ \hline \end{array}$
20. $\begin{array}{r} 79276 \\ \times \quad 15 \\ \hline \\ \hline \end{array}$

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21. If there are 6 boxes of pencils and each box contains 12 pencils, how many pencils are there in total?

22. A businessman sells 88,960 handbags each day. How many handbags will be sold in total in a 31-days month?

23. A Cooperative society had 172 who collected 85 bags of cashews each. How many bags of cashews did the members collect in total?

24. A textile factory produces 96,540 dresses each day. How many dresses will the factory produce in 365 days?

25. A train transports 61,229 bags of cement in each day. How many bags of cement will the train transport in 65 days?

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Chapter Three

Division of whole numbers

Introduction

In this chapter, you will learn to divide whole numbers. The competencies developed will enable you to apply the concept of division in arranging and dividing resources fairly in many aspects of daily life.



Think

Life without the knowledge of division

Division of whole numbers without remainders

When dividing whole numbers, begin with the digit from the left of the number towards the right. A number that divides another number is known as the dividend and a number used to divide is known as the divisor. The obtained answer is known as the quotient.

Division of whole numbers by short method without remainders

Example 1

$68 \div 2 =$

Steps

1. Divide 6 tens by 2 to get 3 tens. Write 3 in the tens place.
2. Divide 8 ones by 2 to get 4 ones. Write 4 in the ones place, to the right of 3 tens.
   Therefore, $68 \div 2 = 34$.

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Example 2

$245 \div 5 =$

Steps

1. Divide 2 by 5, it is not sufficient.
2. Take 24 divide by 5 to get 4 with a remainder of 4. Write 4 in the tens place. Regroup 4 into 40 ones. Add ones: $40 + 5 = 45$.
3. Divide 45 by 5 to get 9. Write 9 in the ones place, to the right of 4.

Therefore, $245 \div 5 = 49$.

Example 3

$2\overline{)284}$

Solution	Steps
$2\overline{)284}$ 142	1. Divide 2 by 2 to get 1. Write 1 in the hundreds place. 2. Divide 8 by 2 to get 4. Write 4 in the tens place, to the right of 1. 3. Divide 4 by 2 to get 2. Write 2 in the ones place, to the right of 4.

Therefore, the answer is 142.

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Example 4

$12\overline{)384}$

Solution	Steps
$12\overline{)384}$ 32	1. Divide 3 by 12, it is not sufficient. Thus, take 38 then divide by 12. You will get 3 with remainder of 2. Write 3 in the tens place. 2. Regroup 2 into 20 ones. Add ones: $20 + 4 = 24$. 3. Divide 24 by 12 to get 2. Write 2 in the ones place to the right of 3.

Therefore, the answer is 32.

Exercise 1

1. $70 \div 7 =$
2. $36 \div 6 =$
3. $54 \div 3 =$
4. $72 \div 3 =$
5. $86 \div 2 =$
6. $96 \div 6 =$
7. $156 \div 4 =$
8. $272 \div 8 =$
9. $981 \div 3 =$
10. $225 \div 5 =$
11. $196 \div 4 =$
12. $175 \div 5 =$
13. $285 \div 19 =$
14. $85 \div 5 =$
15. $378 \div 27 =$
16. $2\overline{)64}$
17. $3\overline{)93}$
18. $9\overline{)999}$
19. $8\overline{)168}$
20. $20\overline{)400}$
21. $7\overline{)217}$
22. $12\overline{)264}$
23. $31\overline{)961}$
24. $75\overline{)825}$

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Division of whole numbers by long method without the remainders

Example 1

$69 \div 3 =$

Solution	Steps
$\begin{array}{r} 23 \\ 3\overline{)69} \\ -\ 6\downarrow \\ \hline 9 \\ -9 \\ \hline 0 \end{array}$	1. Divide 6 by 3 to get 2. Write 2 above the tens place. 2. Multiply 2 by 3 to get 6. Write 6 below 6, then subtract: $6 - 6 = 0$. 3. Drop 9 ones, then divide by 3 to get 3. Write 3 in the ones place to the right of 2. 4. Multiply 3 by 3 to get 9. Write 9 below 9, then subtract: $9 - 9 = 0$.

Therefore, $69 \div 3 = 23$.

Example 2

$80\overline{)8080}$

Solution	Steps
$\begin{array}{r} 101 \\ 80\overline{)8080} \\ -\underline{80}\downarrow \\ 8 \\ -0\downarrow \\ \hline 80 \\ -80 \\ \hline 0 \end{array}$	1. Divide 80 by 80 to get 1. Write 1 above the hundreds place. 2. Multiply 80 by 1 to get 80. Write 80 below 80, then subtract: $80 - 80 = 0$. 3. Drop 8 tens, then divide 8 by 80 to get 0. Write 0 above the tens place, to the right of 1. 4. Multiply 0 by 80 to get 0. Write 0 below 8 then subtract: $8 - 0 = 8$.

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5. Drop 0 ones on the 8 tens to get 80. Divide 80 by 80 to get 1. Write 1 above the ones place, to the right of 0.
6. Multiply 1 by 80 to get 80. Write 80 below 80, then subtract: $80 - 80 = 0$.

Therefore, the answer is 101.

Example 3

$696\overline{)561672}$

Solution

$$\begin{array}{r} 807 \\ 696\overline{)561672} \\ -\,5568 \\ \hline 487 \\ -\,000 \\ \hline 4872 \\ -4872 \\ \hline 0 \end{array}$$

Therefore, the answer is 807.

Exercise 2

1. $3\overline{)723}$
2. $605\overline{)55055}$
3. $78\overline{)23400}$
4. $926\overline{)77784}$
5. $25\overline{)10000}$
6. $6\overline{)660}$
7. $10\overline{)356250}$
8. $60\overline{)66000}$
9. $3512\overline{)231792}$

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10. $9\overline{)126}$

11. $4255\overline{)51060}$

12. $812\overline{)730800}$

13. $8\overline{)824}$

14. $16\overline{)29024}$

15. $7\overline{)847}$

16. $27\overline{)891}$

17. $124\overline{)64852}$

18. $12\overline{)576}$

19. $3\overline{)495}$

20. $32\overline{)20192}$

21. $2512\overline{)87920}$

22. $75\overline{)28950}$

23. $10\overline{)29190}$

24. $34\overline{)782}$

25. $364\overline{)22932}$

26. $84\overline{)924}$

27. $18\overline{)684}$

Division of whole numbers with the remainder

In some cases, the dividend may not be an exact multiple of the divisor. In such cases, division gives a quotient and a remainder.

Division of whole numbers by short method with remainders

Example 1

$6\overline{)82}$

Solution	Steps
$6\overline{)82}$ 13 remainder 4	1. Divide 8 tens by 6 to get 1 ten and the remainder of 2 tens. 2. Write 1 below 8 and then take the remainder of 2 tens and 2 ones to get 22. 3. Divide 22 by 6 to get 3 remainder 4. Write 3 below 2 and 4 as the remainder to the right of 13.

Therefore, the answer is 13 and the remainder is 4.

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Example 2

12)573

Solution	Steps
12)573 47 remainder 9	1. Divide 5 by 12, it is not sufficient. Take 57 then divide by 12 to get 4 and remainder 9. 2. Write 4 below 7, then combine the remainder 9 and 3 ones to form 93. 3. Divide 93 by 12 to get 7 and remainder 9. Write 7 below 3 and 9 as the remainder to the right of 47.

Therefore, the answer is 47 and the remainder is 9.

Example 3

23)14262

Solution

23)14262
620 remainder 2

Therefore, the answer is 620 and the remainder is 2.

Exercise 3

1. 3)457
2. 5)99
3. 9)68
4. 22)938
5. 12)4005
6. 644 ÷ 10 =
7. 610)313252
8. 43)267
9. 111)2461
10. 58)4940
11. 847 ÷ 84 =
12. 340)1720

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13. $139 \div 11 =$

14. $32\overline{)3407}$

15. $234 \div 38 =$

16. $28\overline{)20364}$

Division of whole numbers by long method with a remainder

Example 1

$7\overline{)148}$

Solution	Steps
21 $7\overline{)148}$ $-14\downarrow$ 8 -7 1	1. Divide 14 by 7 to get 2. Write 2 above the tens place. 2. Multiply 2 by 7 to get 14. Write 14 below 14 then subtract: $14 - 14 = 0$. 3. Drop 8 one, then divide by 7 to get 1. Write 1 above the ones place. 4. Multiply 1 by 7 to get 7. Write 7 below 8, then subtract: $8 - 7 = 1$.

Therefore, the answer is 21 and the remainder is 1.

Example 2

$27\overline{)866}$

Solution	Steps
32 $27\overline{)866}$ $-81\downarrow$ 56 -54 2	1. Divide 86 by 27 to get 3. Write 3 above the tens place. 2. Multiply 3 by 27 to get 81. Write 81 below 86 then subtract: $86 - 81 = 5$. 3. Drop the 6 ones to get 56. Divide 56 by 27 to get 2. Write 2 above the ones place, to the right of 3.

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	4. Multiply 2 by 27 to get 54. Write 54 below 56, then subtract: $56 - 54 = 2$.

Therefore, the answer is 32 and the remainder is 2.

Example 3

$8806\overline{)880890}$

Solution

$$
\begin{array}{r}
100 \\
8806\overline{)880890} \\
-8806 \\
\hline
2 \\
-0\downarrow \\
\hline
29 \\
-0\downarrow \\
\hline
290
\end{array}
$$

Therefore, the answer is 100 and the remainder is 290.

Exercise 4

1. $6\overline{)467}$
2. $1279\overline{)255812}$
3. $5\overline{)327}$
4. $25\overline{)815}$
5. $90\overline{)405}$
6. $251\overline{)5397}$
7. $104\overline{)84765}$
8. $43\overline{)267}$
9. $5850\overline{)175594}$
10. $48\overline{)4954}$
11. $32\overline{)14498}$
12. $12\overline{)1235}$

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13. $21\overline{)25240}$

14. $35\overline{)159610}$

15. $164\overline{)527756}$

16. $11\overline{)139}$

17. $1450\overline{)72507}$

18. $14\overline{)535}$

19. $28\overline{)234}$

20. $200\overline{)502808}$

Word problems involving division of whole numbers

Word problems on division shows the use of division in daily life activities.

Example 1

Thirteen farmers shared 1,625 chickens equally. How many chickens did each farmer get?

Solution

Number of farmers = 13

Number of chickens = 1625

Divide 1625 by 13 to get 125

Therefore, each farmer got 125 chickens.

Example 2

A teacher had 305 oranges. If she distributed them equally among 35 pupils, how many oranges did each pupil receive? How many oranges did the teacher remain with?

Solution

Number of oranges = 305

Number of pupils = 35

Divide 305 by 35 to get 8 and the remainder is 25.

Therefore, each student received 8 oranges and the teacher remained with 25 oranges.

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Exercise 5

1. If 133,700 sacks of maize were distributed equally among 50 villages, how many sacks did each village receive?
2. Twenty-four primary schools planted a total of 446,400 trees. How many trees did each school plant if each school planted an equal number of trees?
3. A teacher distributed 414 papers equally among 60 pupils. How many papers did each
   (a) pupil receive?
   (b) teacher remain with?
4. On average, 50 factories produce 80,000 tiles per day. How many tiles does each factory produce on average per day?
5. If 955 sacks of maize were distributed equally among 40 households,
   (a) how many sacks of maize did each household receive?
   (b) how many sacks of maize remained?
6. If 99,000 books were distributed equally among 200 schools, how many books did each school receive?
7. If 100,800 chalks are packed equally in 700 boxes. How many chalks does each box contain?
8. Twenty-four members of the cooperative society shared 986 pineapples. If the pineapples were shared equally, how many pineapples were left over?
9. A businessperson sells 455,000 pens in 14 days. If the sales are the same each day, how many pens does the businessperson sell daily?
10. Classrooms in 30 schools were renovated using 136,890 iron sheets. If each classroom used an equal number of iron sheets, how many iron sheets did each school use?

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Activity Learning more about the division of whole numbers through online resources and programs

Description

Explore various examples of dividing whole numbers using online resources and programs such as GeoGebra and Khan Academy.

Summary

1. A number being divided is known as the dividend, a number that divides another number is known as the divisor, and the number obtained after division is known as the quotient.
2. The quotient may be with or without a remainder.
3. In the division of whole numbers, start dividing the digits of the dividend from the left side towards the right.

Revision Exercise

1. $32352 \div 16 =$
2. $500000 \div 1125 =$
3. $3069 \div 99 =$
4. $363600 \div 300 =$
5. $31\overline{)161}$
6. $760\overline{)174800}$
7. $90\overline{)405000}$
8. $998\overline{)558880}$
9. $95\overline{)344}$
10. $3500\overline{)46163}$
11. $55\overline{)868230}$
12. $1514\overline{)113550}$
13. $23\overline{)575}$
14. $2511\overline{)37665}$
15. $63\overline{)204687}$
16. A total of 984,000 people cast their votes equally at each polling station. If they voted at 820 stations, how many voters were present at each station?

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17. A teacher distributed 508 exercise books among 95 pupils equally. How many exercise books did each pupil receive, and how many did the teacher remain with?

18. From the following table, write all numbers that are divisible by 8 without leaving a remainder.

560	360	124	270	600
94	70	80	208	40
12	63	30	188	120
54	190	49	250	72

19. A total of 814,000 people watched football matches at 37 different venues. If each venue had an equal number of people, how many people were there at each venue?

20. The Government distributed a total of 17,556 bags of fertilizer to 231 farmers. If the bags were allocated equally to each farmer, how many bags of fertilizer did each farmer receive?

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Chapter Four

Measurements of length, mass and volume

Introduction

In this chapter, you will learn the metric measurements of length, mass, and volume. You will also learn to convert, add, and subtract the measurements. The competencies developed will be used in constructions, tailoring, cooking, hospitals, agriculture, designing and many other applications.



Think

Life without measurements

Conversion of metric units of length

The basic standard metric unit of length is metre (m). Other commonly used metric units of length are millimetres (mm), centimetres (cm) and kilometres (km). Length measurements can be converted from one metric unit to another.

The following table shows the relationships of metric units of length.

Millimetre (mm)	centimetre (sm)	decimetre (dm)	metre (m)	decametre (dam)	hectometre (hm)	kilometre (km)
1						
10	1					
100	10	1				
1,000	100	10	1			
10,000	1,000	100	10	1		
100,000	10,000	1,000	100	10	1	
1,000,000	100,000	10,000	1,000	100	10	1

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Conversion of large metric units of length into small units

The measurements of length can be converted from large metric units to small units. The relationships of metric units of length are used in converting the units according to the needs. Use multiplication (×) method when converting larger units to smaller units.

Example 1

Convert 3 m into centimetres.

Solution

Use the relationship between metre and centimetre.

1 m = 100 cm

3 m = 100 × 3 cm

= 300 cm

Therefore, 3 m is equal to 300 cm.

Example 2

Convert 5 km 250 m into metres.

Solution

Use the relationship between kilometre and metre.

1 km = 1000 m

5 km = 1000 × 5 m

= 5000 m

Add 5000 m and 250 m. Thus, 5000 m + 250 m = 5,250 m

Therefore, 5 km 250 m is equal to 5,250 m.

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Example 3

Convert 7 m into decimetres.

Solution

Use the relationship between metre and decimetre.

$1\,m = 10\,dm$

$7\,m = 10 \times 7\,dm$

$= 70\,dm$

Therefore, 7 m is equal to 70 dm.

Exercise 1

1. Convert the following measurements:
   (a) 17 km into metres
   (b) 7 km 500 m into centimetres
   (c) 35 cm into millimetres
   (d) 7 dam into metres
2. A road is 22 kilometres long. Convert this length to decimetres.
3. Convert the following measurements:
   (a) A farm of length 268 metres into millimetres
   (b) A building with a height of 63 dam into metres
   (c) A board with a length of 3 metres and 20 centimetres into millimetres
   (d) A tower with a height of 15 hectometres into metres.

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Conversion of small units of length into large units

Length measurements can be converted from small units to large units. Use multiplication (÷) method when converting small units into large units.

Example 1

Convert 500 cm into metres.

Solution

Use the relationship between metre and centimetre

1 m = 100 cm

$$500\,\text{cm} = \frac{500\,\text{cm} \times 1\,\text{m}}{100\,\text{cm}}$$

$$= 5\,\text{m}$$

Therefore, 500 cm is equal to 5 m.

Example 2

Convert 4500 m into kilometres and metres.

Solution

Use the relationship between kilometre and metre

m 4500 = m 4000 + m 500

Convert 4000 m into kilometre:

1 km = 1000 m

$$4000\,\text{m} = \frac{4000\,\text{m} \times 1\,\text{km}}{1000\,\text{m}}$$

$$= 4\,\text{km}$$

500 m is less than 1 km.

Therefore, 4500 m = 4 km 500 m.

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Exercise 2

1. Convert the following measurements into kilometres:
   (a) 9000 m (b) 15000 m (c) 6000 m (d) 9750 m
2. Convert the following measurements into metres:
   (a) 2000 cm (b) 150 cm (c) 325 cm (d) 2025 cm
3. Convert the following measurements into centimetres:
   (a) 40 mm (b) 100 mm (c) 245 mm (d) 550 mm

Addition of measurements of length

Measurements of the same unit can be added. Mixed measurements are added after converting them into a common unit using the relationship between the given units.

Example 1

8 km 3 hm + 6 km 5 hm =

Solution

km	hm
8	3
+ 6	5
	8

Steps

1. Add hm: 3 hm + 5 hm = 8 hm.
   Write 8 in the hm column.

km	hm
8	3
+ 6	5
14	8

2. Add km: 8 km + 6 km = 14 km.
   Write 14 in the km column.

Therefore, the answer is 14 km 8 hm.

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Example 2

km 7 hm 5 + km 4 hm 8 =

Solution

km	hm
$^{+1}7$	5
+ 4	8
	3

Steps

1. Add hm: 5 hm + 8 hm = 13 hm.
   Convert 13 hm into 1 km and 3 hm.
   Write 3 in the hm column. Take 1 km to the km column.

km	hm
$^{+1}7$	5
+ 4	8
12	3

2. Add km: 1 km + 7 km + 4 km = 12 km.
   Write 12 in the km column.

Therefore, the answer is 12 km 3 hm.

Example 3

km	hm	dam
7	4	9
+ 6	6	3

Solution

km	hm	dam
$^{+1}7$	$^{+1}4$	9
+ 6	6	3
14	1	2

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Exercise 3

1.

cm	mm
13	3
+ 5	4

2.

cm	mm
1	9
+ 3	2

3.

m	dm	mm
3	6	7
+ 7	5	9

4.

km	hm
5	6
+ 2	8

5.

m	cm	mm
4	25	4
+ 6	80	8

6.

km	dam	m
8	9	7
+ 5	8	4

7.

km	m	cm
4	40	26
3	22	68
+ 5	71	23

8.

km	m
9	56
+ 6	48

9.

dm	cm	mm
14	4	6
40	8	6
+ 12	3	1

Subtraction of measurements of length

Subtraction can be performed when the measurements are of the same units. Mixed measurements can be subtracted after converting them using their relationships.

Example 1

m	cm
4	45
− 2	20

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Solution	Steps
m cm 4 45 − 2 20 25	1. Subtract cm: 45 cm − 20 cm = 25 cm. Write 25 cm in the cm column.
m cm 4 45 − 2 20 2 25	2. Subtract m: 4 m − 2 m = 2 m. Write 2 in the m column.

Therefore, the answer is 2 m 25 cm.

Example 2

km m
12 200
− 4 950

Solution	Steps
km m 12 200 − 4 950	1. Subtract metres: 200 m − 950 m, it is not sufficient. Take 1 km from 12 km and convert it into 1000 m.
km m 11 1200 − 4 950 250	2. Add metres: 1000 m + 200 m = 1200 m. Now, subtract metres: 1200 m − 950 m = 250 m. Write 250 in the m column. Remember that 11 km remained in the km column.

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km	m
11	1200
− 4	950
7	250

3. Subtract kilometres:
11 km − 4 km = 7 km. Write 7 in the km column.

Therefore, the answer is 7 km 250 m.

Example 3

km	m	sm
10	160	55
− 4	580	76

Solution

km	m	cm
10	159	155
− 4	580	76
		79

Steps

1. Subtract centimetres:
55 cm − 76 cm, it is not sufficient. Take 1 m from 160 m and convert it to get 100 cm. Add centimetres;
100 cm + 55 cm = 155 cm.
Now, subtract cm:
155 cm − 76 cm = 79 cm. Write 79 in the cm column. Remember that 159 m remained in the m column

km	m	cm
9	1159	155
− 4	580	76
	579	79

2. Subtract metres: 159 m − 580 m, it is not sufficient. Take 1 km from 10 km and convert it into 1000 m. Add metres: 1000 m + 159 m = 1159 m.
Now, subtract metres:
1159 m − 580 m = 579 m. Write 579 in the m column. Remember that 9 km remained in the km column.

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km	m	cm
9	1159	155
− 4	580	76
5	579	79

3. Subtract kilometres:
9 km − 4 km = 5 km. Write 5 in the km column.

Therefore, the answer is 5 km 579 m 79 cm.

Exercise 4

1.

km	m
105	200
− 5	105

2.

km	m	cm
27	240	64
− 14	860	95

3.

m	cm
12	30
− 4	35

4.

km	m	dm
125	120	8
− 85	260	3

5.

dm	cm
10	5
− 3	7

6.

dm	cm	mm
9	4	9
− 3	7	6

7.

cm	mm
9	9
− 2	3

8.

dm	mm
75	2
− 4	5

9.

km	dam	cm
17	6	61
− 10	3	5

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Word problems involving measurements of length

Example 1

The length of a football ground is 100 metres. Convert this length into centimetres.

Solution

Convert 100 m into centimetre as follows:

$1\ m = 100\ cm$

$100\ m = 100\ cm \times 100$

$= 10000\ cm$

Therefore, the length of the football ground is 10,000 cm.

Example 2

Kija walked a distance of 8 kilometres and 700 metres from his home to the open market. Ashura walked a distance of 6,928 metres from school to her home.

(a) Who walked a longer distance?

(b) Find the difference in metres between the distances walked by Kija and Ashura.

Solution

(a) Kija walked 8 km and 700 m. Convert 8 km and 700 m into metres as follows.

$1\ km = 1000\ m$

$8\ km = 8 \times 1000\ m$

$= 8000\ m$

Add 8000 m and 700 m.

$8000\ m + 700\ m = 8700\ m.$

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Ashura walked a distance of 6928 m which is less than 8700 m.

Thus, Kija walked a longer distance than Ashura.

(b) The difference in the distances walked is;

8700 m − 6928 m = 1772 m.

Therefore, the difference in distance walked is 1772 m.

Exercise 5

1. The length of the Standard Four classroom is 6 m and 75 cm. Convert the length of the classroom into centimetres.
2. The distance from Mtakuja market to Lake Tanganyika is 12 km. How many metres are equivalent to this distance?
3. Bilali bought a piece of fabric of length 50 m while Kulwa bought a piece of fabric of length 200 cm. Who bought a longer piece of fabric? Give reasons for your answer.
4. The distance from the school to the dispensary is 800 m. The distance from the school to the market is 1 km. Find the difference between the two distances.
5. The distance from the Standard Four classroom to the football ground is 7,000 cm. The distance from the Standard Four classroom to the Head teacher's office is 70 m. Which distance between the two is longer?
6. Bahati walks 3 km and 500 m every morning, and 1 km and 500 m every evening. Doto walks 5 km and 100 m every morning. Who walks a longer distance?
7. A certain road has a length of 3 kilometres and 500 metres. What is the length of the road in metres?

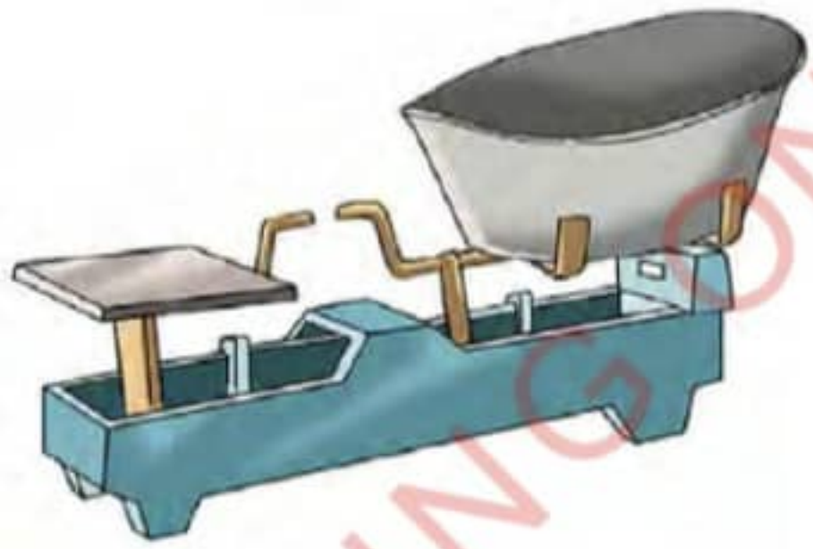
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Measurements of mass

The mass of an object is measured in metric units which include tonnes (t), kilograms (kg), hectograms (hg), dekagrams (dag), grams (g), decigrams (dg), centigrams (cg) and milligrams (mg).

The following pictures show the example of a weighing balance.





Relationships of metric units of mass

The following table shows the relationships of metric units of mass.

milligram (mg)	centigram (cg)	decigram (dg)	gram (g)	dekagram (dag)	hectogram (hg)	kilogram (kg)
1						
10	1					
100	10	1				
1,000	100	10	1			
10,000	1,000	100	10	1		
100,000	10,000	1,000	100	10	1	
1,000,000	100,000	10,000	1,000	100	10	1

1 tonne (t) = 1000 kilograms (kg)

From the table, the relationship between measurements can be established. For example, the following relationships can be easily established:

(a) 1 gram = 1000 milligrams
(b) 1 centigram = 10 milligrams
(c) 1 gram = 100 centigrams
(d) 1 kilogram = 1000 grams

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Conversion of measurements of mass

Large metric units of mass can be converted into small units. Similarly, small metric units of mass can be converted into large units. The conversion is done by considering the relationship between the metric units involved.

Example 1

Convert 4 kg into grams.

Solution

Use the relationship between kilogram and grams

(a) 1 kg = 1000 g

4 kg = 4 × 1000 g

= 4000 g

Therefore, 4 kg = 4000 g.

(b) Convert 7,000 g into kilograms.

1000 g = 1 kg

7 × 1000 g = 7 × 1kg

7000 g = 7 kg

Therefore, 7,000 g = 7 kg

Example 2

Convert the following measurements into kilograms.

(a) 2 tonnes (b) 6,000 g

Solution

(a) Use the relationship between tonnes and kilograms.

1 tonne = 1000 kg

2 × tonne = 2 × 1000 kg

= 2 × 1000 kg

Therefore, 2 tonnes = 2000 kg.

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(b) Use the relationship between grams and kilograms.

$1000\ g = 1\ kg$

$6 \times 1000\ g = 6 \times 1\ kg$

$= 6\ kg$

Therefore, $6{,}000\ g = 6\ kg$.

Exercise 6

1. Convert the following measurements:
   (a) 89 kilograms into grams
   (b) 58,000 grams into hectograms
   (c) 6,500 kilograms into tonnes
   (d) 4 tonnes into kilograms
2. A farmer harvested maize weighing 5,070,000 grams. What is this mass in kilograms?
3. A vehicle has a mass of 15 tonnes. Convert this mass into hectograms.
4. A car carries 1,200 kilograms of beans and maize which weighs 5 tonnes, find the difference between the mass of beans and that of maize in kilograms and grams.
5. Mwanaisha is going to prepare three different sorts of biscuit. The following are the recipes.

Chocolate biscuits	Oat biscuits	Nut biscuits
sugar 120 g	sugar 140 g	sugar 110 g
butter 60 g	butter 170 g	butter 160 g
wheat flour 100 g	oats flour 80 g	wheat flour 230 g
cocoa powder 20 g	powdered milk 30 g	nuts powder 60 g

(a) What is the total mass of the ingredients for each recipe.
(b) How much sugar, butter, nuts and flour ingredients are needed to make all of three types of biscuits.

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Addition of measurements of mass

Metric units of mass of the same measurement can be added. Start by adding smaller units followed by large units. The relationship between the given units should be considered.

Example 1

g	mg
4	225
+ 4	370

Solution	Steps
g mg 4 225 +4 370 595	1. Add milligrams: 225 mg + 370 mg = 595 mg. Write 595 in the mg column.
g mg 4 225 + 4 370 8 595	2. Add grams: 4 g + 4 g = 8 g. Write 8 in the g column.

Therefore, the answer is 8 g 595 mg.

Example 2

t	kg
4	850
+ 3	350

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Solution	Steps
t kg +1 4 850 + 3 350 200	1. Add kilograms: 850 kg + 350 kg = 1200 kg. Convert 1200 kg to get 1 tonne. Write 200 in the kg column. Take 1 tonne to the tonnes column.
t kg +1 4 850 + 3 350 8 200	2. Add tonnes: 1 tonne + 4 tonnes + 3 tonnes = 8 tonnes. Write 8 in the tonnes column.

Therefore, the answer is 8 tonnes 200 kilograms.

Example 3

t	kg
10	470
+ 17	675

Solution

t	kg
+1 10	470
+ 17	675
28	145

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Exercise 7

1.

t	kg
6	750
+ 5	250

2.

kg	dag
4	75
+ 8	25

3.

g	mg
2	175
+ 3	945

4.

hg	dg
19	375
+ 14	765

5.

t	dag
4	500
+ 7	660

6.

kg	g
3	800
+ 3	434

7.

kg	g
3	460
+ 1	630

8.

t	cg	mg
12	216	9
+ 5	801	6

9.

t	hg	g
50	90	10
25	70	30
+ 20	55	30

10.

hg	g	cg
40	80	30
71	14	40
+ 20	22	70

11.

hg	dg
14	365
9	475
+ 3	300

12.

t	kg	g
9	896	735
+ 8	250	289

Subtraction of measurements of mass

Metric units of mass of the same measurement can be subtracted. Start by subtracting smaller units followed by larger units. In regrouping the units, the relationship between the given units should be considered.

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Example 1

tonne	kg
12	740
− 6	420

Solution	Steps
t kg 12 740 − 6 420 320	1. Subtract kilograms: 740 kg − 420 kg = 320 kg. Write 320 in the kg column.
t kg 12 740 − 6 420 6 320	2. Subtract tonnes: 12 tonnes − 6 tonnes = 6 tonnes. Write 6 in the tonnes column.

Therefore, the answer is 6 tonnes 320 kg.

Example 2

tonne	kg	g
6	200	550
− 2	300	750

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Solution

t	kg	g
6	199	1550
−2	300	750
		800

Steps

1. Subtract grams:
   550 g − 750 g, it is not sufficient.
   Take 1 kg from 200 kg and convert it into 1000 g.
   Add grams:
   1000 g + 550 g = 1550 g.
   Then, subtract grams:
   1550 g − 750 g = 800 g. Write 800 in the g column. Remember that, 199 kg remained in the kg column

t	kg	g
5	200	1550
− 2	300	750
	899	800

2. Subtract kilograms:
   199 kg − 300 kg; is not sufficient.
   Take 1 tonne from 6 tonnes and convert it into 1000 kg.
   Add kilograms:
   1000 kg + 199 kg = 1199 kg.
   Then, subtract kilograms:
   1199 kg − 300 kg = 899 kg. Write 899 in the kg column. Remember that, 5 tonnes remained in the tonnes column..

t	kg	g
5	1199	1550
− 2	300	750
3	899	800

3. Subtract tonnes:
   5 tonnes − 2 tonnes = 3 tonnes.
   Write 3 in the tonnes column.

Therefore, the answer is 3 tonnes 899 kg 800 g.

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Exercise 8

1.

t	kg
6	220
− 4	114

2.

t	mg
26	370
− 13	250

3.

kg	g
14	239
−12	910

4.

t	kg
4	54
− 2	340

5.

kg	dag
20	600
− 6	560

6.

kg	g
16	860
−13	900

7.

t	kg	cg
10	143	64
−	337	27

8.

g	mg
40	720
−15	570

9.

hg	g
15	85
−12	90

10.

kg	cg	mg
70	40	60
− 15	12	38

11.

t	g	mg
6	800	660
− 3	280	830

12.

hg	g	cg
4	31	12
− 1	81	4

Word problems involving measurements of mass

Word problems in measurements of mass show how the metric units are used in real-life situations.

Example

A sack of rice weighs 54 kg while a sack of wheat weighs 49,000 g.

(a) Which of the two sacks is heavier?

(b) Find the difference in their masses.

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Solution

(a) Mass of rice $= 54$ kg

Mass of wheat $= 49000$ g

Convert the mass of the sack of rice into kilograms as follows.

$1 \text{ kg} = 1000 \text{ g}$

$54 \text{ kg} = 54 \times 1000 \text{ g}$

$= 54000 \text{ g}$

The sack of rice weighs 54000 g while that of wheat weighs 49000 g. Therefore, the sack of rice is heavier than that of wheat.

(b) The difference in their masses is $54000 \text{ g} - 49000 \text{ g} = 5000 \text{ g}$.

Therefore, the difference in mass is 5000 g.

Exercise 9

1. Which is heavier between 2,000 g of salt and 20 kg of cotton?
2. A fisherman caught fish weighing 3 kg and 500 g. What is the mass of the fish in grams?
3. Elia weighs 32 kg, Zawadi weighs 300 hg and Furaha weighs 30,750 g.
   (a) Who has the greatest mass?
   (b) Find the difference in mass between Elia and Furaha in grams.
4. Buholo bought 2 kg of millet flour, 1 kg of sugar, and 500 g of powdered milk. How many grams of items did she buy?
5. Find the total mass of five loaves of bread in grams if each loaf has a mass of half a kilogram.
6. The mass of a chicken is 3 kg. After preparing it for cooking, its mass was 2 kg 750 g. How many grams did its mass decrease? Give reasons for the decrease in mass.

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Measurements of volume

The basic metric unit of volume is the litre (L). Other commonly used metric unit of volume is millilitre (mL). The following is the relationship between the two units.

1 litre (L) = 1,000 millilitres (mL)

Conversion of measurements of volume

The metric measurements of volume can be converted based on the relationship between the given units.

Example 1

How many millilitres are in 9 litres?

Solution

Use the relationship between millilitres and litres.

To convert litres to millilitres, multiply the number of litres by 1000.

$$1 \text{ litre} = 1000 \text{ mL}$$

$$9\,\text{L} = 9 \times 1000\,\text{mL}$$

Therefore, there are 9000 millilitres in 9 litres.

Example 2

How many litres are in 18000 millilitres?

Solution

To convert millilitres to litres, divide the number of millilitres by 1000.

$$1000 \text{ mL} = 1 \text{ litre}$$

Multiply by 18 both sides to obtain

$$18 \times 1000 \text{ mL} = 18 \text{ L}$$

$$18000 \text{ mL} = 18 \text{ L}$$

$$= 18 \text{ L}$$

Therefore, there are 18 L in 18000 mL.

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Exercise 10

1. How many millilitres are in the following volumes?
   (a) 9 litres (b) 12 litres (c) 3 litres (d) 5 litres
2. Convert the following volumes into litres.
   (a) 14,000 mL (b) 6,000 mL (c) 2,000 mL (d) 8,500 mL
3. A bucket has a volume of 20 litres. What is this volume in millilitres?
4. A water tank is filled with 6 buckets of 20 litres each. How many millilitres of water does the tank hold?

Addition of measurements of volume

Metric units of volume with the same measurements can be added. Start by adding smaller units followed by larger units.

Example 1

	L	mL
	3	600
+	4	450

Solution		Steps
	L mL	1. Add millilitres:
	+1	600 mL + 450 mL = 1050 mL. Convert
	3 600	1050 mL to get 1 L and 50 mL.
+	4 450	Write 50 in the mL column and take 1 litre
	50	from the litre column
	L mL	2. Add litres: 1 L + 3 L + 4 L = 8 L. Write 8
	+1	in the column of L.
	3 600	
+	4 450	
	8 50	

Therefore, the answer is 8 L 50 mL.

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Example 2

6 L 550 mL + 3 L 160 mL =

Solution

	L	mL
	6	550
+	3	160
	9	710

Exercise 11

1.

	L	mL
	415	205
+	27	8

2.

	L	mL
	600	40
+	350	250

2.

	L	mL
	78	426
+	55	542

4.

	L	mL
	4	750
	316	250
+	15	101

5.

	L	mL
	17	800
+	3	366

6.

	L	mL
	20	920
	14	112
+	13	126

Subtraction of measurements of volume

Metric units of volume with the same measurements can be subtracted. Start by subtracting smaller units followed by larger units.

Example 1

L 80 mL 370 – L 45 mL 160 =

Solution

	L	mL
	80	370
–	45	160
		210

Steps

1. Subtract millilitres:
   370 mL – 160 mL = 210 mL.
   Write 210 in the mL column.

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L	mL
80	370
− 45	160
35	210

2. Subtract litres: 80 L − 45 L = 35 L.
Write 35 in the L column.

Therefore, the answer is 35 L 210 mL.

Example 2

L	mL
15	350
− 10	370

Solution

L	mL
14	1350
− 10	370
4	980

Exercise 10

1.

L	mL
66	89
− 52	48

2.

L	mL
500	50
− 300	250

3.

L	mL
68	423
− 41	544

4.

L	mL
45	513
− 16	701

5.

L	mL
46	48
− 33	35

6.

L	mL
72	490
−37	541

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7.

L	mL
100	100
− 15	900

8.

L	mL
330	800
− 130	600

9.

L	mL
75	426
− 53	532

10.

L	mL
11	41
− 8	74

11.

L	mL
20	93
− 16	37

12.

L	mL
90	180
− 45	382

Word problems involving measurements of volume

Word problems involving measurements of volume highlight the use of metric units of volume in real-life situations. They reflect the activities performed every day.

Example

A cow produces 20 litres of milk every day. How many millilitres of milk will the cow produce in 3 days?

Solution

1 Day: 20 litres

3 days $= 3 \times 20\,\text{L} = 60\ \text{L}$

Convert 60 L into millilitres as follows.

$1\,\text{L} = 1000\,\text{mL}$

$60\,\text{L} = 60 \times 1000\,\text{mL}$

$= 60000\,\text{mL}$

So, the cow will produce 60,000 millilitres of milk in 3 days.

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Exercise 13

1. A gas cylinder has a volume of 18 litres. If 13 litres are used, how many litres will remain?
2. How many 500 mL water bottles are needed to fill a 20-litre bucket?
3. A 1 litre bottle contains 750 mL of milk. How much more is needed to fill the bottle?
4. Lina, Sanka, and Imani filled a water bucket. If Lina put in 5 litres, Sanka 2 litres and Imani 3 litres, what is the total volume of water in millilitres that they all put in together?
5. Mbale drank 3 litres of water and Senga drank 2,000 millilitres of water. Find the difference in the amount of water they drank.
6. A water well produces 24,758 litres and 580 millilitres of water per day. If 17,900 litres and 846 millilitres are used every day, how much water remains?

Activity — Learning measurements of length, mass and volume through reliable online materials

Description

Use various reliable programs such as Khan Academy to learn more about measurements of length, mass and volume.

Summary

1. The basic unit of length is metre (m)
2. The basic unit of mass is gram (g)
3. The basic unit of volume is litre (L)

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Revision Exercise

1.

L	mL
75	378
− 53	532

2.

L	mL
330	600
+ 130	800

3.

L	mL
731	116
+ 123	882

4.

km	m
12	349
+ 4	651

5.

m	cm
240	64
+ 660	95

6.

km	m	cm
14	700	00
− 13	860	95

7.

t	kg
12	555
− 8	227

8.

kg	g	mg
19	552	701
− 11	910	900

9.

t	kg	g	mg
15	821	800	697
+ 83	260	333	222

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10. The length of a ruler is 100 centimetres. What is this length in metres?

11. A container weighs 2 tonnes and 300 kilograms. Convert this mass into kilograms.

12. A factory produces 345 litres of sunflower oil every day. How many millilitres of oil will the factory produce in one week?

13. A maize sack weighs 18250 grams. Convert this mass into kilograms.

14. Two cows produce milk as follows: the first cow produces 2 litres and 353 millilitres and the second cow produces 5 litres and 452 millilitres. Find the total volume of milk produced by the two cows.

15. Which measurement is the largest among 7500 grams, 3 tonnes, 4000 kilograms, and 2350 milligrams? Give reasons for your answer?

16. A bucket of water has a volume of 10 litres. If 5534 millilitres of water have been added to the bucket, how many more millilitres of water are needed to fill the bucket?

17. A trader purchased products with the following masses: 76320 grams of flour, 61000 grams of rice, and 7200 grams of salt. What is the total mass of the products purchased?

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Chapter Five

Perimeters of plane figures

Introduction

In this chapter, you will learn to find the perimeter of a rectangle, square and triangle. You will also learn to solve problems involving the concept of perimeters of plane figures. The competencies developed will be applied in building constructions, estimating of distances covered and many other real-life applications.



Think

Mathematics without figures

Concept of perimeter

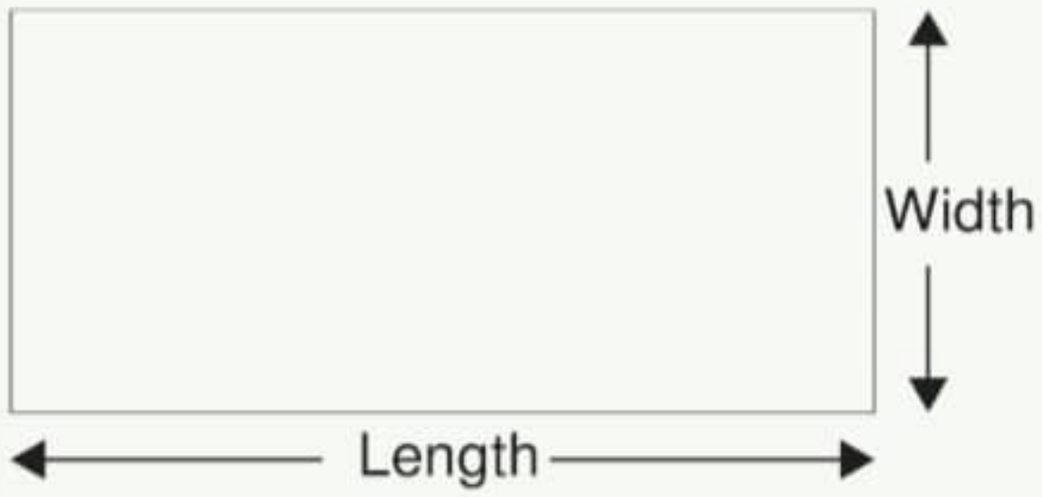
A perimeter is the length of the boundaries of a plane figure. Different ways are used to find the perimeter of a rectangle, square and triangle. The perimeter can be expressed in various metric units of length.

The perimeter of a rectangle

The perimeter of a rectangle is the sum of the lengths of its sides.

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Activity 1 Measuring the perimeter of a rectangle



Steps

1. Put the 0 mark on a tape measure at one corner of the rectangle.
2. Measure the length of the rectangle's boundary until you reach the starting point.
3. Record the measurement of the length obtained.
4. Measure and record the length of each side of the rectangle using a ruler.
5. Add the lengths of all sides of the rectangle in step 4.
6. Compare the results from step 3 with those from step 5.
7. What is the perimeter of the rectangle.

In Activity 1, the total length of the boundaries obtained is the perimeter of the rectangle. Thus, the perimeter of a rectangle is obtained by adding the lengths of all its four sides

The formula for finding the perimeter of a rectangle is given by;

$$\begin{aligned}\text{Perimeter of a rectangle} &= \text{length} + \text{width} + \text{length} + \text{width} \\ &= \text{length} + \text{length} + \text{width} + \text{width} \\ &= 2 \times \text{length} + 2 \times \text{width}\end{aligned}$$

Therefore, the perimeter of a rectangle $= 2 \times \text{length} + 2 \times \text{width}$.

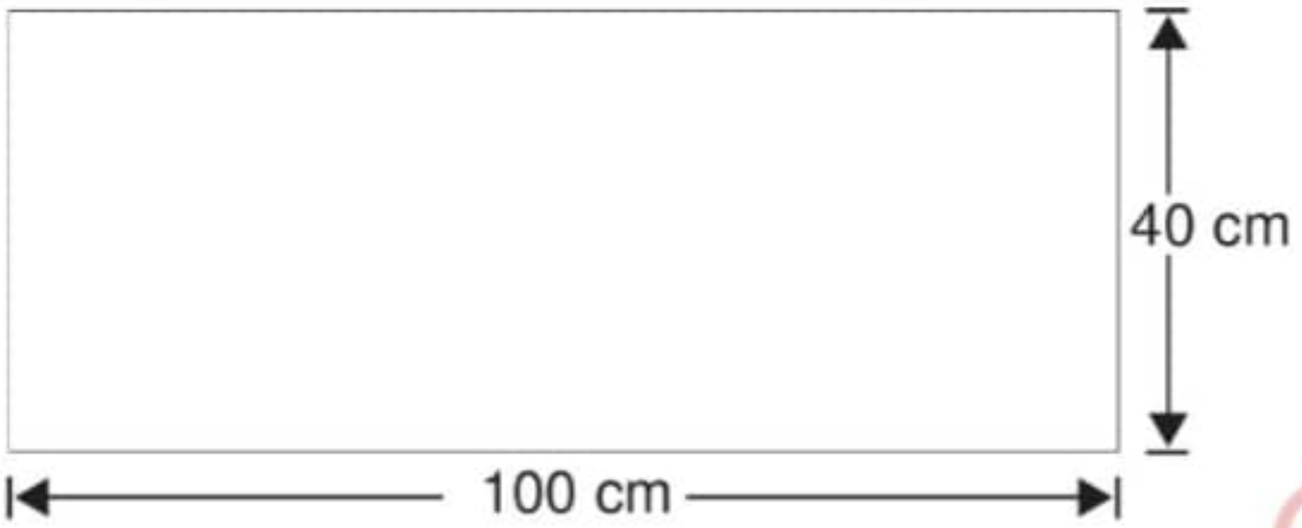
If l = length and w = width, then the perimeter will be;

The perimeter of a rectangle $= 2 \times l + 2 \times w$

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Example 1

Find the perimeter of the following rectangle.



Solution

Perimeter of the rectangle = length + width + length + width

= 100 cm + 40 cm + 100 cm + 40 cm

= 280 cm

Or

The rectangle has two sides with equal length and two sides with equal width. The perimeter is given by

Perimeter of the rectangle = 2 × length + 2 × width

= 2 × 100 cm + 2 × 40 cm

= 200 cm + 80 cm

= 280 cm

Therefore, the perimeter of the rectangle is 280 cm.

Example 2

Find the perimeter of a rectangle of length 13 m and width 11 m.

Solution

Length = 13 m, width = 11 m

Perimeter of a rectangle = 2 × length + 2 × width

= 2 × 13 m + 2 × 11 m

= 26 m + 22 m

= 48 m

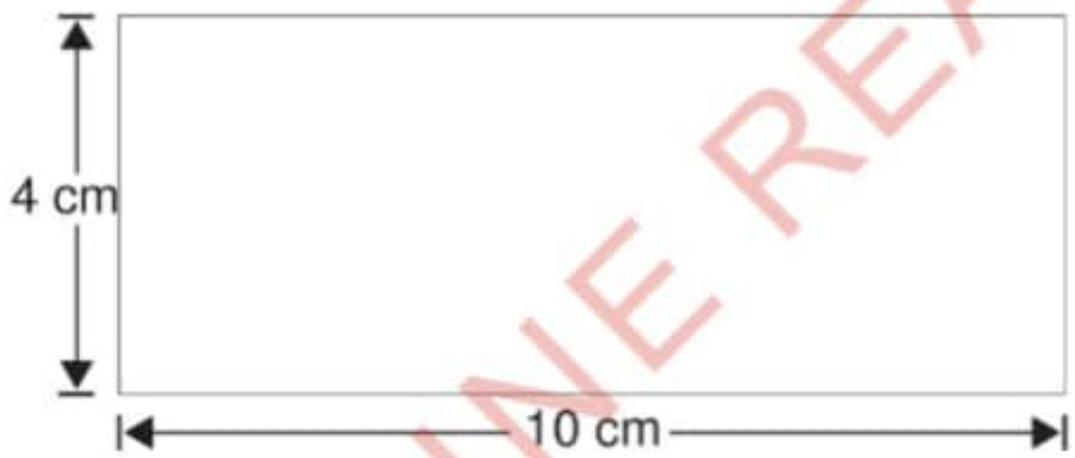
Therefore, the perimeter of the rectangle is 48 m.

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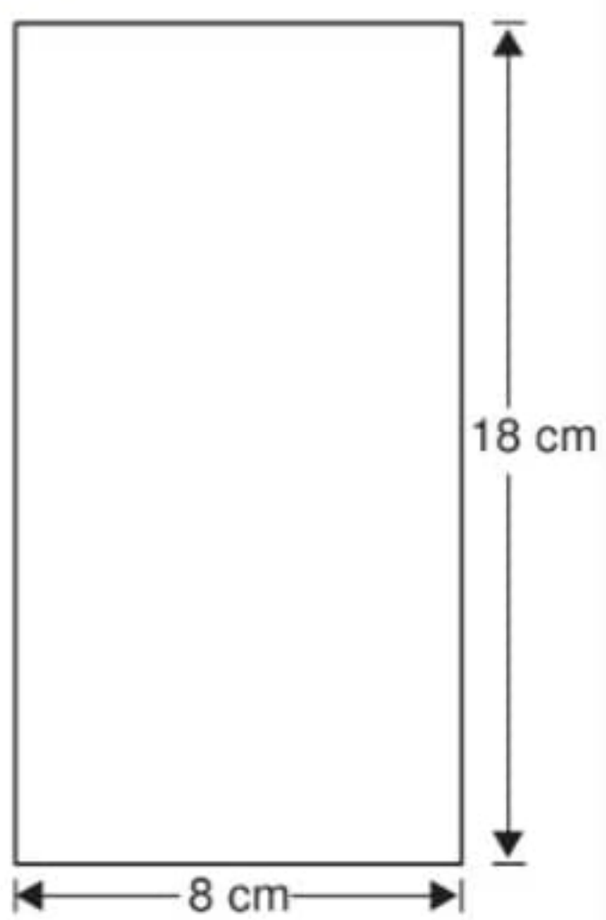
Exercise 1

1. Measure the length of an exercise book and record it in centimetres. Also, measure the width of the same exercise book and record it in centimetres.
   (a) Find the total length of all four sides of the exercise book
   (b) What does the length of four sides of the exercise book represent?
2. The classroom floor has a length of 20 metres and a width of 12 metres. Find its perimeter.
3. Find the perimeter of each of the following figures:

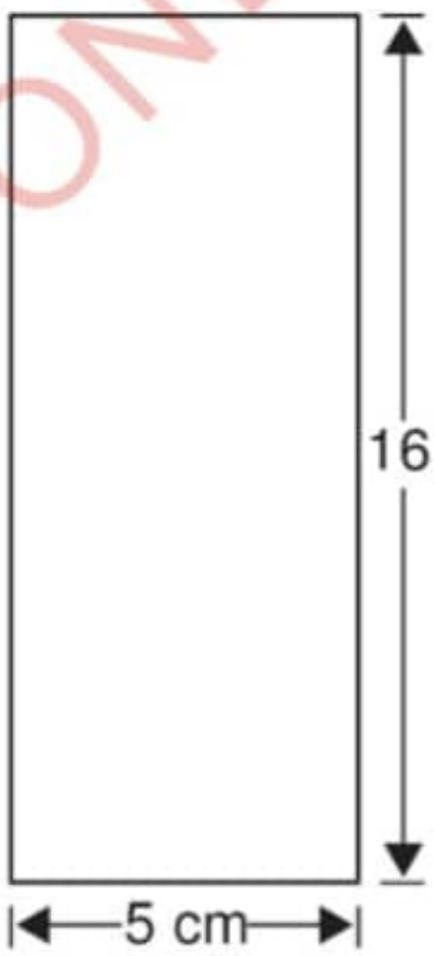
(a)



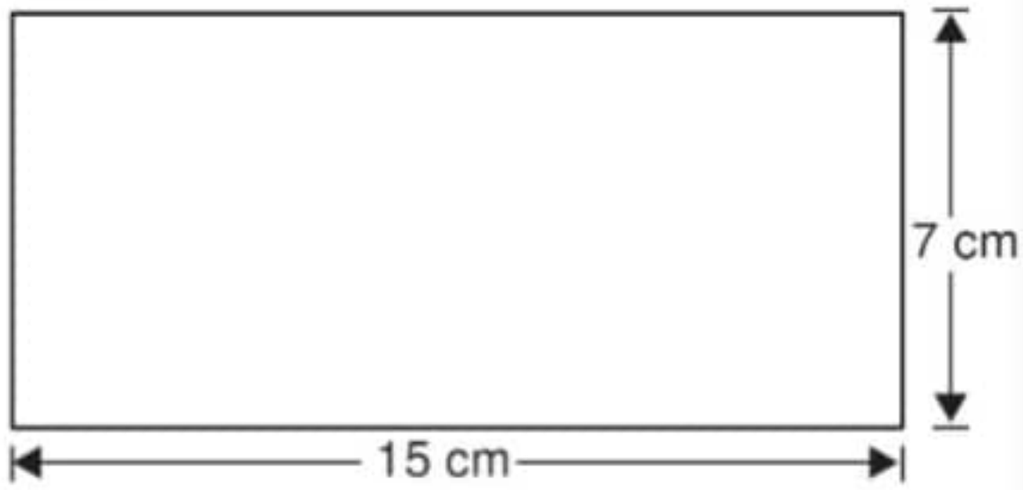
(b)



(c)



(d)



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The perimeter of a square

The perimeter of a square is the sum of the lengths of all its sides. All sides of a square have equal length. Therefore, its perimeter can be found by multiplying the length of one side by 4. Thus,

$$\text{Perimeter of a saquare} = \text{length} + \text{length} + \text{length} + \text{length}$$
$$= 4 \times \text{length}$$

Therefore, the perimeter of a square $= 4 \times l$.

Example 1

Find the perimeter of the following square.



Solution

$$\text{Perimeter of a square} = \text{length} + \text{length} + \text{length} + \text{length}$$
$$= 20\ \text{cm} + 20\ \text{cm} + 20\ \text{cm} + 20\ \text{cm}$$
$$= 80\ \text{cm}$$

Alternative solution

The square has four sides of equal length. It follows that,

$$\text{Perimeter of a square} = 4 \times \text{length}$$
$$= 20\ \text{cm} \times 4$$
$$= 80\ \text{cm}$$

Therefore, the perimeter of the square is 80 cm.

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Example 2

The length of a square garden is 13 metres. Find its perimeter.

Solution

Length of the square garden $= 13$ m

Perimeter of the garden $= 4 \times$ length

$= 4 \times 13$ m

$= 52$ m

Therefore, the perimeter of the garden is 52 m.

Exercise 2

1. Find the perimeter of each of the following squares:

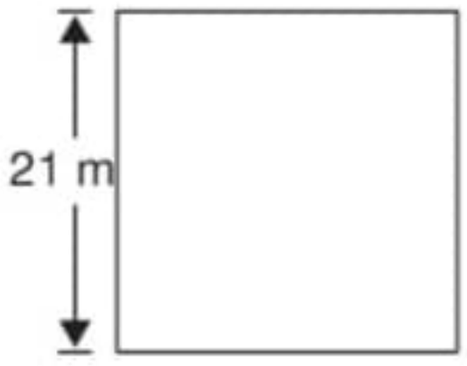
(a)



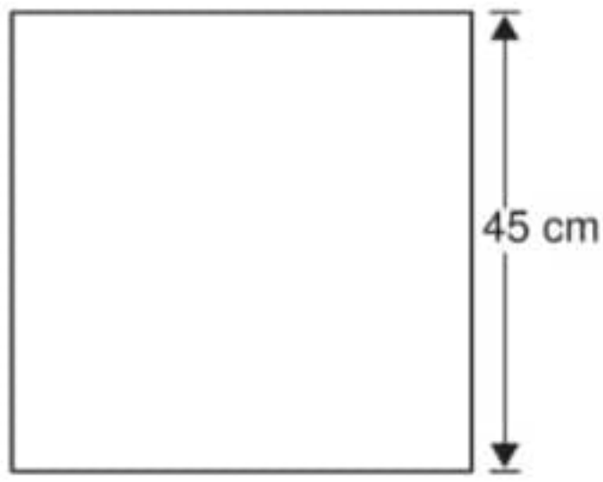
(b)



(c)



(d)



2. Find the perimeter of a square whose side has a length of 18 m.
3. The square has a side of 37 cm long. Find its perimeter.

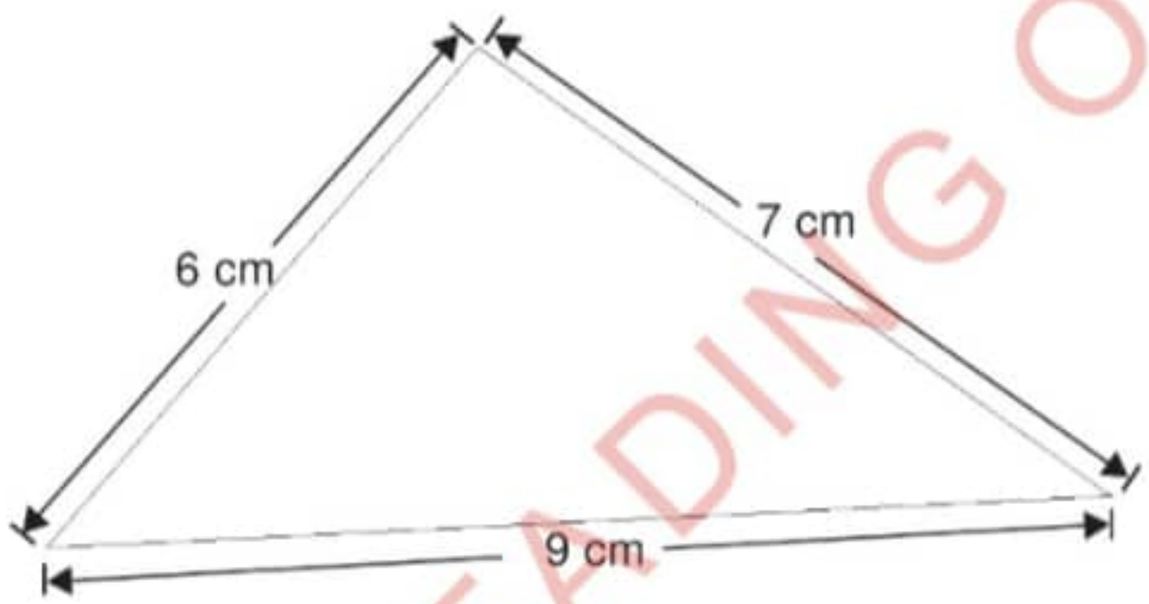
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The perimeter of a triangle

The perimeter of a triangle is the sum of the lengths of its three sides. Thus, the perimeter of a triangle is obtained by adding the lengths of its sides.

Example 1

Find the perimeter of the following triangle.



Solution

Perimeter of a triangle = sum of the lengths of all sides
= 6 cm + 7 cm + 9 cm
= 22 cm

Therefore, the perimeter of the triangle is 22 cm.

Example 2

A triangle has sides measuring 14 metres, 16 metres and 11 metres. Find its perimeter.

Solution

Perimeter of a triangle = sum of the lengths of all sides
= 14 m + 16 m + 11 m
= 41 m

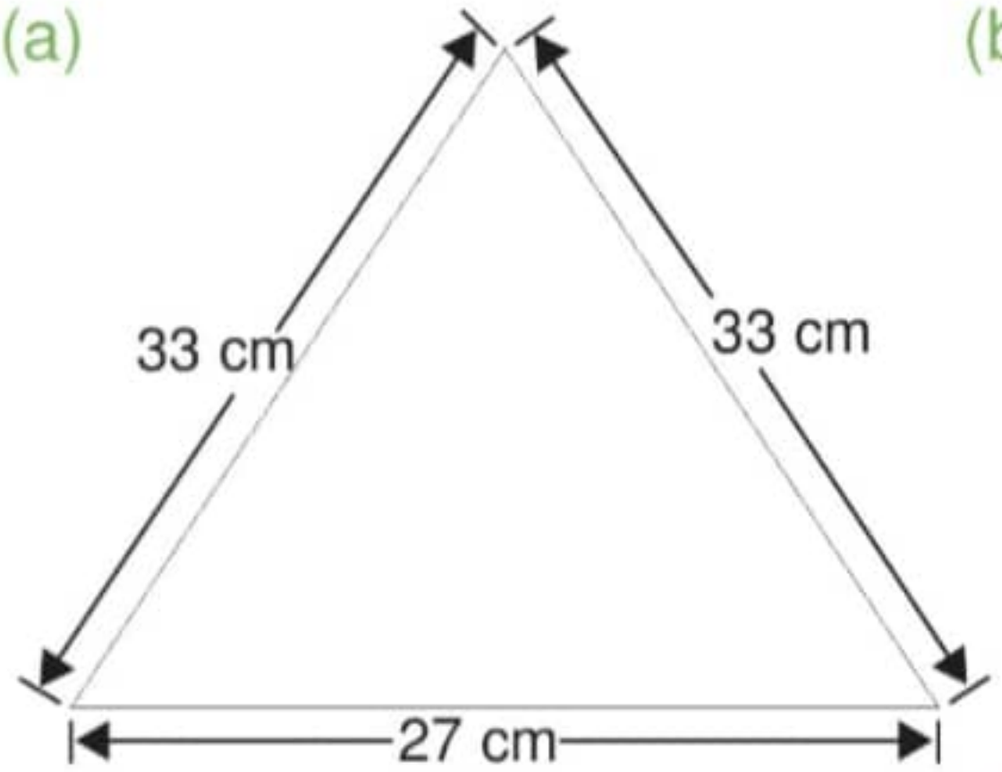
Therefore, the perimeter of the triangle is 41 m.

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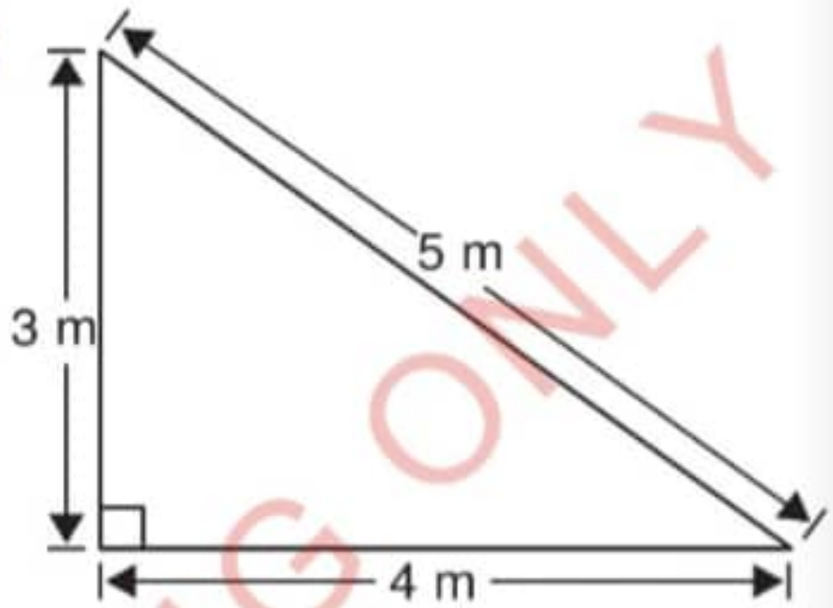
Exercise 3

1. Find the perimeter of each of the following triangles:

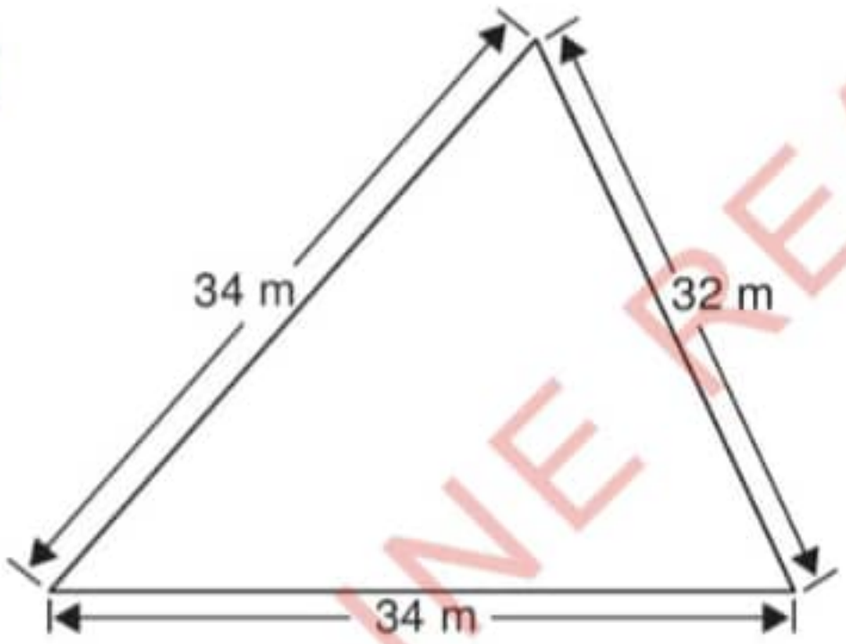
(a)



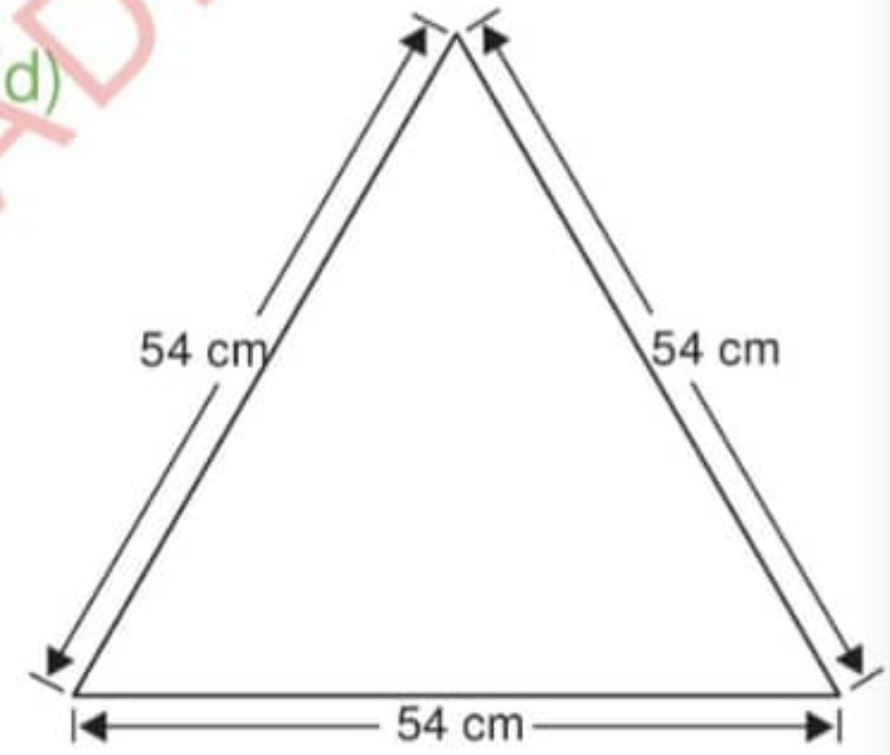
(b)



(c)



(d)



2. Find the perimeter of a triangle whose sides are 26 m, 24 m and 10 m.

3. A triangle has sides of lengths; 84 cm, 85 cm and 13 cm. Find its perimeter.

Word problems involving perimeter

Word problems involving perimeters are mathematical questions written in one or more sentence. The concept of perimeter is used to solve such real-life problems.

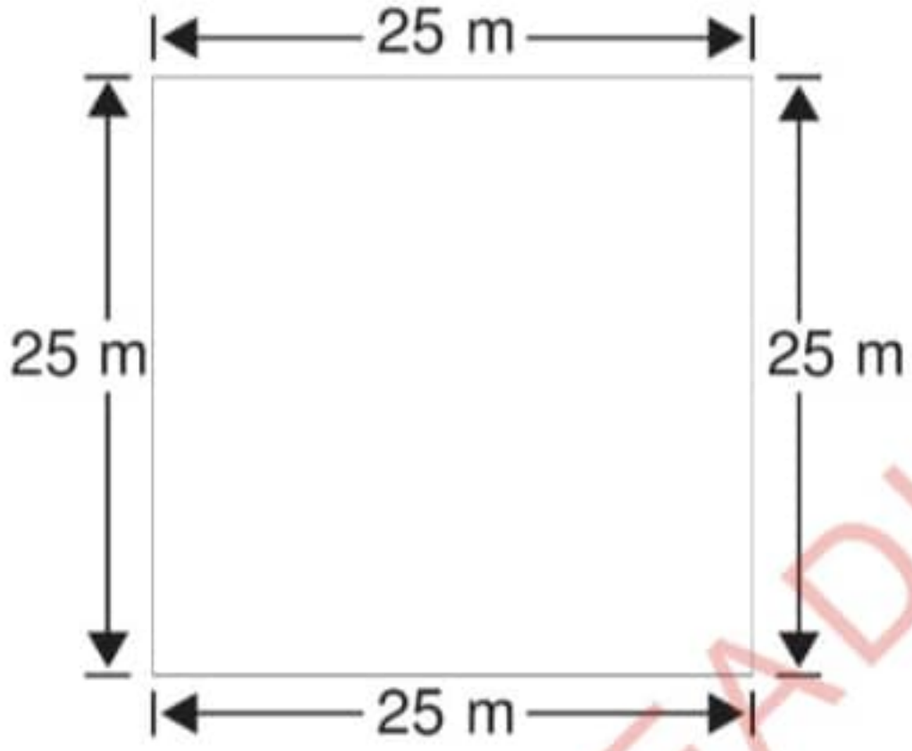
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Example 1

A square flower garden is 25 m long. Find its perimeter.

Solution

Length of the garden $= 25\,\text{m}$.



Perimeter of the garden $= 25\,\text{m} + 25\,\text{m} + 25\,\text{m} + 25\,\text{m}$

$= 100\ \text{m}$

Therefore, the perimeter of the flower garden is 100 metres.

Example 2

The football ground is 100 metres long and 64 metres wide. Find its perimeter.

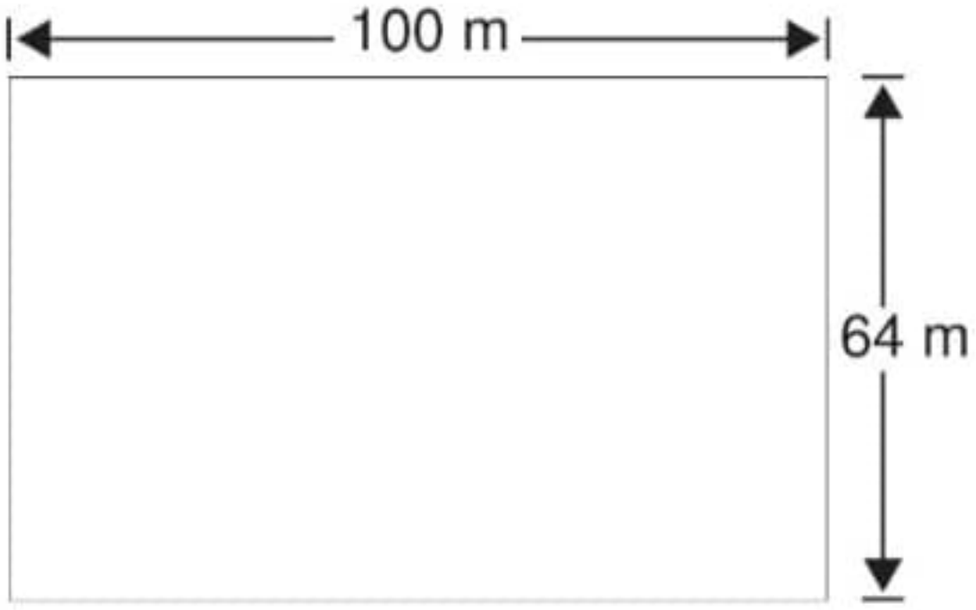
Solution

Length of the football ground = 100 m

Width of the football ground = 64 m

The football ground has a rectangular shape as shown in the following figure.

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Perimeter of the ground $= 2 \times \text{length} + 2 \times \text{width}$

$= 2 \times 100\text{ m} + 2 \times 64\text{ m}$

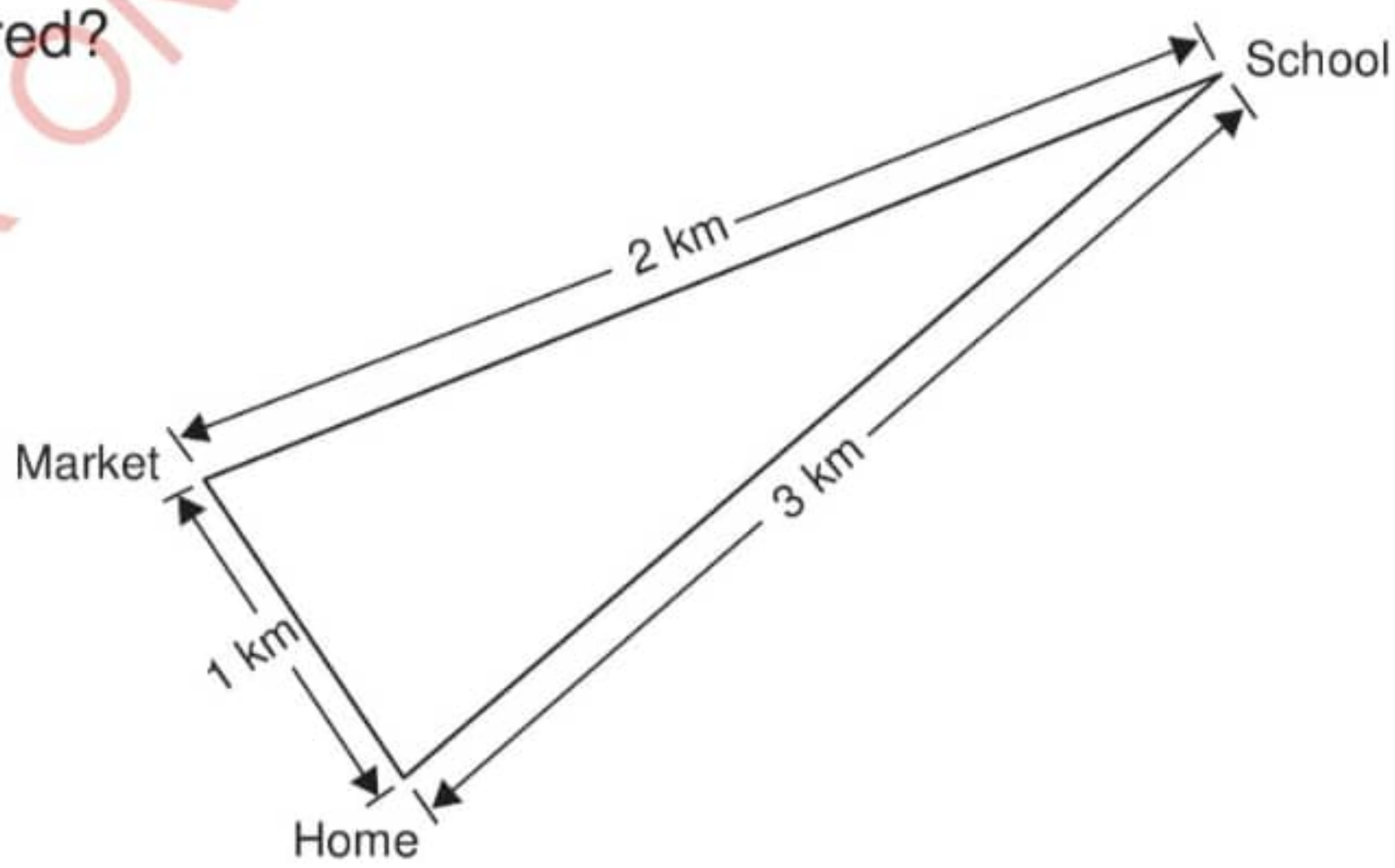
$= 200\text{ m} + 128\text{ m}$

$= 328\text{ m}$

Therefore, the perimeter of the football ground is 328 m.

Example 3

A pupil rode a bicycle a distance of 3 km from home to school. After school sessions, he went to the market to buy some fruits. The distance from the school to the market is 2 km. If the pupil used a short cut road to ride 1 km back home as shown in the following figure, what is the total distance covered?



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Solution

The distance covered is the length of the entire route. This distance is given by the perimeter of the route.

Perimeter of the route = distance from home to school + distance from school to the market + distance from the market to home

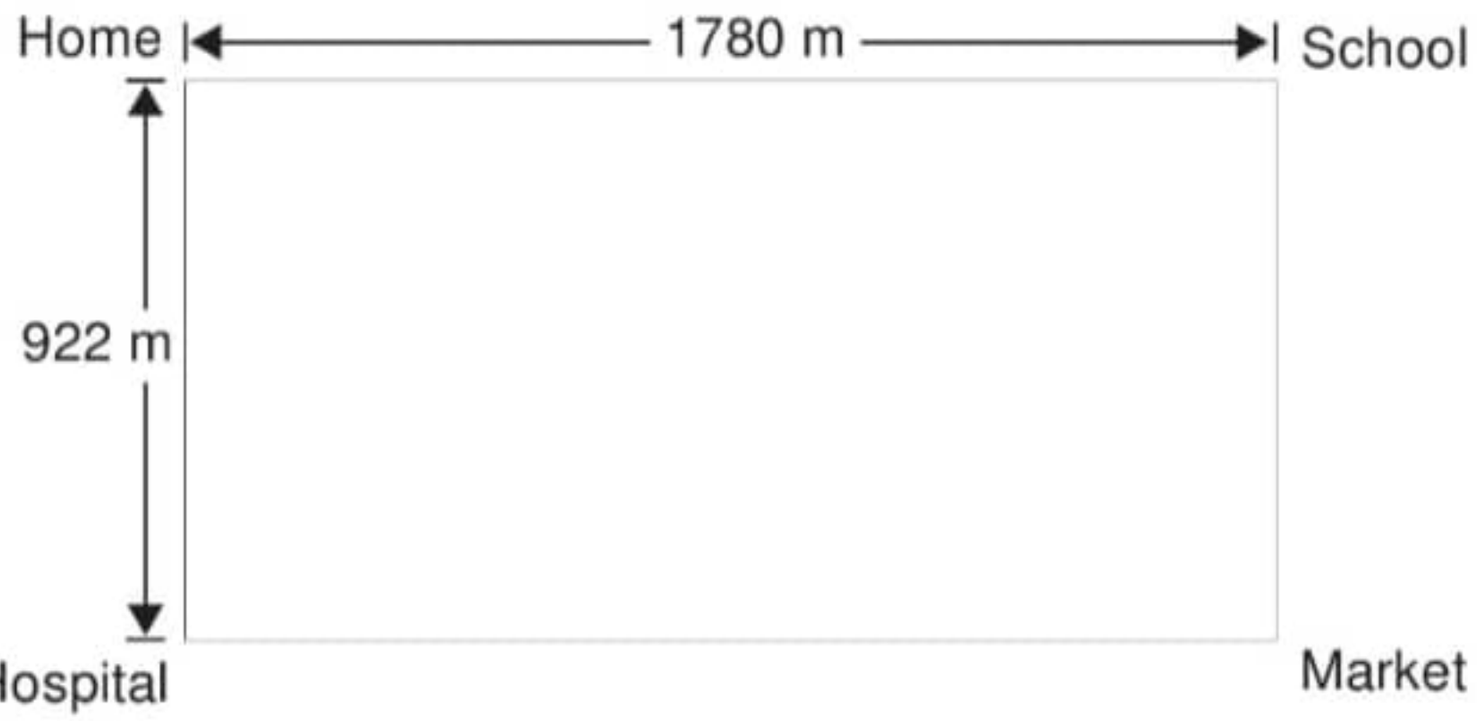
$$= 3\text{ km} + 2\text{ km} + 1\text{ km}$$

$$= 6\text{ km}$$

Therefore, the pupil covered a distance of 6 km.

Exercise 4

1. A teacher has a square shaped garden. If the garden has length of 20 metres, find its perimeter.
2. A triangular field has sides measuring 15 metres, 20 metres and 25 metres. Find the perimeter of the field.
3. A square table has a length of 8 metres. Find its perimeter.
4. Juma walked from his home to school. After school, he walked to the market, from the market to the hospital, and back home as shown in the following figure:



Find the total distance walked by Juma.

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5. The sides of a triangular pizza are 36 cm, 9 cm and 27 cm long. Find the perimeter of the pizza.
6. A businessman sold a carpet that is 12 metres long and 8 metres wide. Find its perimeter.
7. The length of one side of a triangular pool is 40 metres. If the lengths of the sides of the pool are equal, find its perimeter.
8. A field has a length of 65 metres and a width of 40 metres. Find its perimeter.

Activity 2 — Recognising perimeters of plane figures using reliable ICT resources

Description

Use online programs and computer packages such as GeoGebra and Khan Academy to learn more on perimeters of plane figures and their applications.

Summary

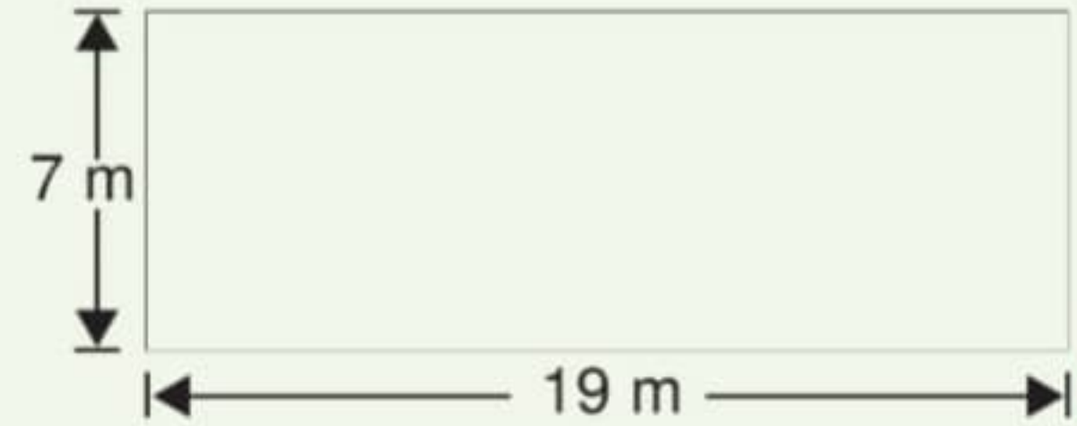
1. Perimeter is the length of the boundary of a plane figure.
2. Perimeter of a rectangle $= 2 \times$ length $+ 2 \times$ width.
3. Perimeter of a square $= 4 \times$ length.
4. Perimeter of a triangle = The sum of the lengths of all sides.

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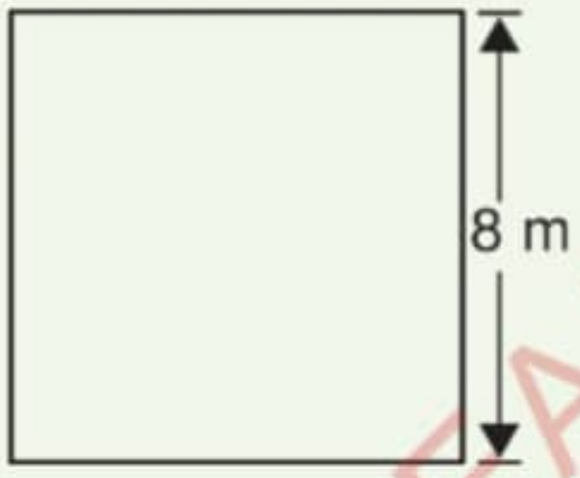
Revision Exercise

1. Find the perimeter of each of the following figures.

(a)

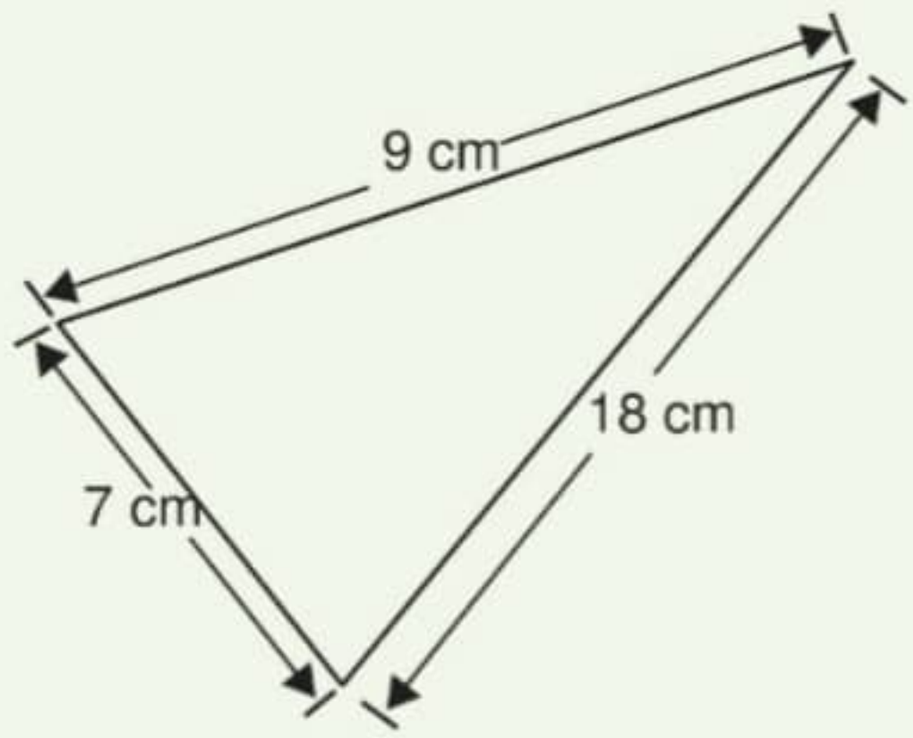


(b)



(c)

(d)



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2. The length of one side of a square is 23 cm. Find its perimeter.
3. A school garden is rectangular in shape. If its length is 1200 m and its width is 80 m, find its perimeter.
4. Find the perimeter of a triangle if each of its sides is 17 cm long.
5. Find the perimeter of a square whose side length is 16 m.
6. Jamila ran one round of a rectangular field of length 100 m and width 70 m. What is the total distance covered by Jamila?
7. Find the length of the rope that can be used to surround a window that is 3 m high and 2 m wide.
8. The perimeter of square A is 4 metres. Two same squares make a rectangle B. Baraka and Neema calculated the perimeter of the rectangle B. Baraka got a perimeter of 8 metres, while Neema got 6 metres. Who got the correct answer? Give reason for your answer.



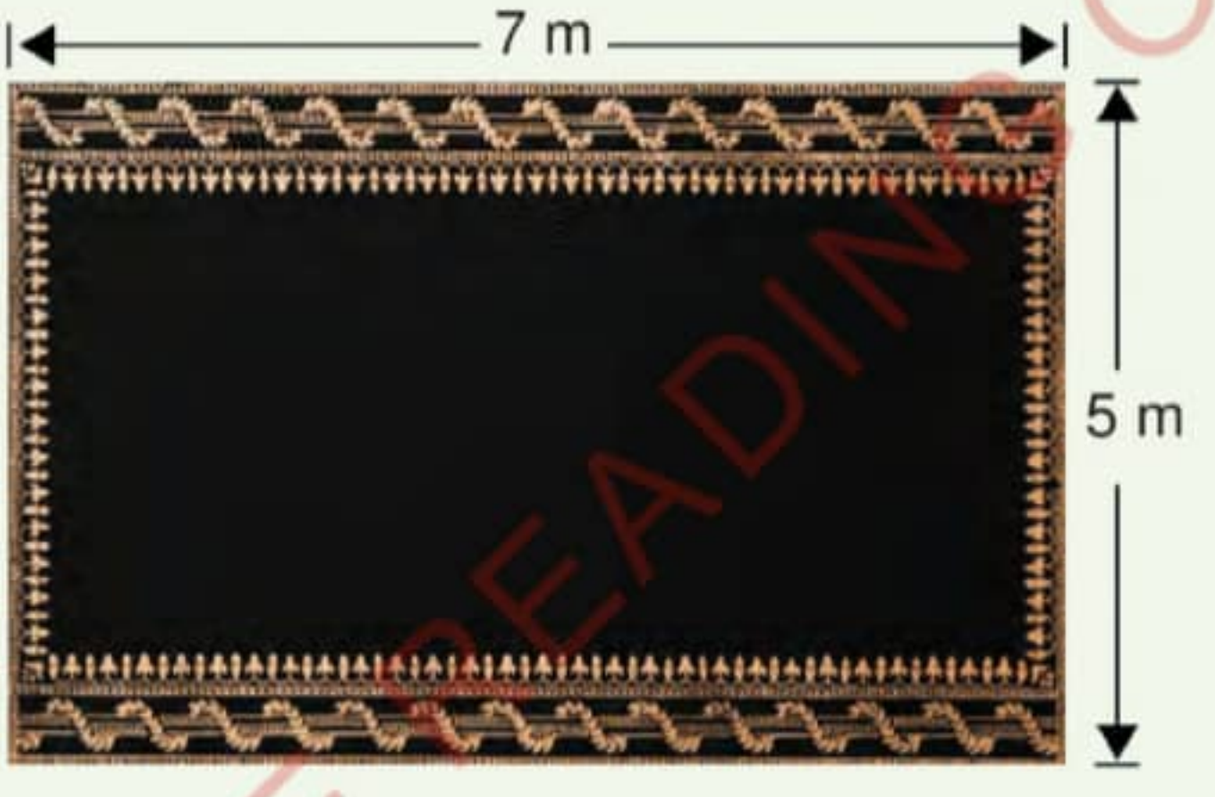
9. A rectangular shed has a length of 28 metres and a width of 4 metres. Find its perimeter.
10. Furaha has a square garden of a side length of 7 m, he calculated its perimeter. What is the perimeter obtained by Furaha in centimetres?

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11. Musa drew a rectangle with a length of 80 mm and a width of 50 mm. What is its perimeter in centimetres?

12. The tailor bought a piece of fabric with a length of 2 metres and a width of 1 metre. What is the perimeter of that piece of fabric in centimetres?

13. A carpet in the teachers' office is 7 metres long and 5 metres wide. Find its perimeter.



14. The garden has a triangular shape measured: 10 m, 7 m and 6 m. Find its perimeter.

15. Four pupils cut a piece of paper in a square shape. The piece of paper had a perimeter of 16 centimetres. If they cut the paper into four equal square pieces, what is the perimeter of each square.

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Chapter Six

Area of plane figures

Introduction

In this chapter, you will learn how to find the area of a rectangle, square and triangle. The competencies developed will enable you to apply the concept of area in building constructions, textile and dressmaking and many other real-life applications.



Think

Constructions without plane figures

The concept of area

The area is the surface region of the shape. The area can be determined by dividing the shape into smaller, equal squares and counting them. The area can also be obtained by using a formula and measurements. The area is measured in metric units such as square metres (m^2), square centimetres (cm^2) and square millimetres (mm^2).

Area of a rectangle

The area of a rectangle is the entire surface region of the rectangle.

Activity 1 Determining the area of a rectangle

Steps

1. Draw a rectangle of any size of your choice.
2. Divide the width of a rectangle into small strips of 1 cm each.

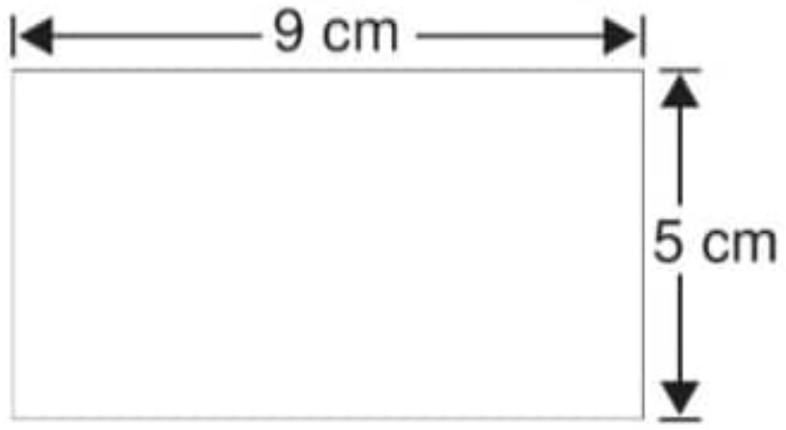
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3. Divide the length of the rectangle into small strips of 1 cm each.
4. Identify squares with side length of 1 cm obtained in Step 3.
5. Count the small squares in Step 4 and write their number.
6. How many squares have you obtained?
7. What is the area of each square in square centimetres.
8. What is the area of the rectangle in square centimetres?

In Activity 1, you have learnt that the area of the rectangle is obtained by counting small squares in that rectangle. Thus, the area of the rectangle of your choice is equal to the number of small squares obtained. Also, the area of a rectangle is obtained by counting numbers of small squares in the length side and width side. If the number of small squares in the length side is multiplied by that of width side, the answer will be the total number of small squares in the rectangle. Thus, the area of the rectangle is length × width. That is the area of a rectangle = length × width.

Example 1

Find the area of the following rectangle.



Solution

Area of a rectangle = length × width
= 9 cm × 5 cm
= 45 cm^2.

Therefore, the area of the rectangle is 45 cm^2.

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Example 2

A book is 30 cm long and 20 cm wide. Find the surface area of the book.

Solution

The book has a rectangular surface.

length = 30 cm and width = 20 cm

Area of a rectangle = length × width

= 30 cm × 20 cm

= 600 cm^2

Therefore, the surface area of the book is 600 cm^2.

Example 3

The width of the rectangle is 40 cm. If its length is 90 centimetres, find its area.

Solution

length = 90 cm

width 40 cm

Area of a rectangle = length × width

= 90 cm × 40 cm

= 3600 cm^2

Therefore, the area of a rectangle is 3600 cm^2.

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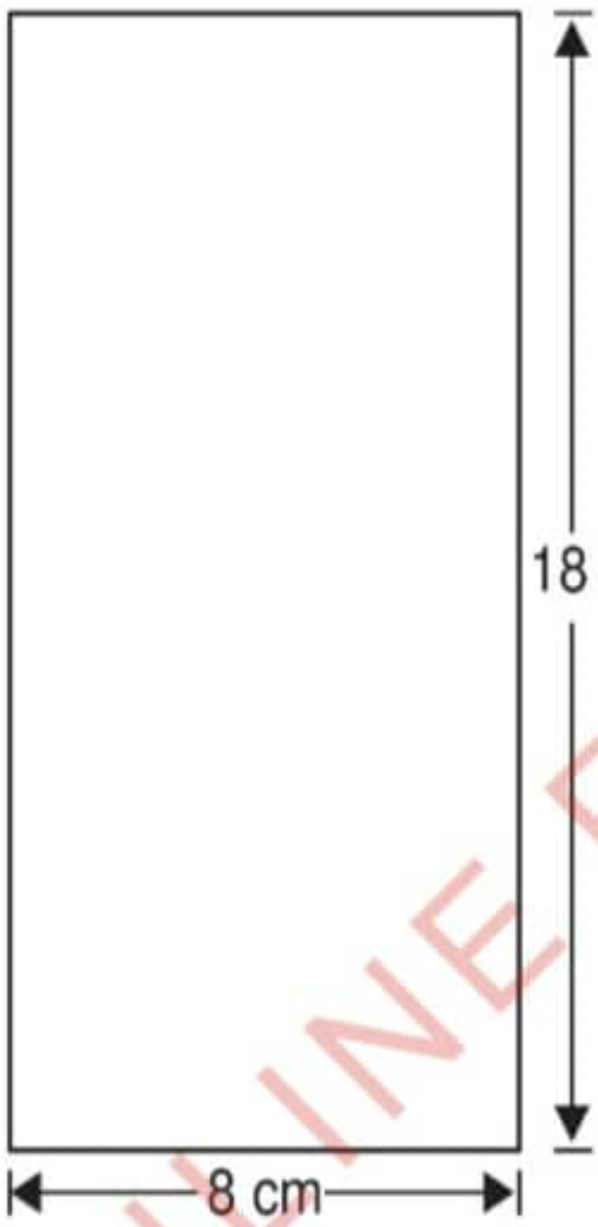
Exercise 1

1. Find the area in each of the following:

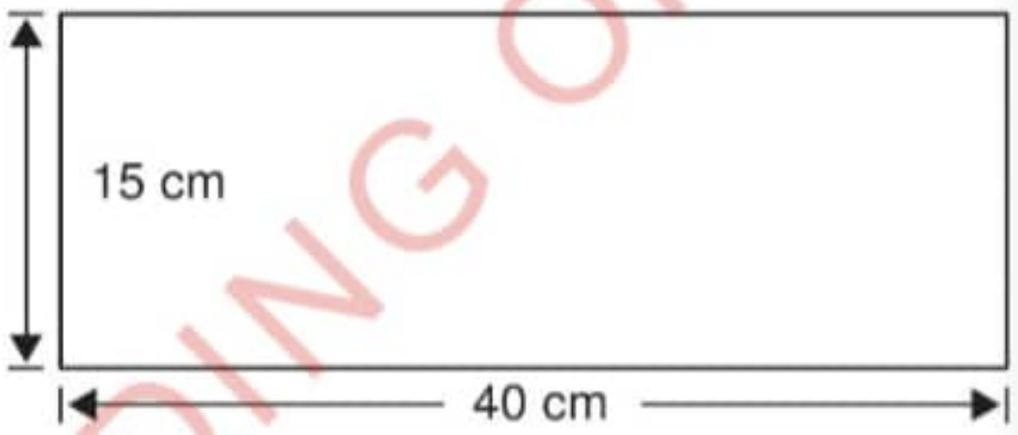
(a)



(b)



(c)



(d)

2. A rectangle has a length of 10 cm and a width of 9 cm. Find the area of the rectangle.
3. A teacher's table is 80 cm long and 60 cm wide. Find its surface area.
4. A classroom has a length of 7 metres and a width of 6 metres. Find the area of its floor.
5. Juma has a plot measuring 35 m long and 26 m wide. Find the area of the plot.
6. A garden has a length of 15 metres and a width of 8 meters. Find the area of the garden.

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7. The length of the rectangle is 25 metres and the width is 4 metres. Find its area.
8. The width of the school farm is 20 metres. If its length is 41 metres, find its area.
9. The length of a rectangle is 215 metres and its width is 92 metres. Find its area in centimetres.
10. The length of a rectangular garden is 84 metres. If its width is 45 metres, find its area.

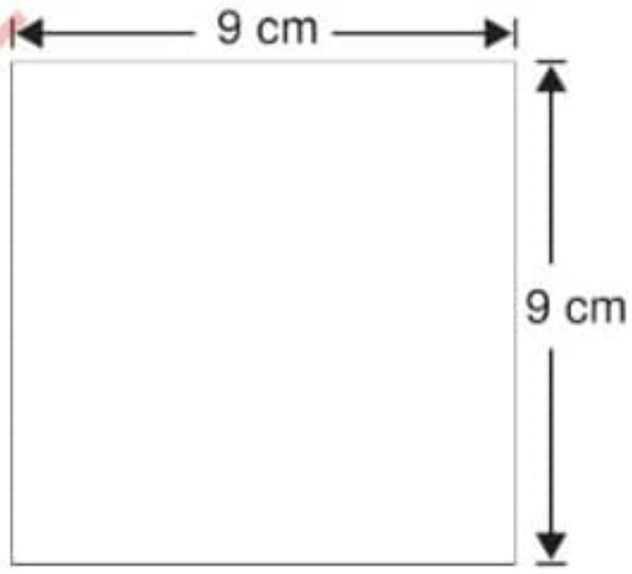
Area of a square

The area of a square is the entire surface region of the square. The area of a square is obtained by multiplying the length by the width of the square. Recall that all sides of a square have equal length.

Thus, Area of a square = length × width or length × length.

Example 1

Find the area of the following square.



Solution

Length of a side = 9 cm

Area of the square = 9 cm × 9 cm

= 81 cm^2.

Therefore, the area of the square is 81 cm^2.

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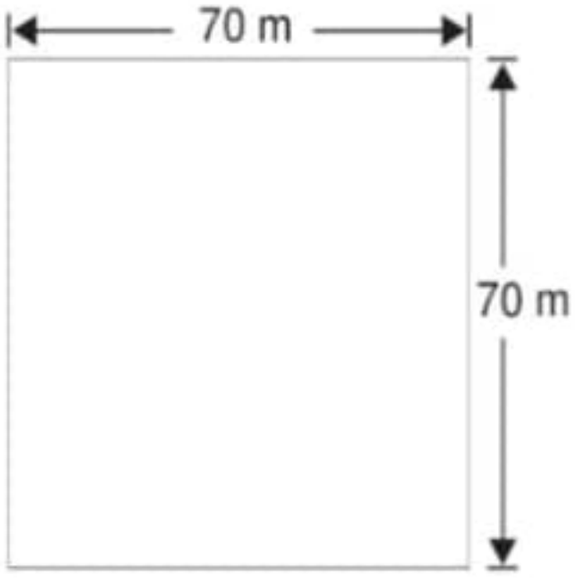
Example 2

A square side field has a length of 70 metres. Find the area of the field.

Solution

Side length = 70 m

The square is drawn as follows.



Thus, area of a square = length × length.

$= 70\text{ m} \times 70\text{ m}$

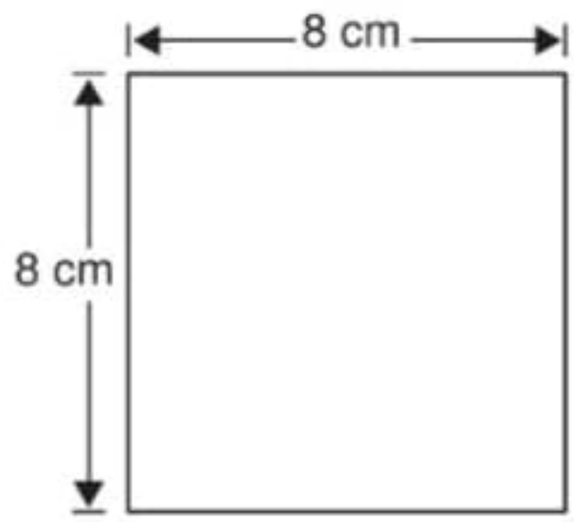
$= 4900\text{ m}^2$.

Therefore, the area of the field is 4900 m^2.

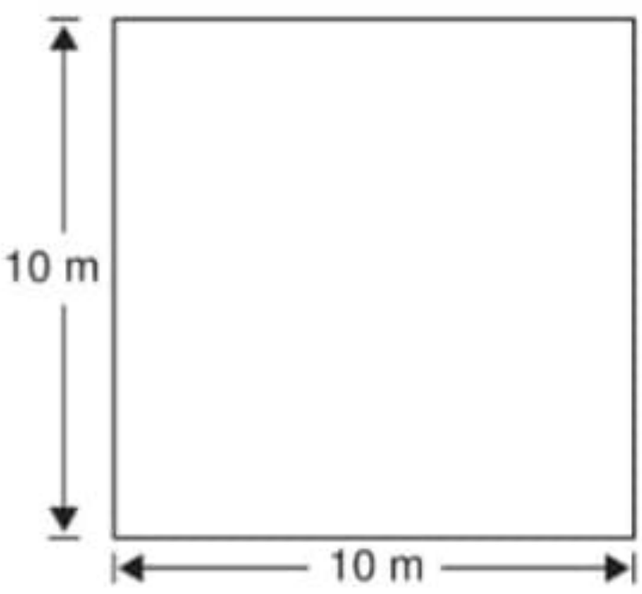
Exercise 2

1. Find the area in each of the following shapes:

(a)



(b)



2. The length of one side of a square is 40 centimetres. Find the area of the square.

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3. One side of a square garden is 21 metres long. Find the area of the garden.
4. The floor of a square classroom has one side of length 9 metres. Find the area of the floor.
5. The length of one side of a square farm is 170 metres. Find the area of the farm.
6. A window with a square shape has a length of 140 centimetres. Find the surface area of the window.

Area of a triangle

The area of a triangle is the surface region of the triangle.

Activity 2 Determining the area of a triangle

Steps

1. Draw a square LMNO on graph paper.
2. Draw a line segment to join the points L and N to form two triangles LMN and LON.
3. Count the complete small squares and half small squares in the triangle LON.
4. How many complete small squares do you get?
5. How many half-small squares do you get? (Remember that two half squares make one complete square).
6. What is the total number of complete squares in the triangle LON?
7. Is there any relationship between the triangles LON and LMN?
8. What is the relationship between the triangles LON, LMN and the area of the square LMNO?

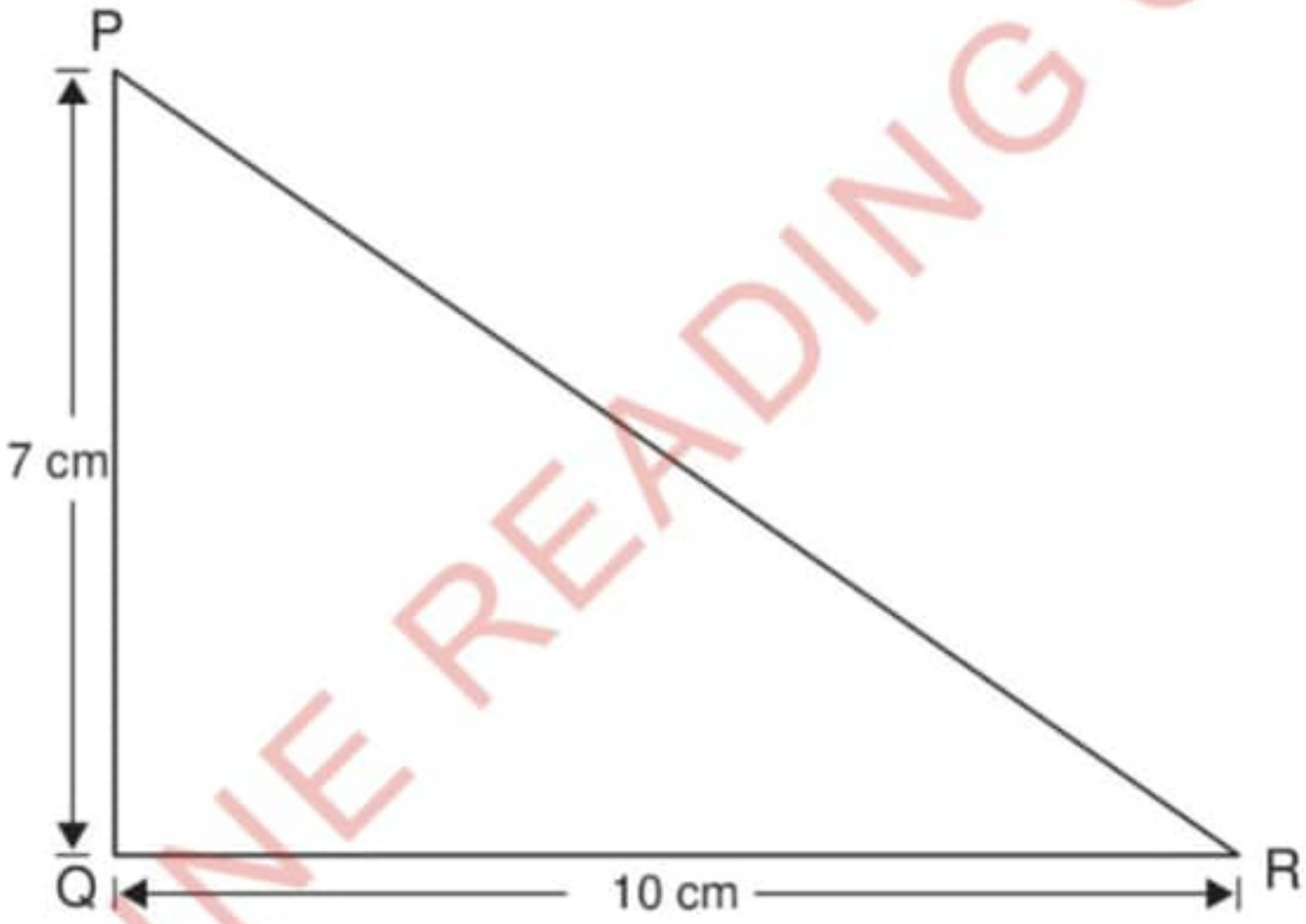
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In Activity 2 you have learnt that the area of a triangle is obtained by counting small squares. That is the area of a triangle is obtained by taking half of the area of the square or rectangle.

Therefore, the area of a triangle $= \frac{1}{2} \times$ height $\times$ base.

Example 1

Find the area of the triangle PQR.



Solution

Base $(\overline{QR}) = 10$ cm

Height $(\overline{QP}) = 7$ cm

$$\text{Area of a triangle} = \frac{1}{2} \times \text{height} \times \text{base}$$

$$= \frac{1}{2} \times 10\text{ cm} \times 7\text{cm}$$

$$= 35\text{ cm}^2$$

Therefore, the area of a triangle PQR is 35 cm^2.

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Example 2

The height of a triangle is 15 metres. If its base is 30 metres, find the area of the triangle.

Solution

Height of a rectangle $= 15\text{m}$

Base of a rectangle $= 30\text{m}$

$$\text{Area of a rectangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

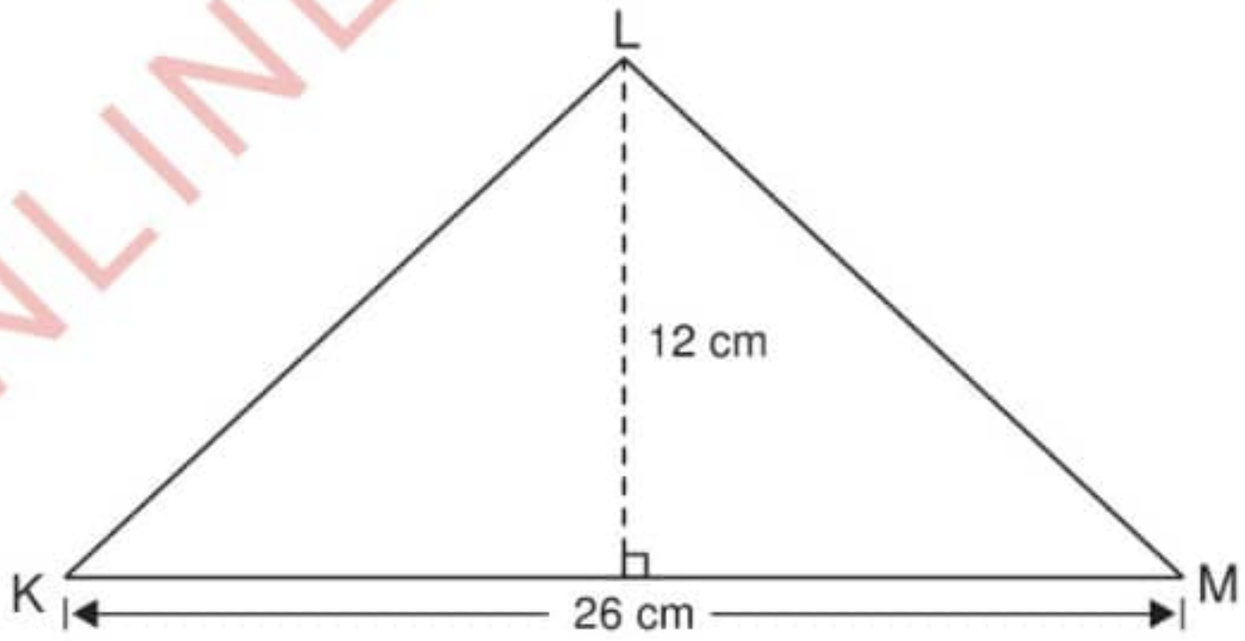
$$= \frac{1}{2} \times 30\text{ m} \times 15\text{ m}$$

$$= 225\text{ m}^2$$

Therefore, the area of the triangle is 225 m^2.

Example 3

Find the area of the triangle KLM.



Solution

Base $\left(\overline{KM}\right) = 26\text{cm}$

Height $\left(\overline{LN}\right) = 12\text{ cm}$

$$\text{Area of the triangle KLM} = \frac{1}{2} \times \text{height} \times \text{base}$$

$$= \frac{1}{2} \times 12\text{ cm} \times 26\text{ cm}$$

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$$= \frac{12 \times 26}{2} \text{ cm}^2$$

$$= \frac{312}{2} \text{ cm}^2$$

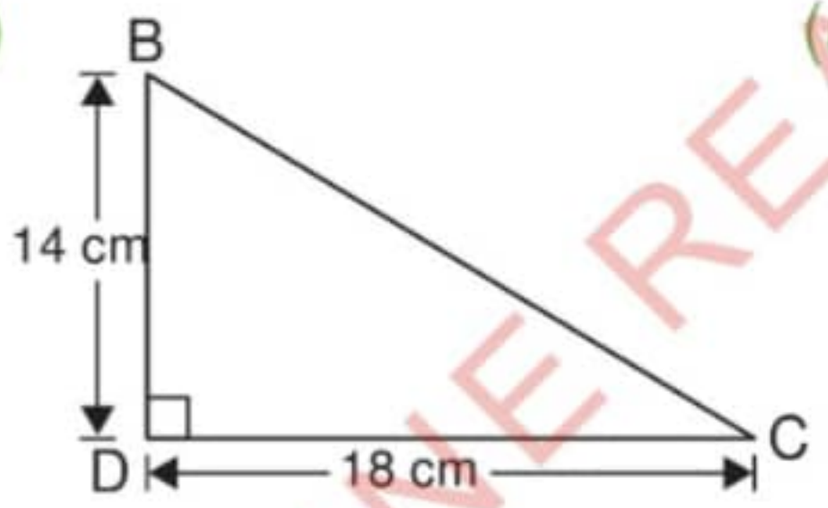
$$= 156 \text{ cm}^2$$

Therefore, the area of the triangle KLM is 156 cm^2.

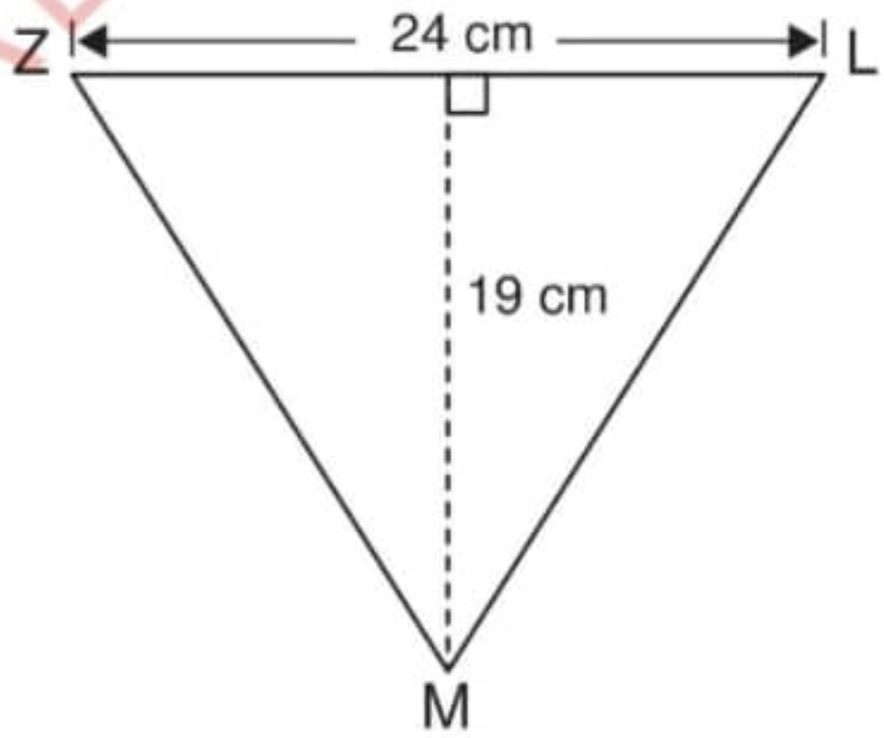
Exercise 3

1. Find the area in each of the following shapes:

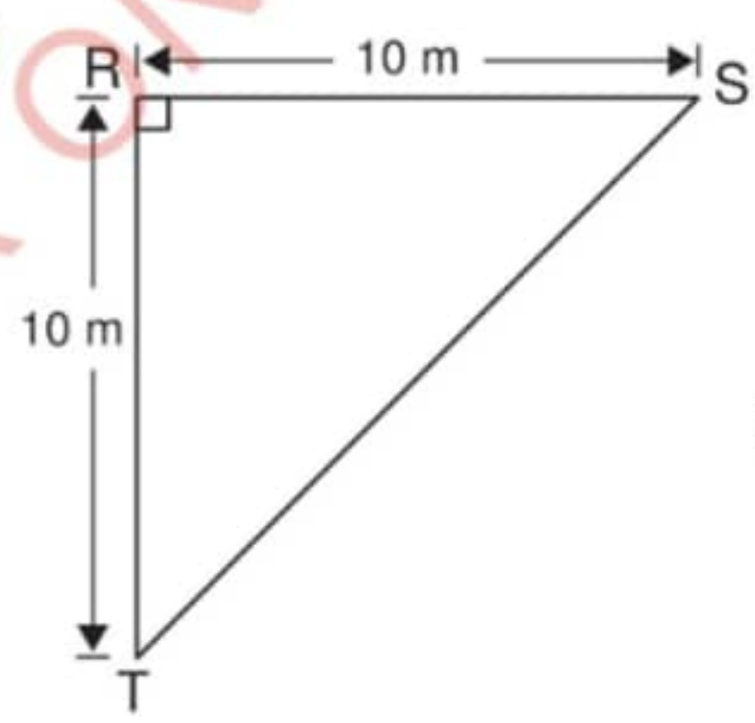
(a)



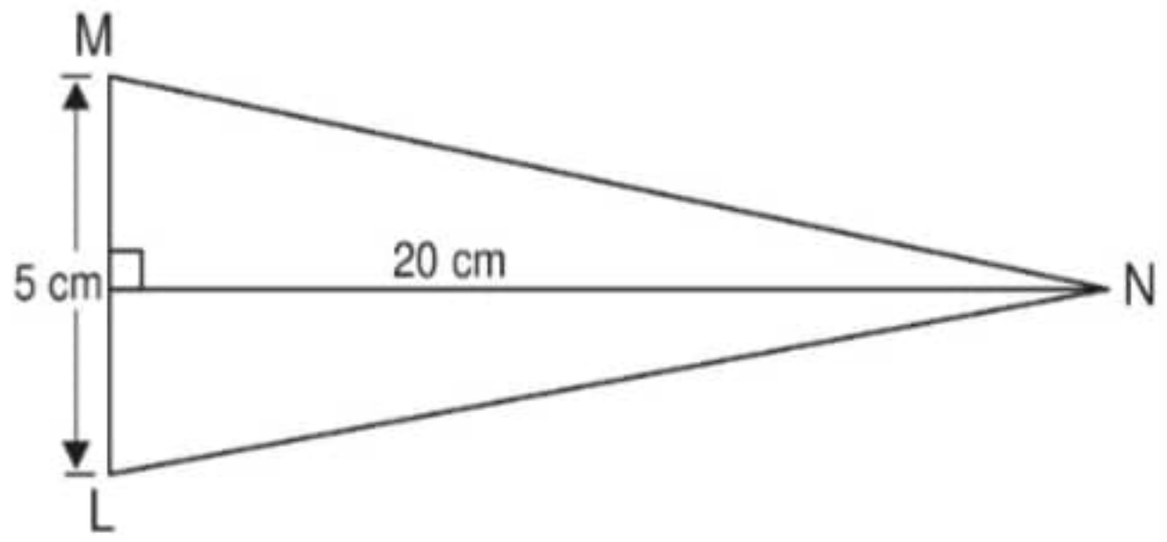
(a)



(c)



(c)



2. A triangular table surface has a height of 6 metres and a base of 3 metres. Find the area of the table surface.

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3. Find the area of a triangular road sign if its base is 40 centimetres and its height is 55 centimetres.
4. A triangle has a height of 96 cm and a base of 24 cm. Find its area.
5. Find the area of a triangular garden with a height of 9 m and a base of 6 m.
6. The base of a triangle is 40 centimetres. If its height is 30 centimetres find the area of the triangle.
7. A triangle has a height of 15 metres. If its base is 26 metres. Find the area of the triangle.
8. An engineer wants to construct a building with a triangular roof of the height 27 metres and the base of 36 metres. Find the surface area of the roof.

Activity 3 Using reliable online resources to learn about the area of rectangles, squares, and triangles

Description

Use online resources and programs such as GeoGebra and Khan Academy to learn more about the area of rectangles, squares, and triangles.

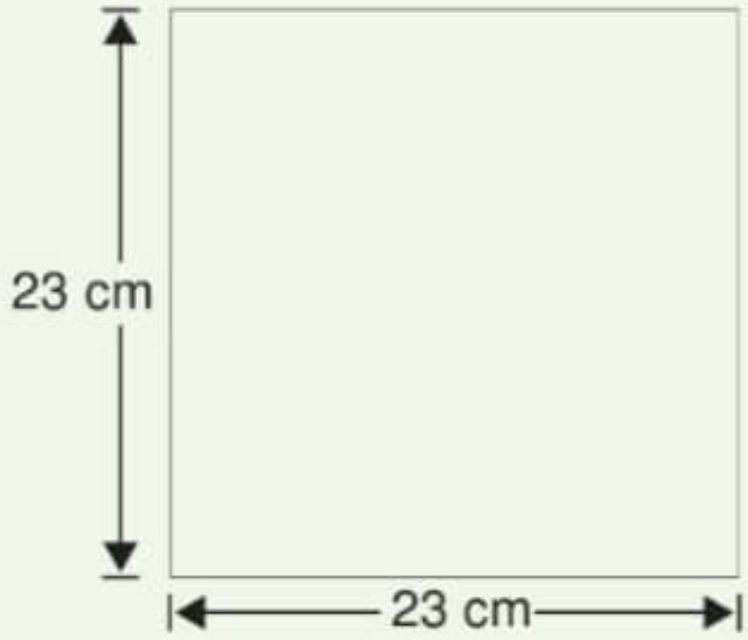
Summary

1. The formula for finding the area of a triangle;
   Area of a rectangle = length × width.
2. The formula for finding the area of a square;
   Area of a square = length × length.
3. The formula for finding the area of a triangle;
   $\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}.$

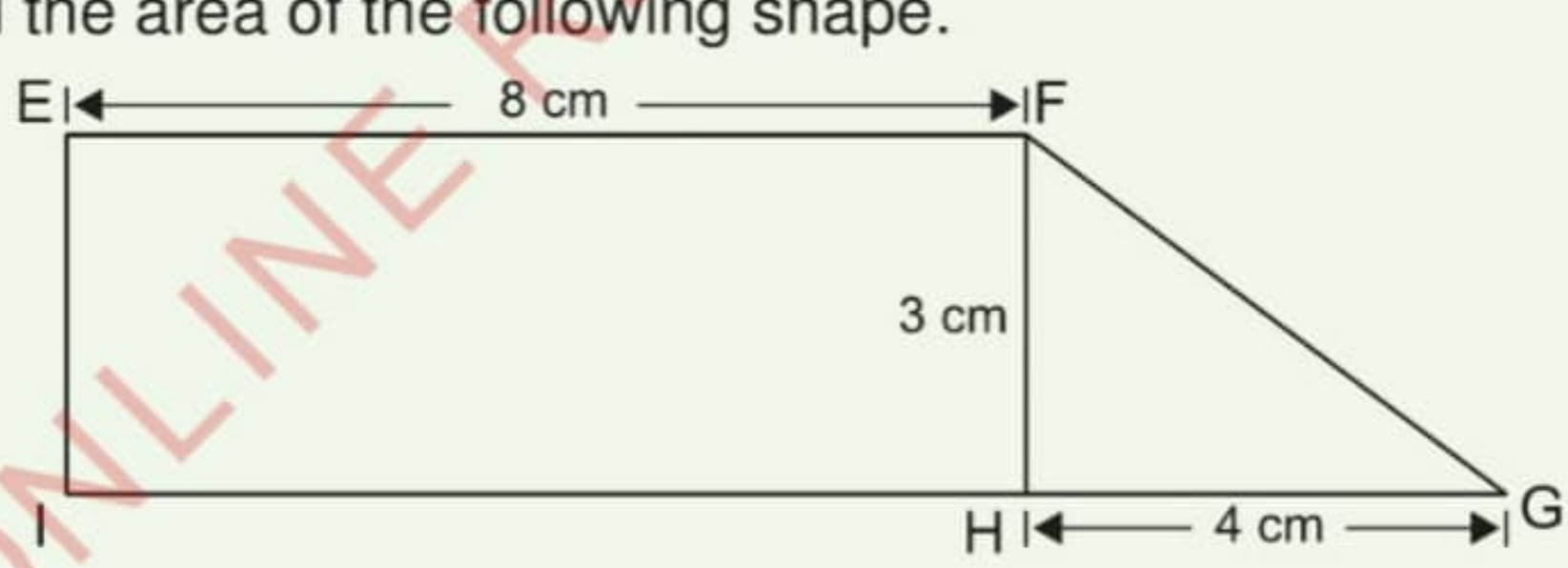
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Revision Exercise

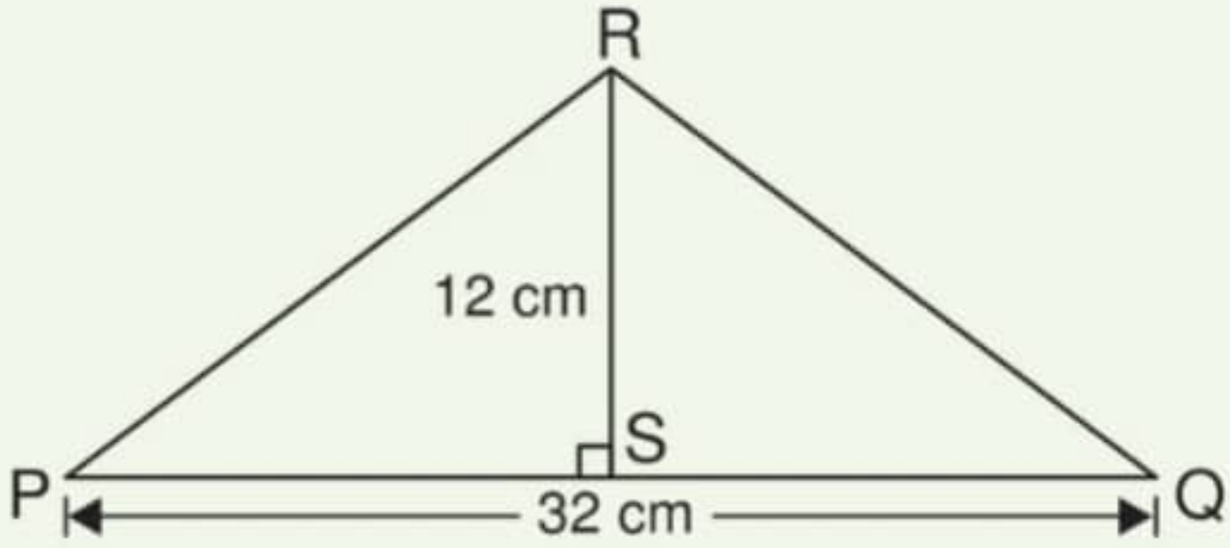
1. Find the area of a rectangle whose length is 19 centimetres and width is 16 centimetres.
2. Find the area of the following shape.



3. The length of one side of a square is 41 centimetres. Find the area of the square.
4. Find the area of the following shape.

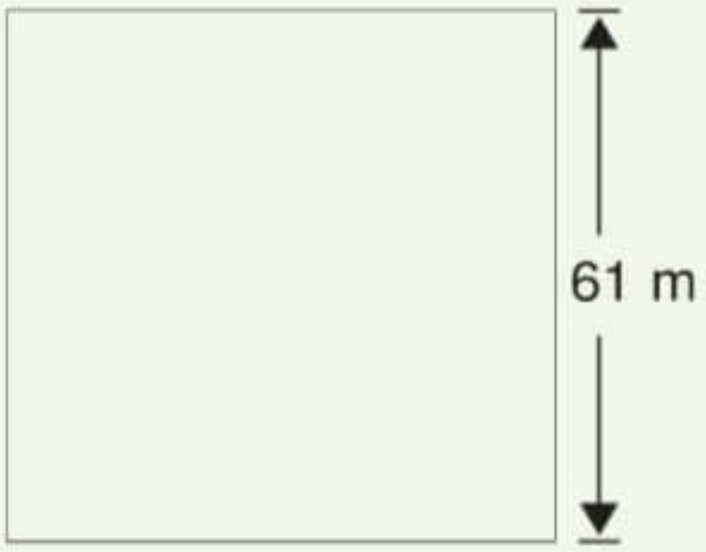


5. A triangle has a height of 25 metres and a base of 32 metres. Find its area.
6. Find the area of the shape PQR.

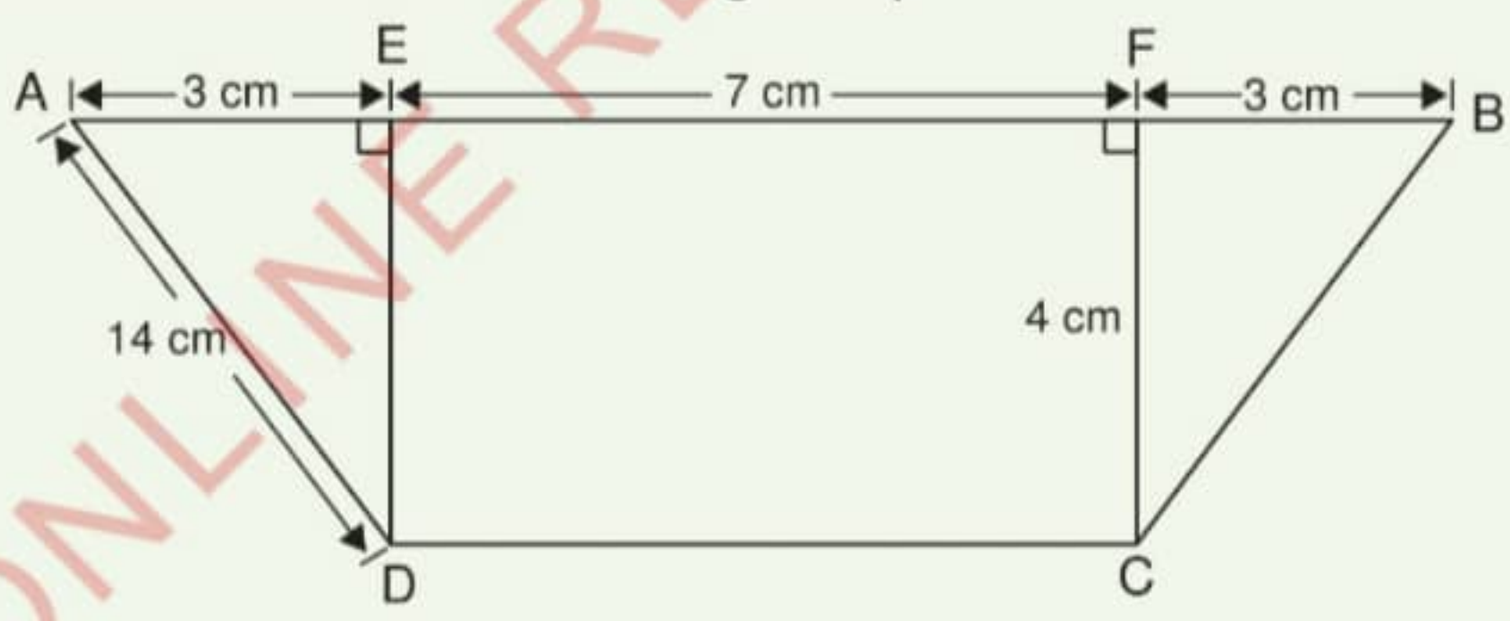


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7. Find the area of the following shape.



8. A rectangle has a length of 18 centimetres and a width of 14 centimetres. Find the area of the rectangle.
9. A square-shaped garden has a width of 27 metres. Find the area of the garden.
10. Find the area of a triangle with a base of 70 cm and a height of 84 cm.
11. Find the area of the following shape.



12. The floor of a classroom has a length of 14 metres and a width of 6 metres. Find the area of the floor.
13. The area of a triangle is 1485 square centimetres. If its base has a length of 66 centimetres, find the length of its height.
14. The area of a rectangle is 1134 square metres. If its length is 42 centimetres, find its width.
15. The area of the school farm is 3,072 square metres. If its width is 40 metres, find its length.

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Chapter Seven

Fractions

Introduction

In this chapter, you will learn to compare, multiply and divide fractions with different denominators. You will also learn to multiply and divide fractions by whole numbers. The competencies developed will enable you to apply the concept of fractions in different contexts.



Think

Life without application of fractions

Comparing fractions

Fractions can be compared using various symbols. The equal sign (=), less than sign (<) and greater than sign (>) will be used to compare fractions in this aspect.

Equivalent fractions

Equivalent fractions are compared using the equal sign (=). Any fraction with the same value as the whole has a value of 1. For example,

$\frac{2}{2} = \frac{3}{3} = \frac{4}{4} = \frac{5}{5} = \frac{6}{6} = 1.$

Note: When a fraction is multiplied by 1, its value does not change.

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Example 1

Write other four fractions that are equal to $\frac{1}{2}$.

Solution

$$\frac{1}{2} = \frac{1}{2} \times \frac{2}{2} = \frac{2}{4}$$

$$\frac{1}{2} = \frac{1}{2} \times \frac{3}{3} = \frac{3}{6}$$

$$\frac{1}{2} = \frac{1}{2} \times \frac{4}{4} = \frac{4}{8}$$

$$\frac{1}{2} = \frac{1}{2} \times \frac{5}{5} = \frac{5}{10}$$

Therefore, the other four fractions equivalent to $\frac{1}{2}$ are $\frac{2}{4}, \frac{3}{6}, \frac{4}{8}$ and $\frac{5}{10}$.

Example 2

Write other four fractions that are equal to $\frac{3}{4}$.

Solution

$$\frac{3}{4} = \frac{3}{4} \times \frac{2}{2} = \frac{6}{8}$$

$$\frac{3}{4} = \frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$$

$$\frac{3}{4} = \frac{3}{4} \times \frac{4}{4} = \frac{12}{16}$$

$$\frac{3}{4} = \frac{3}{4} \times \frac{5}{5} = \frac{15}{20}$$

Therefore, the other four fractions equivalent to $\frac{3}{4}$ are $\frac{6}{8}, \frac{9}{12}, \frac{12}{16}$, and $\frac{15}{20}$.

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Example 3

Use the equal sign (=) to recognise equivalent fractions from the following list:

$$\frac{8}{18}, \frac{2}{5}, \frac{4}{9}, \frac{18}{45}$$

Solution

Relate fractions by multiplying the fraction with a smaller denominator by a fraction with the same numerator and denominator as follows:

$$\frac{2}{5} = \frac{2}{5} \times \frac{9}{9} = \frac{18}{45}$$

$$\frac{4}{9} = \frac{4}{9} \times \frac{2}{2} = \frac{8}{18}$$

Therefore, $\frac{2}{5} = \frac{18}{45}$ and $\frac{4}{9} = \frac{8}{18}$.

Exercise 1

1. For each of the following fractions, list other three fractions which are equal.

   (a) $\frac{1}{7}$ (b) $\frac{1}{11}$ (c) $\frac{2}{9}$

   (d) $\frac{3}{15}$ (e) $\frac{1}{4}$ (f) $\frac{2}{3}$

2. In each sequence, fill in the missing fractions:

   (a) $\frac{1}{4}, \frac{2}{8}, —, \frac{4}{16}$. (b) $\frac{1}{8}, \frac{2}{16}, —, \frac{4}{32}$.

   (c) $\frac{3}{5}, —, \frac{9}{15}, —$. (d) $\frac{3}{16}, —, \frac{9}{48}$.

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3. Write equivalent fractions for each item.

(a) $\frac{2}{3}, \frac{1}{2}, \frac{4}{6}, \frac{5}{10}$

(b) $\frac{1}{4}, \frac{1}{3}, \frac{4}{12}, \frac{5}{20}$

(c) $\frac{6}{8}, \frac{1}{5}, \frac{3}{4}, \frac{5}{25}$

(d) $\frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{25}{100}, \frac{2}{16}, \frac{3}{18}$

Fractions with different values

Fractions with different values can be identified using a fraction chart. Study the following chart.

1											
$\frac{1}{2}$						$\frac{1}{2}$					
$\frac{1}{3}$				$\frac{1}{3}$				$\frac{1}{3}$			
$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$			$\frac{1}{4}$		
$\frac{1}{5}$		$\frac{1}{5}$			$\frac{1}{5}$			$\frac{1}{5}$			$\frac{1}{5}$
$\frac{1}{6}$			$\frac{1}{6}$			$\frac{1}{6}$			$\frac{1}{6}$		
$\frac{1}{7}$				$\frac{1}{7}$				$\frac{1}{7}$			
$\frac{1}{8}$					$\frac{1}{8}$					$\frac{1}{8}$	
$\frac{1}{9}$						$\frac{1}{9}$					
$\frac{1}{10}$							$\frac{1}{10}$				
$\frac{1}{11}$								$\frac{1}{11}$			
$\frac{1}{12}$									$\frac{1}{12}$		

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In a fraction chart, the shaded area represents the size of the fraction. Therefore, the shaded area in the chart can be used to compare fractions.

Example 1

Which fraction is greater between $\frac{1}{2}$ and $\frac{1}{4}$?

Solution

Examining the fraction chart, it can be seen that the shaded area representing $\frac{1}{2}$ is greater than the shaded area representing $\frac{1}{4}$.

Therefore, $\frac{1}{2}$ is greater than $\frac{1}{4}$.

Example 2

Which fraction is smaller between $\frac{2}{6}$ and $\frac{2}{5}$?

Solution

Examining the fraction chart, it can be seen that the shaded area representing $\frac{2}{6}$ is smaller than the shaded area representing $\frac{2}{5}$.

Therefore, $\frac{2}{6}$ is smaller than $\frac{2}{5}$.

Exercise 2

Use the fraction chart to answer the following questions:

1. Which fraction is greater among the following pairs?

   (a) $\frac{2}{3}$ and $\frac{5}{6}$ (b) $\frac{1}{8}$ and $\frac{2}{5}$ (c) $\frac{2}{5}$ and $\frac{2}{10}$

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2. Which item has the smallest fraction?

(a) $\frac{5}{11}$ and $\frac{2}{3}$ (b) $\frac{4}{12}$ and $\frac{6}{10}$ (c) $\frac{6}{7}$ and $\frac{3}{9}$.

3. For each of the following items, arrange the fractions from the smallest to the largest.

(a) $\frac{1}{3}, \frac{2}{3}, \frac{1}{2}$ (b) $\frac{3}{4}, \frac{7}{12}, \frac{2}{3}$ (c) $\frac{7}{8}, \frac{5}{6}, \frac{3}{4}$

4. For each of the following items, arrange the fractions from the largest to the smallest.

(a) $\frac{5}{6}, \frac{3}{4}, \frac{2}{3}$ (b) $\frac{7}{12}, \frac{5}{8}, \frac{3}{7}$ (c) $\frac{7}{10}, \frac{1}{2}, \frac{3}{5}$

Comparing fractions using < and > symbols

Fractions with different values can be compared using the symbols < (less than) or > (greater than).

Example 1

Use the symbols < or > to compare $\frac{1}{4}$ and $\frac{2}{5}$.

Solution

$\frac{1}{4}$ is smaller than $\frac{2}{5}$. Thus, use the symbol <.

Therefore, $\frac{1}{4} < \frac{2}{5}$.

Example 2

Use the symbols < or > to compare $\frac{2}{3}$ and $\frac{3}{7}$.

Solution

$\frac{2}{3}$ is greater than $\frac{3}{7}$, Thus, use the symbol >.

Therefore, $\frac{2}{3} > \frac{3}{7}$.

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Exercise 3

Use the symbols < or > to compare the fractions for each of the following:

1. $\frac{1}{2}$ and $\frac{1}{3}$
2. $\frac{1}{3}$ and $\frac{1}{4}$
3. $\frac{3}{7}$ and $\frac{5}{7}$
4. $\frac{7}{12}$ and $\frac{5}{12}$
5. $\frac{5}{10}$ and $\frac{6}{10}$
6. $\frac{1}{9}$ and $\frac{1}{12}$
7. $\frac{1}{5}$ and $\frac{1}{6}$
8. $\frac{3}{5}$ and $\frac{2}{3}$
9. $\frac{4}{8}$ and $\frac{5}{9}$
10. $\frac{7}{9}$ and $\frac{9}{10}$

Multiplication of fractions with different denominators

When multiplying fractions, the numerators and denominators are multiplied separately to obtain the answer.

Example 1

$\frac{1}{5} \times \frac{3}{7} =$

Solution

Multiply the numerators by the numerator and the denominator by the denominator:

$$\frac{1}{5} \times \frac{3}{7} = \frac{1 \times 3}{5 \times 7} = \frac{3}{35}$$

Therefore, $\frac{1}{5} \times \frac{3}{7} = \frac{3}{35}$.

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Example 2

$\frac{1}{3} \times \frac{1}{4} =$

Solution

Multiply the numerator by the numerator and the denominator by the denominator:

$$\frac{1}{3} \times \frac{1}{4} = \frac{1 \times 1}{3 \times 4} = \frac{1}{12}$$

Therefore, $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$.

Alternative method: Multiplication by using rectangular diagrams

Multiplication of fractions by fractions can be done using rectangular diagrams as follows.



Steps

1. Draw three horizontal rectangles and four vertical rectangles in the larger rectangle.
2. Shade the rectangles $\frac{1}{4}$ vertically and $\frac{1}{3}$ horizontally.
3. Count the number of squares that have been shaded twice: you get 1 square which is the numerator.
4. The total number of squares, both shaded and unshaded, is 12, which is the denominator.

   Therefore, $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

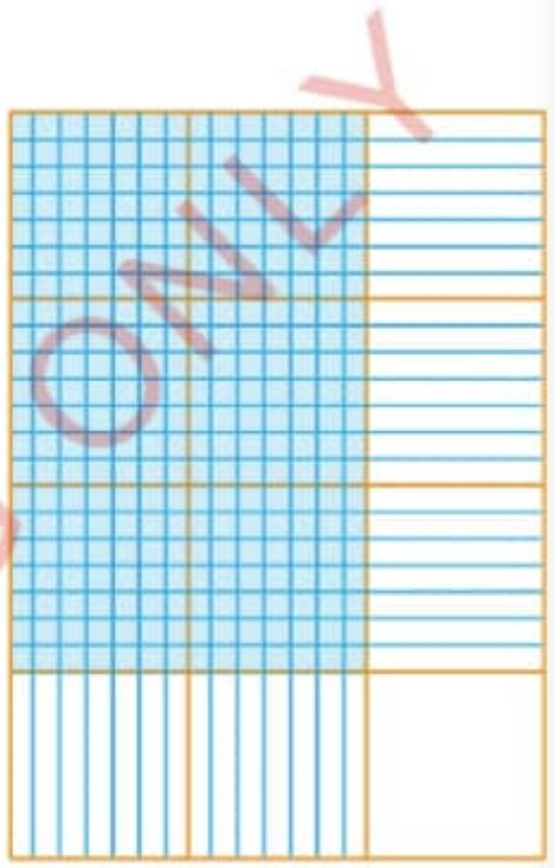
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Example 3

$\frac{3}{4} \times \frac{2}{3} =$

Steps

1. Draw four horizontal rectangles and three vertical rectangles within one larger rectangle.
2. Shade $\frac{3}{4}$ horizontally and $\frac{2}{3}$ vertically.
3. Count the number of squares that have been shaded twice: you get 6 squares units which is the **numerator.**
4. The total number of squares, both shaded and unshaded is 12, which is the **denominator.**



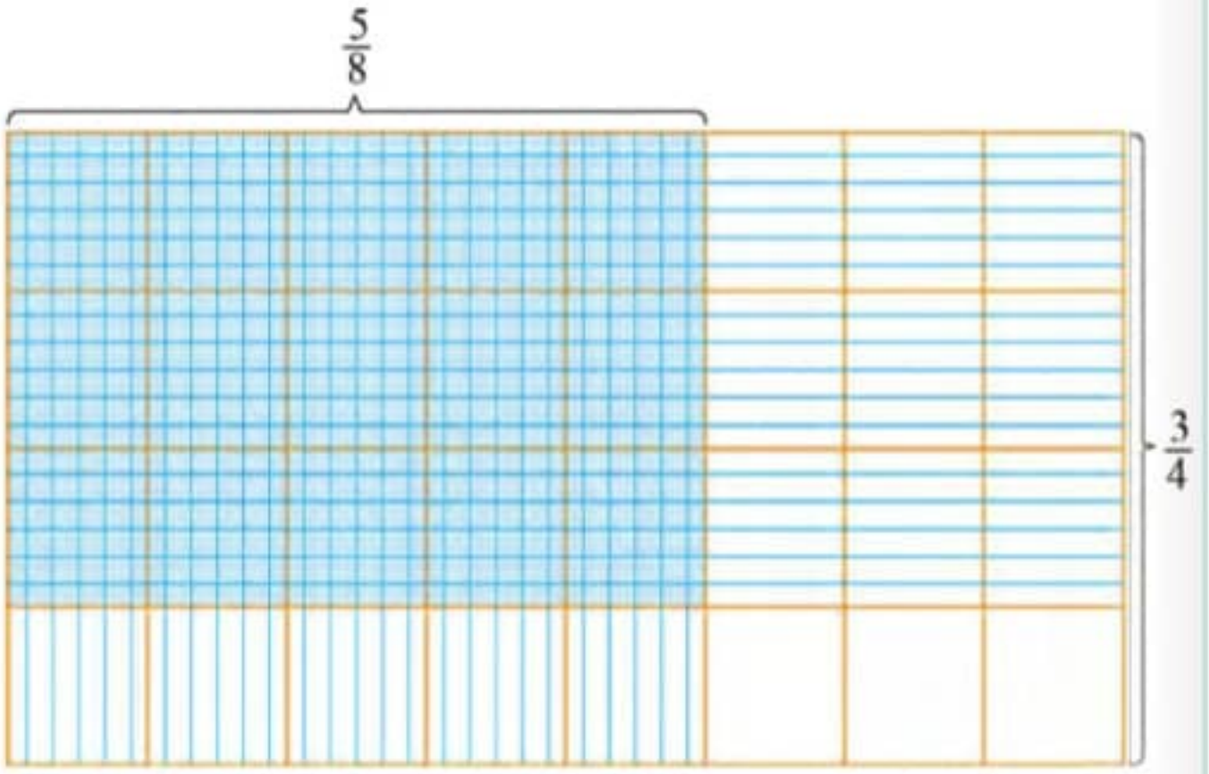
Therefore, $\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$.

Example 4

$\frac{5}{8} \times \frac{3}{4} =$

Steps

1. Draw a rectangle and divide it into 4 equal parts horizontally and 8 equal parts vertically as shown in the following diagram.



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2. Shade $\frac{3}{4}$ horizontally.
3. Shade $\frac{5}{8}$ vertically.
4. The squares that are shaded twice are the product of $\frac{5}{8} \times \frac{3}{4}$.
5. Count the number of squares that have been shaded twice: you get 15 squares, which is the numerator.
6. Count all shaded and unshaded squares. The total is 32 which is the denominator.

Therefore, $\frac{5}{8} \times \frac{3}{4} = \frac{15}{32}$.

Exercise 4

1. $\frac{2}{5} \times \frac{3}{7} =$
2. $\frac{6}{9} \times \frac{1}{7} =$
3. $\frac{2}{4} \times \frac{3}{5} =$
4. $\frac{3}{8} \times \frac{5}{10} =$
5. $\frac{13}{15} \times \frac{2}{4} =$
6. $\frac{7}{12} \times \frac{1}{6} =$
7. $\frac{4}{5} \times \frac{6}{11} =$
8. $\frac{1}{4} \times \frac{2}{7} =$
9. $\frac{2}{3} \times \frac{7}{10} =$
10. $\frac{8}{9} \times \frac{1}{3} =$
11. $\frac{2}{3} \times \frac{4}{7} =$
12. $\frac{11}{13} \times \frac{1}{5} =$
13. $\frac{3}{14} \times \frac{1}{2} =$
14. $\frac{5}{9} \times \frac{4}{6} =$
15. $\frac{7}{8} \times \frac{1}{5} =$
16. $\frac{2}{11} \times \frac{4}{5} =$
17. $\frac{3}{5} \times \frac{5}{9} =$
18. $\frac{3}{4} \times \frac{6}{10} =$

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Multiplication of fractions by whole numbers

This involves converting the whole number into fractions then followed by the rule of multiplying fractions by fraction.

Example 1

$\frac{1}{5} \times 4 =$

Solution

When multiplying a whole number by a fraction, write the whole number as a fraction:

$$4 = \frac{4}{1}$$

Then, multiply numerators and denominators separately:

$$\frac{1}{5} \times 4 = \frac{1}{5} \times \frac{4}{1}$$

$$= \frac{4}{5}$$

Therefore, $\frac{1}{5} \times 4 = \frac{4}{5}$.

Example 2

$\frac{2}{3} \times 12 =$

Solution

$$\frac{2}{3} \times 12 = \frac{2}{3} \times \frac{12}{1}$$

$$= \frac{24}{3}$$

$$= 8$$

Therefore, $\frac{2}{3} \times 12 = 8$.

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Exercise 5

1. $6 \times \frac{3}{4} =$
2. $\frac{2}{3} \times 9 =$
3. $\frac{5}{6} \times 30 =$
4. $\frac{1}{2} \times 24 =$
5. $27 \times \frac{1}{9} =$
6. $15 \times \frac{2}{5} =$
7. $12 \times \frac{1}{6} =$
8. $\frac{2}{10} \times 4 =$
9. $\frac{5}{9} \times 18 =$
10. $21 \times \frac{1}{7} =$
11. $\frac{1}{3} \times 36 =$
12. $\frac{1}{8} \times 32 =$
13. $5 \times \frac{1}{6} =$
14. $20 \times \frac{1}{4} =$
15. $\frac{1}{10} \times 50 =$
16. $3 \times \frac{3}{11} =$
17. $13 \times \frac{1}{13} =$
18. $40 \times \frac{1}{20} =$

Word problems involving multiplication of fractions

Example

A pupil drinks $\frac{3}{4}$ litres of milk each day. How many litres of milk will the pupil drink in six days?

Solution

$$\frac{3}{4} \times 6 = \frac{3}{4} \times \frac{6}{1}$$

$$= \frac{3 \times 6}{4 \times 1} = \frac{18}{4}$$

$$= \frac{9}{2}$$

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$= \frac{9}{2}$

Therefore, the pupil will drink $\frac{9}{2}$ litres of milk in six days.

Exercise 6

1. Eliud had 20 sacks of millet. He gave. $\frac{1}{2}$ of those sacks to his brother. How many sacks of millet did his brother receive?
2. A certain guest house has 48 beds. If $\frac{1}{12}$ of those beds are metal, how many beds are metal?
3. Kangaroo ran $\frac{1}{3}$ of a distance of 15 kilometres. What distance did it run?
4. A certain primary school is expected to register 160 pupils. If $\frac{7}{8}$ of the pupils were registered, how many pupils were not registered?
5. A grandfather drinks half a litre of milk each day. How many litres of milk does the grandfather drink in four weeks?

Division of fractions by whole numbers

When dividing a fraction by a whole number, first convert the whole number into a fraction and change it to a fraction with 1 as the denominator. Then, use the rule of multiplying fractions.

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Example 1

The teacher divided half of a circular sheet of paper into three equal pieces and shared them equally with three pupils. What fraction of the whole circular sheet did each pupil get?

Steps

1. The problem description can be written using numbers and the division symbol as follows:
   $\frac{1}{2} \div 3 =$
2. If half of a piece of paper is divided into three equal parts, there will be a total of six pieces.



3. One piece out of 6 in the fraction is written as $\frac{1}{6}$. Thus, $\frac{1}{2} \div 3 = \frac{1}{6}$.

   Therefore, each pupil got $\frac{1}{6}$ of the circular sheet.

Alternative solution

1. Convert 3 into a fraction, $\frac{3}{1}$.
2. Change the denominator to the numerator and the numerator to the denominator, $\frac{1}{3}$.
3. Multiply: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$.

Therefore, each pupil got $\frac{1}{6}$ of the whole circular sheet.

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Example 2

$\frac{5}{3} \div 15 =$

Solution

Follow the steps shown in Example 1.

$$\frac{5}{3} \div 15 = \frac{5}{3} \div \frac{15}{1}$$

$$= \frac{5}{3} \times \frac{1}{15}$$

$$= \frac{5}{45}$$

$$= \frac{1}{9}$$

Therefore, the answer is $\frac{5}{45}$ or $\frac{1}{9}$.

Exercise 7

1. $\frac{4}{5} \div 10 =$
2. $\frac{8}{11} \div 16 =$
3. $\frac{12}{13} \div 8 =$
4. $\frac{12}{7} \div 6 =$
5. $\frac{121}{90} \div 55 =$
6. $\frac{81}{10} \div 9 =$
7. $\frac{125}{3} \div 25 =$
8. $\frac{3}{20} \div 28 =$
9. $\frac{9}{17} \div 108 =$
10. $\frac{396}{5} \div 22 =$
11. $\frac{1}{64} \div 24 =$
12. $\frac{6}{20} \div 12 =$
13. $\frac{11}{5} \div 33 =$
14. $\frac{5}{2} \div 46 =$
15. $\frac{230}{31} \div 76 =$
16. $\frac{15}{7} \div 8 =$
17. $\frac{32}{5} \div 9 =$
18. $\frac{53}{8} \div 7 =$
19. $\frac{6}{13} \div 15 =$
20. $\frac{15}{16} \div 38 =$

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Division of fractions with different denominators

When dividing fractions by fraction interchange its denominator and numerator, then apply the rule of multiplying fractions by fraction. Similarly, fractions can be divided using diagrams.

Example 1

$\frac{3}{8} \div \frac{4}{6} =$

Steps

1. Converting the numerator and denominator of the fraction $\frac{4}{6}$, the divisor becomes $\frac{6}{4}$.
2. Multiply: $\frac{3}{8} \times \frac{6}{4} = \frac{18}{32}$

$= \frac{9}{16}$

Therefore, the answer is $\frac{9}{16}$.

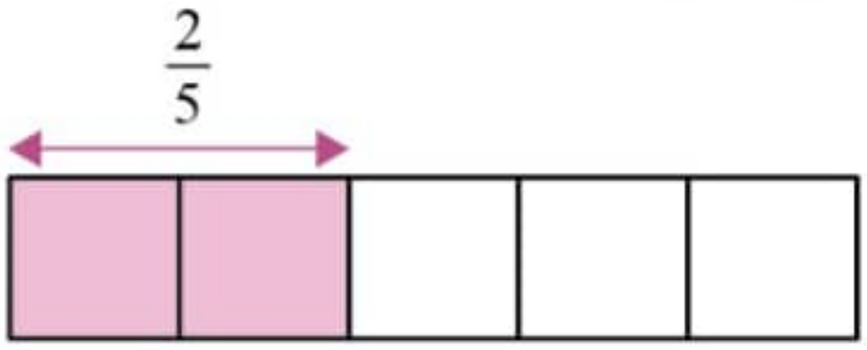
Example 2

Use a diagram to find the value of:

$\frac{2}{5} \div \frac{1}{10} =$

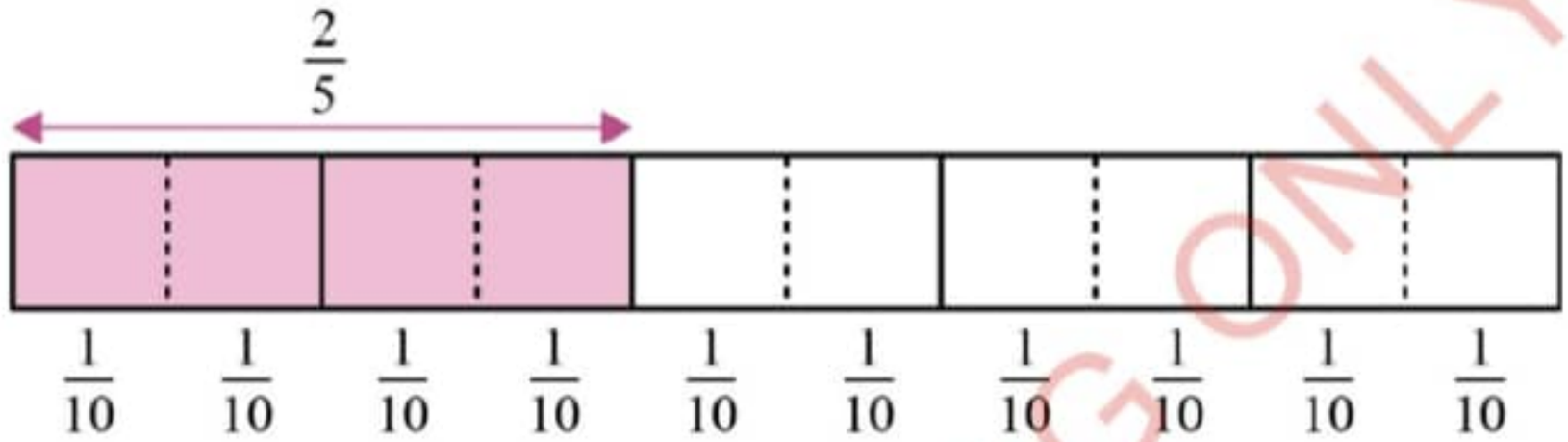
Steps

1. Show $\frac{2}{5}$ on a diagram by drawing five equal boxes as shown in the following diagram:



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2. $\frac{2}{5} \div \frac{1}{10}$ is equivalent to saying how many $\frac{1}{10}$ fit into $\frac{2}{5}$?
Divide all the boxes in the diagram to get 10 parts as shown in the following diagram:



3. In step 2, the diagram shows that in $\frac{2}{5}$, there are four $\frac{1}{10}$.
Therefore, the answer is 4.

Exercise 8

1. $\frac{5}{12} \div \frac{3}{4} =$
2. $\frac{12}{48} \div \frac{3}{24} =$
3. $\frac{4}{13} \div \frac{8}{39} =$
4. $\frac{24}{1000} \div \frac{6}{8} =$
5. $\frac{17}{3} \div \frac{21}{34} =$
6. $\frac{25}{100} \div \frac{3}{8} =$
7. $\frac{6}{122} \div \frac{3}{4} =$
8. $\frac{32}{56} \div \frac{25}{4} =$
9. $\frac{3}{121} \div \frac{5}{7} =$
10. $\frac{2}{9} \div \frac{25}{51} =$
11. $\frac{104}{112} \div \frac{13}{42} =$
12. $\frac{32}{117} \div \frac{4}{39} =$
13. $\frac{27}{30} \div \frac{9}{10} =$
14. $\frac{1}{256} \div \frac{9}{80} =$
15. $\frac{1}{60} \div \frac{25}{27} =$
16. $\frac{125}{12} \div \frac{15}{4} =$
17. $\frac{65}{480} \div \frac{17}{7} =$
18. $\frac{12}{45} \div \frac{18}{35} =$

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Activity — Practice on multiplication and division of fractions

Description

Use reliable online programs such as Khan Academy to learn more fractions.

Revision Exercise

1. Write three equivalent fractions for each of the following fractions:

   (a) $\frac{2}{5}$ (b) $\frac{1}{12}$ (c) $\frac{4}{11}$

2. Which fraction is greater between the following pairs:

   (a) $\frac{3}{9}$ and $\frac{5}{9}$ (b) $\frac{1}{6}$ and $\frac{1}{5}$ (c) $\frac{4}{8}$ and $\frac{3}{4}$

3. Which fraction is smaller between the following pairs:

   (a) $\frac{3}{4}$ and $\frac{2}{3}$ (b) $\frac{7}{10}$ and $\frac{3}{5}$ (c) $\frac{2}{3}$ and $\frac{3}{7}$

4. Compare the following fractions using the symbols =, <, or >.

   (a) $\frac{1}{10}$ and $\frac{2}{9}$ (b) $\frac{3}{4}$ and $\frac{12}{16}$ (c) $\frac{2}{3}$ and $\frac{1}{2}$

   (d) $\frac{6}{10}$ and $\frac{5}{8}$ (e) $\frac{5}{6}$ and $\frac{10}{12}$ (f) $\frac{4}{11}$ and $\frac{7}{11}$

5. (a) $\frac{2}{5} \times \frac{3}{4} =$ (b) $\frac{2}{3} \times \frac{3}{5} =$ (c) $\frac{4}{9} \times \frac{3}{4} =$

   (d) $\frac{1}{4} \times \frac{5}{6} =$ (e) $\frac{5}{8} \times \frac{1}{5} =$ (f) $\frac{3}{7} \times \frac{4}{13} =$

6. (a) $6 \times \frac{2}{5} =$ (b) $12 \times \frac{2}{7} =$ (c) $18 \times \frac{4}{8} =$

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(d) $21 \times \frac{5}{9} =$ (e) $13 \times \frac{1}{17} =$ (f) $10 \times \frac{2}{10} =$

7. (a) $\frac{2}{5} \div \frac{3}{4} =$ (b) $\frac{2}{3} \div \frac{4}{8} =$ (c) $\frac{4}{9} \div \frac{3}{4} =$

(d) $\frac{4}{15} \div \frac{4}{7} =$ (e) $\frac{4}{5} \div \frac{2}{6} =$ (f) $\frac{5}{20} \div 15 =$

8. A class has a total of 100 pupils. If $\frac{1}{4}$ of these pupils are boys, how many of them are girls?

9. Masalu had 88 mangoes. If he sold $\frac{7}{8}$ of those mangoes, how many mangoes were sold?

10. An institute distributed 445 Mathematics books to 3 schools. If one school got $\frac{2}{5}$ of all the books, how many books did this school get?

11. In one boarding school, each pupil eats $\frac{1}{8}$ kilogram of meat. How many pupils will eat the meat weighing 64 kilograms?

12. How many $\frac{1}{1000}$ are there in $\frac{1}{10}$?

13. The households of a village shared 2 tonnes of maize. If each household was given $\frac{}{20}$ of a tonne, how many households are in that village?

14. How many pieces of boards each $\frac{5}{4}$ metres long can be cut from a board that is 75 metres long?

15. The parents divided $\frac{1}{2}$ of the land to their children. If each child got $\frac{1}{12}$ of the land, how many children did they have?

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Chapter Eight

Decimals

Introduction

In this chapter, you will learn how to convert fractions into decimals, add and subtract decimals. The competencies developed will enable you to apply the concept of decimals in metric measurements, constructions, exchange of money, sports, medicine, health and fitness and many other real-life applications.



Think

Life without application of decimal numbers

Concept of decimals

Decimals are numbers that consist of a whole number and a fractional part separated by a dot called decimal point. Decimals are system of counting that originates from the base-ten system. Study the following diagram that explain the concept of decimals.



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This shape is divided into ten equal parts. Each part represents $\frac{1}{10}$ of the whole shape. The shaded portion is $\frac{1}{10}$ and is written as 0.1 in decimal form. The number 0.1 is obtained by dividing 1 by 10. Decimals and fractions that can be derived from the former diagram include:

Fraction	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{5}{10}$	$\frac{6}{10}$	$\frac{7}{10}$	$\frac{8}{10}$	$\frac{9}{10}$
Decimals	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

In decimals, the dot point separates the whole number from the number that is a fraction of ten.

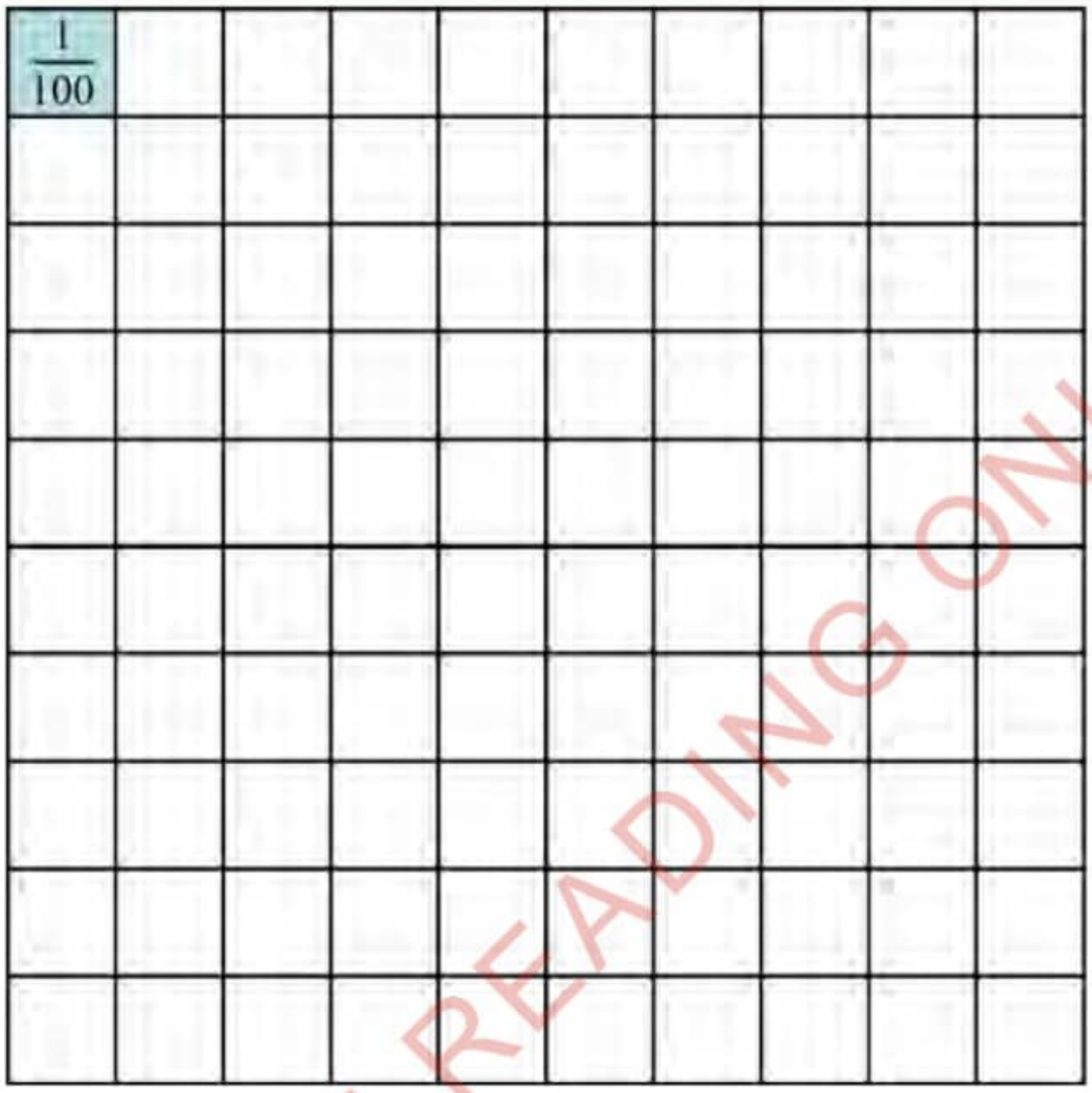
Reading decimals

Decimal numbers have one or more than one decimal place. Decimal numbers are read from left to right. For example;

Decimal number	How the number is read
0.7	zero point seven
1.8	one point eight
25.6	twenty-five point six
250.1	two hundred fifty point one

Decimals with two places can be explained using the following diagram.

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The shaded part is $\frac{1}{100}$ of the whole shape and is written as 0.01.

The number 0.01 is obtained by dividing 1 by 100. Other decimals can be obtained from the diagram. For example;

(a) $\frac{3}{100} = 0.03$ (b) $\frac{12}{100} = 0.12$ (c) $\frac{94}{100} = 0.94$

Example of decimal numbers with more than two decimal places are:

(a) $\frac{2}{1000} = 0.002$ (b) $\frac{14}{1000} = 0.014$ (c) $\frac{127}{1000} = 0.127$

Numbers with more than one decimal place are read from left to right. The digits to the left of the decimal point are read as whole numbers. The digits to the right of the decimal point are read individually. For example;

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Decimals numbers	How the number is read
0.05	zero point zero five
0.13	zero point one three
15.34	fifteen point three four
525.25	five hundred and twenty-five point two five
131.026	one hundred and thirty-one point zero two six
725.307	seven hundred and twenty-five point three zero seven

Exercise 1

Read the following numbers:

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
18.9	2.07	716.881	113.04	631.8	2.004	24.99	3.00	0.019

Writing decimals in numerals and in words

Decimals can be written in numerals and in words.

Example 1

Write the following number in words:

(a) 2.1 (b) 0.08 (c) 13.26 (d) 121.712

Answer

	Decimal in numerals	Decimal numbers in words
(a)	2.1	two point one
(b)	0.08	zero point zero eight
(c)	13.26	thirteen point two six
(d)	121.712	one hundred and twenty-one point seven one two

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Example 2

Write the following numbers in numerals:

(a) Zero point zero five

(b) Eleven point six

(c) Twenty-three point three seven

(d) One hundred fifty-seven point four two seven

Answer

	Number in words	Number in numerals
(a)	zero point zero five	0.05
(b)	eleven point six	11.6
(c)	twenty-three point three seven	23.37
(d)	one hundred and fifty-seven point four two seven	157.427

Exercise 2

1. Write the following numbers in words:

   (a) 68.0 (b) 0.904 (c) 203.405 (d) 0.29

2. Write the following numbers in numerals:

   (a) Two hundred and nine point zero

   (b) Zero point zero zero six

   (c) Fifty-one point eight two

   (d) Eight hundred and fifty-one point nine two four

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Conversion of fractions into decimals

Decimals are obtained by dividing the numerator by the denominator of a fraction.

Example 1

Convert $\frac{6}{10}$ into decimals.

Steps	Solution
1. Divide the numerator by the denominator: $6 \div 10$, it is not sufficient. Write 0 above the digit 6 in the answer position, then put a decimal point. 2. Multiply: $0 \times 10 = 0$. Write 0 below 6 then subtract: $6 - 0 = 6$. 3. Write zero in front of 6 to make it 60, then divide: $60 \div 10 = 6$. Write 6 in the answer position after the decimal point. 4. Multiply: $6 \times 10 = 60$. Write 60 below 60 then subtract: 60 60 = 0.	$10\overline{)\,6}$ → 0.6 −0 60 −60 0

Therefore, $\frac{6}{10} = 0.6$.

Example 2

Convert $\frac{1}{2}$ into decimals.

Steps	Solution
1. Divide the numerator by the denominator: $1 \div 2$, it is not sufficient. Write 0 above the digit 1 in the answer position, then put a decimal point. 2. Multiply: $0 \times 2 = 0$. Write 0 below 1then subtract: $1 - 0 = 1$. 3. Write a zero in front of 1 to make it 10, then divide: $10 \div 2 = 5$. Write 5 in the answer position after the decimal point.	$2\overline{)\,1}$ → 0.5 −0 10 −10 0

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4. Multiply: $5 \times 2 = 10$. Write 10 below 10 then subtract,: $10 - 10 = 0$.

Therefore, $\frac{1}{2} = 0.5$

Example 3

Convert $\frac{1}{100}$ into decimal.

Steps	Solution
1. Divide the numerator by the denominator: $1 \div 100$, it is not sufficient. Write 0 above the digit 1 in the answer position, then put a decimal point. 2. Multiply: $0 \times 10 = 0$. Write 0 below 1 then subtract: $1 - 0 = 1$. 3. Write a zero in front of the 1 to make it 10, then divide: $10 \div 100$, it is not sufficient. Write 0 to the right of the decimal point in the answer position, then write 0 in front of 10 to make it 100. 4. Divide 100 by 100 to get 1. Write 1 in the answer position to get 0.01. 5. Multiply: $1 \times 100 = 100$. Write 100 below 100 then subtract:$100 - 100 = 0$..	$100\overline{)1}$ = 0.01 -0 10 -0 100 -100 0

Therefore, $\frac{1}{100} = 0.01$.

Exercise 3

Convert the following fractions into decimals:

1. (a) $\frac{3}{10}$ (b) $\frac{8}{10}$ (c) $\frac{7}{10}$ (d) $\frac{12}{10}$
2. (a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) $\frac{3}{2}$ (d) $\frac{7}{5}$

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3. (a) $\frac{3}{100}$ (b) $\frac{15}{100}$ (c) $\frac{76}{100}$ (d) $\frac{83}{100}$

4. (a) $\frac{1}{25}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{5}{4}$

5. (a) $\frac{6}{100}$ (b) $\frac{108}{100}$ (c) $\frac{176}{100}$ (d) $\frac{27}{100}$

Addition of decimal numbers

Decimal numbers are added by considering the value of each digit. There are various methods applied in adding decimals.

Addition of decimal numbers with one decimal point

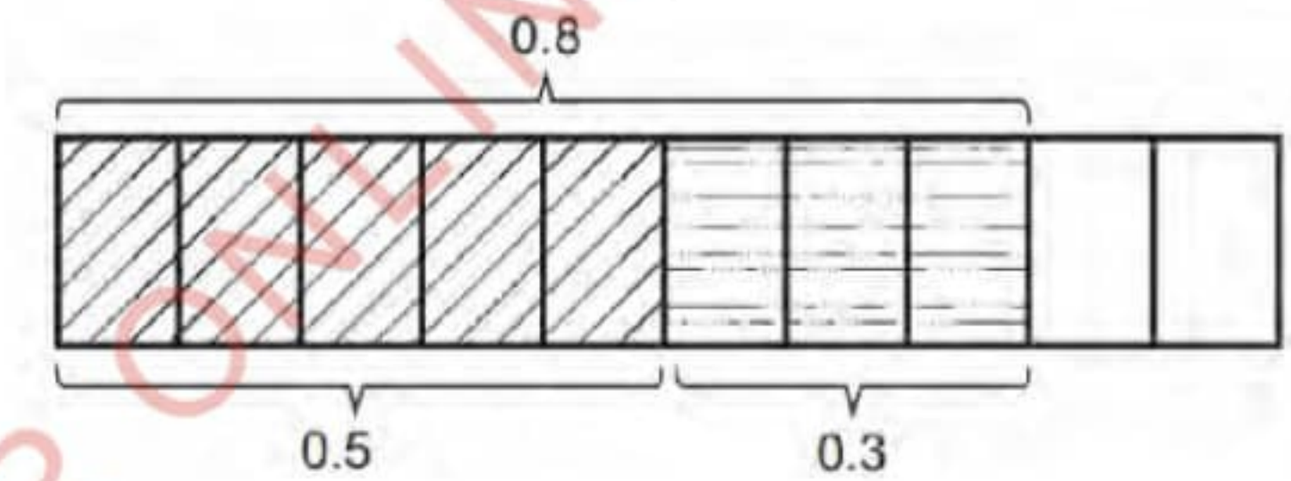
When adding decimals, numbers in the decimal part are added first followed by whole numbers.

Example 1

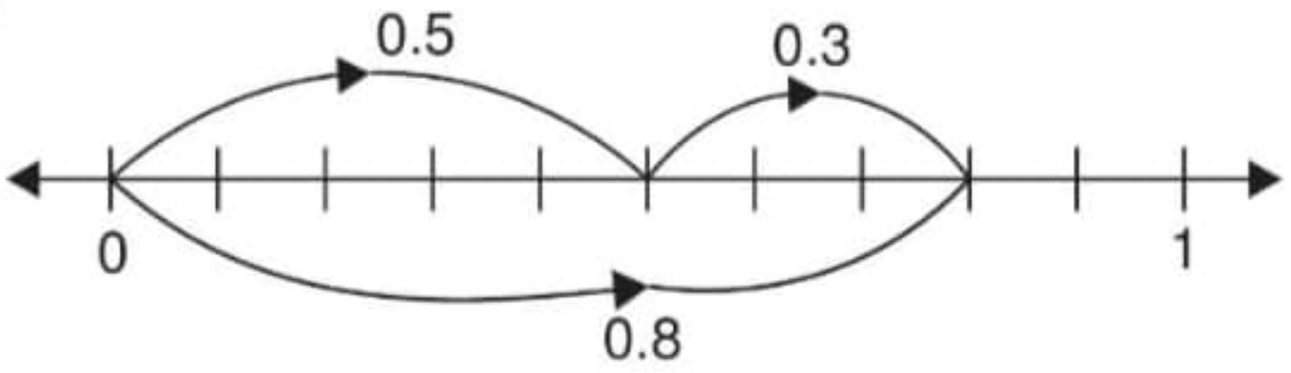
0.5 + 0.3

Solution

Use the following diagram.



Alternatively;



From the diagram, 0.8 has 0 whole number and 8 as part of 10.

Using this diagram, $0.5 + 0.3 = 0.8$.

Therefore, $0.5 + 0.3 = 0.8$.

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Example 2

$0.5 + 0.2 =$

Steps	
1. Add the digits to the right of the decimal point, $5 + 2 = 7$. Write 7 in the decimal place.	$\begin{array}{r} 0.5 \\ +\ 0.2 \\ \hline 0.7 \end{array}$
2. Add the whole numbers: $0 + 0 = 0$. Write 0 in the place for the whole number.	

Therefore, $0.5 + 0.2 = 0.7$.

Example 3

$$\begin{array}{r} 1.7 \\ +\ 2.6 \\ \hline \\ \hline \end{array}$$

Steps	
1. Add the digits to the right of the decimal point: $7 + 6 = 13$. Write 3 in the decimal place. The remaining whole number is 1. Shift 1 to the whole numbers position.	$\begin{array}{r} {\scriptstyle +1} \\ 1.7 \\ +2.6 \\ \hline 4.3 \end{array}$
2. Add the whole numbers: $1 + 1 + 2 = 4$. Write 4 in the whole number place.	

Therefore, $1.7 + 2.6 = 4.3$.

Exercise 4

1. $0.4 + 0.7 =$	2. $0.6 + 0.3 =$	3. $126.7 + 2.1 =$
4. $6.4 + 7.9 =$	5. $23.4 + 5.2 =$	6. $36.7 + 5.4 =$
7. $34.9 + 98.5 =$	8. $86.5 + 12.2 =$	9. $137.1 + 28.7 =$

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10. $250.8+22.4=$ 11. $325.4+86.8=$ 12. $58.0+2.6=$

13. $224.2+4.5=$ 14. $79.3+46.1=$ 15. $188.3+15.9=$

16. $92.6+29.8=$

Addition of decimal numbers with more than one decimal place

Numbers with more than one decimal place can be added by starting with decimal parts then whole numbers.

Example 1

$2.45+0.31=$

Steps	Solution
1. Add the decimals: $45+31=76$. Write 76 in decimal place.	$\begin{array}{r} 2.45 \\ +\ 0.31 \\ \hline 2.76 \end{array}$
2. Add the whole numbers: $2+0=2$. Write 2 in the whole number place.	

Therefore, $2.45+0.31=2.76$.

Example 2

$7.98+2.65=$

Steps	Solution
1. Add the decimals: $98+65=163$. Write 63 in the decimal place. Shift 1 to the whole numbers position. 2. Add the whole numbers: $1+7+2=10$. Write 10 in the whole number place.	$\begin{array}{r} {\scriptstyle +1} \\ 7.98 \\ +\ 2.65 \\ \hline 10.63 \end{array}$

Therefore, $7.98+2.65=10.63$.

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Exercise 5

1. $0.35+0.24=$
2. $1.56+0.33=$
3. $2.36+1.22=$
4. $26.72+38.28=$
5. $38.64+6.27=$
6. $21.527+9.294=$
7. $37.26+2.83=$
8. $22.781+19.209=$
9. $3.68+72.44=$
10. $95.25+6.7=$
11. $33.201+121.726=$
12. $94.18+24.32=$
13. $8.3+13.88=$
14. $76.09+21.05=$
15. $286.04+13.179=$

Subtraction of decimal numbers

In subtracting decimals considers the value or place values of the digits before or after the decimal point.

Subtraction of decimal numbers with one decimal place

When subtracting decimal numbers with one decimal place, first subtract decimal parts followed by whole numbers.

Example 1

$5.3-3.2=$

Steps	Solution
1. Subtract decimals: 3 – 2 = 1. Write 1 in the decimal place. 2. Subtract whole numbers: 5 – 3 = 2. Write 2 in the whole number place.	$\begin{array}{r} 5.3 \\ -\,3.2 \\ \hline 2.1 \end{array}$

Therefore, $5.3-3.2=2.1$.

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Example 2

$26.5 - 18.6 =$

Steps	Solution
1. Subtract decimals: 5 – 6, it is not sufficient. Take 1 group of ones from the 6 ones and change into 10. Add decimals: 10 + 5 = 15. Subtract decimals: 15 – 6 = 9. Write 9 in the decimals place. Remember in tens place 1 tens remains. 2. Subtract whole number: 5 – 8, it is not sufficient. Add ones: 10 + 5 = 15. Subtract whole numbers: 15 – 8 = 7. Write 7 in the ones place. Remember in tens place, 1 tens remains. 3. Subtract whole numbers: 1 – 1 = 0.	26.5 −18.6 7.9

Therefore, $26.5 - 18.6 = 7.9$.

Exercise 6

1. $0.8 - 0.2 =$
2. $0.9 - 0.7 =$
3. $2.4 - 0.8 =$
4. $3.6 - 2.8 =$
5. $4.7 - 0.9 =$
6. $22.7 - 4.0 =$

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7. $35.6 - 24.2 =$

8. $64.5 - 34.8 =$

9. $125.2 - 0.7 =$

10. $226.5 - 122.5 =$

11. $27.4 - 13.3 =$

12. $12.6 - 1.4 =$

13. $17.8 - 9.7 =$

14. $45.1 - 22.8 =$

15. $125.2 - 94.8 =$

16. $0.6 - 0.3 =$

17. $30.8 - 15.3 =$

18. $120.2 - 98.7 =$

19. $55.1 - 54.9 =$

20. $235.9 - 0.7 =$

Subtraction of decimal numbers with more than one decimal place

In subtracting decimal numbers, start with decimal parts followed by whole numbers.

Example 1

$0.84 - 0.21 =$

Steps	Solution
1. Align the numbers vertically. Start by subtracting the numbers on the decimal side: $4 - 1 = 3$. Write 3 in the decimal place. 2. Subtract decimals: $8 - 2 = 6$. Write 6 in the decimal place to the left of 3. 3. Subtract whole numbers: $0 - 0 = 0$. Write 0 in the whole number place.	$\begin{array}{r} 0.84 \\ -0.21 \\ \hline 0.63 \end{array}$

Therefore, $0.84 - 0.21 = 0.63$.

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Example 2

$12.464 - 8.642 =$

Solution

$$\begin{array}{r} 12.464 \\ -\ 8.642 \\ \hline 3.822 \\ \hline \end{array}$$

Therefore, $12.464 - 8.642 = 3.822$.

Exercise 7

1. $0.62 - 0.21 =$
2. $0.25 - 0.12 =$
3. $0.86 - 0.34 =$
4. $0.82 - 0.36 =$
5. $3.05 - 2.45 =$
6. $0.33 - 0.29 =$
7. $2.4 - 0.26 =$
8. $95.501 - 88.342 =$
9. $5.27 - 4.25 =$
10. $11.008 - 0.007 =$
11. $12.54 - 6.7 =$
12. $47.76 - 34.25 =$
13. $76.468 - 56.342 =$
14. $364.20 - 72.66 =$
15. $96.31 - 28.29 =$
16. $800.426 - 345.907 =$
17. $45.08 - 2.424 =$
18. $237.78 - 116.572 =$
19. $102.064 - 41.596 =$
20. $312.007 - 98.06 =$

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Word problems involving decimals

Decimal numbers are used in everyday activities in situation where more precision is required.

Example 1

A farmer divided the farm into three areas and planted various crops in each area. He planted 0.35 of the farm with sorghum, 0.25 with coffee and 0.20 with maize. Find the total area used to plant those crops.

Solution

Area planted with sorghum = 0.35.

Area planted with coffee = 0.25.

Area planted with maize = 0.20.

Total area planted with crops $0.35+0.25+0.20=0.80$.

Therefore, the total area used to plant these crops is 0.80.

Example 2

The length of the rectangle is 102.75 centimetres and its width is 82.8 centimetres. Find the difference between the length and the width of the rectangle.

Solution

Length of the rectangle = 102.75 cm

Width of the rectangle = 82.8 cm

Difference between length and width:

$102.75\text{ cm} - 82.8\text{ cm} = 19.95\text{ cm}$

Therefore, the difference between the length and the width of the rectangle is 19.95 cm.

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Exercise 8

1. Amina bought 2.15 kilograms of onions and 4.72 kilograms of potatoes. How many kilograms did Amina buy in total?
2. Rashidi weighs 70.5 kilograms and Joyce weighs 68.15 kilograms. Find their total weight.
3. One farmer stores 80.9 tonnes of crops in his warehouse. Another farmer stores 132.708 tonnes of crops in his warehouse. How many tonnes of crops do they store in total?
4. Juliana sold 851.5 kilograms of maize and 925.25 kilograms of millet. Find the difference between the weight of millet and maize.
5. Two dairy cows produce milk as follows: the first cow produces 18.38 litres of milk and the second produces 22.34 litres of milk. Find the difference of the volume of milk produced by these cows.
6. A train is pulling four carriages weighing 112.37 tons, 115.74 tons, 108.39 tonnes and 114.28 tonnes. How many total tonnes of carriages is the train pulling?
7. A carpenter used planks with length of 0.85 metres, 0.99 metres, and 1.75 metres to make a table. Find the total length of the planks used.
8. A businessman sold 275.85 kilograms of rice and 152.125 kilograms of wheat flour. How many total kilograms of rice and wheat flour did the businessman sell?

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9. The weight of the beans is 3.275 tonnes and the weight of rice is 5.45 tonnes. Find the difference between the weight of the rice and the weight of the beans.
10. The tanks have capacities of 375.75 litres, 424.25 litres and 522.85 litres. Find the total capacity of these tanks.

Activity **Learn more about how to perform addition and subtraction of decimals using reliable online lessons.**

Description

Use various programs such as Khan Academy to learn more about recognizing decimals with different place value and how to correctly perform addition and subtraction of decimals.

Summary

1. Decimals are read from left to right. The digits to the left of the decimal point are read as a whole number. The digits to the right of the decimal point are read one by one.
2. When adding or subtracting decimal numbers, consider the value of the digits or the place value of the digits before and after the decimal point.

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Revision Exercise

1. Write the following numerals in words:
   (a) 0.9 (b) 3.21 (c) 1.008 (d) 459.367

2. Write the following in numerals.
   (a) Ten point zero
   (b) Eighty-seven point three two
   (c) Zero point zero zero three
   (d) Nine hundred and ninety-nine point nine nine nine

3. Convert the following fractions into decimals:
   (a) $\frac{3}{8}$ (b) $\frac{8}{50}$ (c) $\frac{11}{20}$ (d) $\frac{70}{112}$

4. $7.2+9.73=$

5. $0.08-0.02=$

6. $148.28+312.09=$

7. $39.128-31.08=$

8. $88.738+11.004=$

9. $253.092+11.367=$

10. $0.899-0.371=$

11. $283.42-154.761=$

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12. A child weighed 15.75 kilograms in April. If in May the weight of the child increased by 1.25 kilograms, find the weight of the child in May.

13. A pupil spent $\frac{6}{10}$ of his time studying and $\frac{2}{10}$ of his time playing. What is the total time in decimals spent by the pupil in studying and playing.

14. The pass rate of pupils in Mathematics was 0.85 and in Kiswahili it was 0.88. Find the difference in the pass rate of pupils in these subjects.

15. On the first day, the mother cooked 1.25 kilograms of meat. On the second day, she cooked 1.5 kilograms of meat. How much total meat did the mother cook on those two days?

16. Businessmen bought sugar as follows: 122.5 kilograms, 88.25 kilograms and 227.175 kilograms. Find the total number of kilograms of sugar bought.

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Chapter Nine

Measurements of time

Introduction

In this chapter, you will learn how to convert, add and subtract measurements of time. You will also learn to convert measurements of time. The competencies developed will enable you to plan schedules, keep track of time in different activities and make good decisions about time in everyday life.



Think

Life without time management

Conversion of measurements of time

The measurements of time can be converted by considering the relationship between the units involved.

Measurements of time

The following table shows the relationships of different measurements of time

Measurement	Relationship
1 minute	60 seconds
1 hour	60 minutes
1 hour	3600 seconds
1 day	24 hours
1 week	7 days
1 year	52 weeks
1 year	12 months

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leap year	366 days
short year	365 days

Note that, division is applied when changing a small unit to a large unit, whereas multiplication is used when changing a large unit to a small unit.

Example 1

How many minutes are there in 720 seconds?

Solution

$1 \text{ minute} = 60 \text{ seconds}$

$720 \text{ seconds} = \frac{720}{60} \text{ minutes}$

$= 12 \text{ minutes}$

Therefore, there are 12 minutes in 720 seconds.

Example 2

A pupil takes 15 minutes to walk from home to school. Convert this time into hours.

Solution

$1 \text{ hour} = 60 \text{ seconds}$

$15 \text{ minutes} = \frac{15}{60} \text{ hours}$

Therefore, $15 \text{ minutes} = \frac{15}{60} \text{ hours}$.

Example 3

How many minutes are there in 72 hours?

Solution

$1 \text{ hour} = 60 \text{ seconds}$

$72 \text{ hours} = 72 \times 60 \text{ minutes}$

$= 4320 \text{ minutes}$

Therefore, there are 4320 minutes in 72 hours.

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Example 4

How many weeks are there in 91 days?

Solution

$$1 \text{ week} = 7 \text{ days}$$

$$91 \text{ days} = \frac{91}{7} \text{ weeks}$$

$$= 13 \text{ weeks}$$

Therefore, there are 13 weeks in 91 days.

Exercise 1

1. How many weeks and days are there in each of the following?
   (a) 48 days (b) 60 days (c) 19 days (d) 112 days
2. Convert the following time into days:
   (a) 12 weeks 1 day (b) 36 weeks 6 days
   (c) 72 weeks 4 days (d) 80 weeks
3. How many seconds are there in 2 hours and 30 minutes?
4. How many days are there in three leap years?
5. A bridge was constructed in 34 days. Convert this time into weeks and days?
6. A bus travelled for 72 hours. How many days did the bus travel?
7. The spinach in the vegetable garden took 30 days to mature. Convert this time into weeks and days?
8. A teacher has two periods of 40 minutes for each lesson. What is the total time of the periods in seconds?

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Addition of measurements of time without regrouping

Measurement of time with the same units can be added by considering the relationships of their units.

Example 1

hours	minutes
2	25
+ 3	15

Steps	Solution
1. Add minutes: $25+15=40$. Write 40 in the minutes column. 2. Add hours: $2+3=5$. Write 5 in the hours column.	hours minutes 2 25 + 3 15 5 40

Therefore, the answer is 5 hours 40 minutes.

Example 2

hours	minutes
12	28
+ 4	19

Solution

hours	minutes
12	28
+ 4	19
16	47

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Exercise 2

1.

hours	minutes
3	15
+4	25

2.

hours	minutes
2	15
+ 4	20

3.

hours	minutes
7	20
+ 1	35

4.

hours	minutes
2	30
+1	10

5.

hours	minutes
6	30
+ 2	30

6.

hours	minutes
8	10
+ 1	10

7.

hours	minutes
5	30
+ 1	15

8.

hours	minutes
5	00
+ 2	59

9.

hours	minutes
9	05
+2	30

10.

hours	minutes
3	15
+1	35

11.

hours	minutes
6	09
+ 1	14

12.

hours	minutes
4	12
+ 1	17

Addition of measurements of time with regrouping

Addition of measurements of time with regrouping is done by converting the units of time depending on how they are related.

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Example 1

hours	minutes
3	45
+ 4	35

Steps

1. Add minutes: $45+35=80$. Convert 80 minutes into 1 hour and 20 minutes. Write 20 in the column of minutes. Take 1 hour to the column of hours.
2. Add hours: $1+3+4=8$. Write 8 in the column of hours.

Solution

hours	minutes
+1	
3	45
+ 4	35
8	20

Therefore, the answer is 8 hours 20 minutes.

Example 2

hours	minutes
4	32
+ 2	28

Steps

1. Add minutes: $32+28=60$. Convert 60 minutes into 1 hour and 00 minutes. Write 00 in the minutes column. Take 1 hour to the column of hours.
2. Add hours: $1+4+2=7$. Write 7 in the hours column.

Solution

hours	minutes
+1	
4	32
+ 2	28
7	00

Therefore, the answer is 7 hours.

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Example 3

hours	minutes
6	37
+ 3	28

Solution

hours	minutes
+1	
6	37
+ 3	28
10	05

Exercise 3

1.

hours	minutes
7	25
+ 2	55

2.

hours	minutes
1	37
+ 4	24

3.

hours	minutes
4	40
+ 3	50

4.

hours	minutes
2	20
+ 5	60

5.

hours	minutes
2	12
+ 1	49

6.

hours	minutes
9	15
+ 1	50

7.

hours	minutes
5	30
+ 1	45

8.

hours	minutes
4	01
+ 2	59

9.

hours	minutes
1	30
+ 2	30

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10.

hours	minutes
3	15
+ 2	45

11.

hours	minutes
3	05
+ 8	55

12.

hours	minutes
4	12
+ 3	50

13.

hours	minutes
2	10
+ 3	54

14.

hours	minutes
8	20
+ 1	45

15.

hours	minutes
6	30
+ 2	40

Subtraction of measurements of time

Subtraction of measurements of time considers the relationship between the given units. For example, if it involves hours and minutes; subtract the minutes first, then hours.

Example 1

hours	minutes
6	25
− 5	15

Steps	Solution
1. Subtract minutes: 25 − 15 = 10. Write 10 in the minutes column. 2. Subtract hours: 6 − 5 = 1. Write 1 in the hours column.	hours minutes 6 25 − 5 15 1 10

Therefore, the answer is 1 hour 10 minutes.

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Example 2

hours	minutes
8	37
− 3	22

Solution

hours	minutes
8	37
− 3	22
5	15

Example 3

hours	minutes
8	37
− 4	28

Solution

hours	minutes
8	37
− 4	28
4	09

Exercise 4

1.

hours	minutes
3	40
− 1	25

2.

hours	minutes
5	55
− 3	13

3.

hours	minutes
9	52
− 1	37

4.

hours	minutes
4	17
− 2	11

5.

hours	minutes
4	32
− 1	19

6.

hours	minutes
9	55
− 6	50

7.

hours	minutes
5	42
− 1	22

8.

hours	minutes
8	23
− 1	12

9.

hours	minutes
5	50
− 2	30

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10.

hours	minutes
3	55
− 1	45

11.

hours	minutes
8	15
− 5	05

12.

hours	minutes
4	35
− 1	20

13.

hours	minutes
7	30
− 2	15

14.

hours	minutes
6	48
− 3	40

15.

hours	minutes
2	45
− 1	05

Subtraction of measurements of time with regrouping

Measurements of time can be subtracted with regrouping. This should be done by considering the relationship existing between the units of time.

Example 1

hours	minutes
3	15
− 1	45

Steps

1. Subtract minutes: 15 − 45, it is not sufficient. Take 1 hour from the 3 hours, then convert it into 60 minutes. Add minutes: 60 + 15 = 75. Now, subtract minutes: 75 − 45 = 30. Write 30 in the minutes column. Remember that, two 2 hours remained in the hours column.
2. Subtract hours: 2 − 1 = 1. Write 1 in the hours column.

Solution

hours	minutes
2	75
− 1	45
1	30

Therefore, the answer is 1 hour 30 minutes.

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Example 2

hours	minutes
8	00
− 3	53

Steps

1. Subtract minutes: 0 − 53, it is not sufficient. Take 1 hour from the 8 hours, then convert it into 60 minutes. Add minutes: $60 + 00 = 60$ minutes. Now, subtract minutes: $60 - 53 = 7$. Write 07 in the minutes column.
2. Subtract hours: $7 - 3 = 4$. Write 4 in the hours column

Solution

hours	minutes
7	60
− 3	5
4	07

Therefore, the answer is 4 hours 7 minutes.

Example 3

hours	minutes
7	27
− 3	58

Solution

hours	minutes
7	27
− 3	58
3	29

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Exercise 5

1.

hours	minutes
5	25
− 2	35

2.

hours	minutes
2	12
− 1	49

3.

hours	minutes
9	15
− 6	50

4.

hours	minutes
5	30
− 1	45

5.

hours	minutes
5	00
− 1	59

6.

hours	minutes
5	35
− 2	40

7.

hours	minutes
3	15
− 1	45

8.

hours	minutes
8	15
− 5	55

9.

hours	minutes
4	12
− 1	50

10.

hours	minutes
7	11
− 3	35

11.

hours	minutes
5	14
− 3	24

12.

hours	minutes
6	05
− 3	36

13. Today is the first day for Mr. Doto to be at the scout camp. Below is his timetable while at the camp. Study the timetable, then answer the questions.

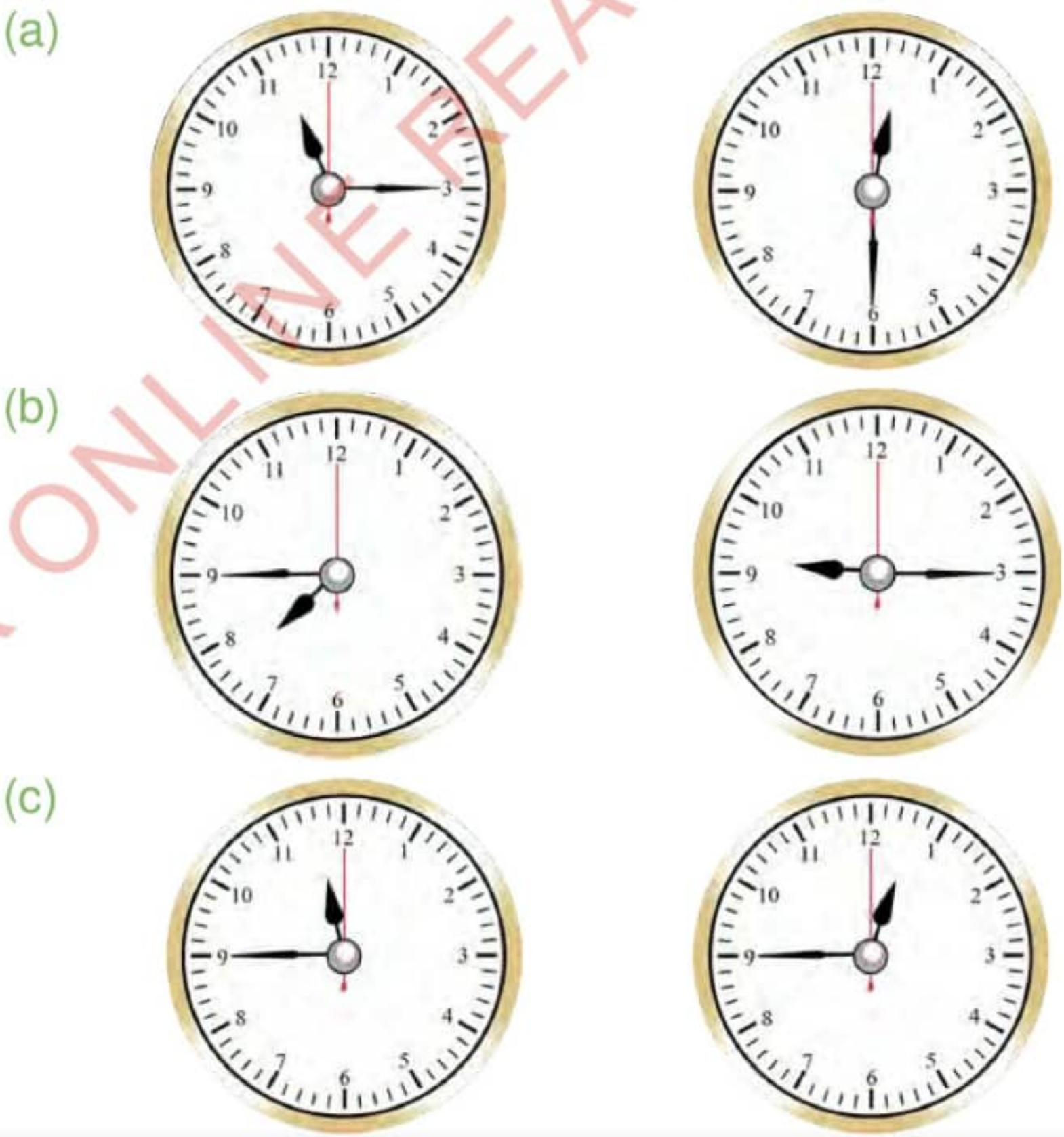
Activity	Start time	End time
breakfast	7:15 morning	8:10 morning
physical exercises	8:10 morning	9:00 morning
theoretical training	9:00 morning	10:45 morning
practical training	10:45 morning	12:15 afternoon

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lunch	12:15 afternoon	1:45 afternoon
practical training	1:45 afternoon	4:15 evening
entertainment	4:15 evening	6:00 evening

(a) How much time does Doto spend on doing physical exercises?
(b) How much time does Doto spend on having lunch?
(c) How much time does practical training take?
(d) Which activities take more than 1 hour 30 minutes?
(e) Which activity takes the shortest duration?
(f) Which activity takes the longest duration?

14. Study each of the following clock faces, then find the time spent:



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Word problems involving measurements of time

Word problems involving measurements of time show how they are used in real life.

Example 1

Nuru spends 1 hour and 20 minutes to wash dishes. She also spends 1 hour 35 minutes to water the garden. What is the total time spent by Nuru on these activities?

Solution

Time for washing dishes = 1 hour 20 minutes

Time for watering the garden = 1 hour 35 minutes

Add the time for the activities as follows:

	hours	minutes
	1	20
+	1	35
	2	55

Therefore, Nuru spends 2 hours 55 minutes to do all activities.

Example 2

A timetable of a certain school starts at 8:45 in the morning and ends at 4:40 in the evening. What is the daily duration of the school timetable?

Solution

Convert the time into 24 hours format.

Starting time = 0845

Ending time = 1640

Duration = Ending time – Starting time

Subtract the measurements as follows:

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	hours	minutes
	16	40
−	8	45
	7	55

Therefore, the duration of the school timetable is 7 hours 55 minutes.

Example 3

A train leaves Dodoma at 0530 in the morning and arrives in Dar es Salaam at 0845 in the morning. How much time does the train take to travel from Dodoma to Dar es Salaam?

Solution

Departing time = 0530
Arrival time = 0845
Duration = Arrival time − Departure time
Subtract the measurements as follows:

	hours	minutes
	08	45
−	05	30
	3	15

Therefore, the train takes 3 hours 15 minutes to travel from Dodoma to Dar es Salaam.

Exercise 6

1. Rehema is a standard four pupil. After lessons, she spends 3 hours and 15 minutes studying. If she starts studying at 7:15 in the evening, what time does she finish studying?

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2. An athletic competition started at 9:45 in the morning and ended at 12:25 in the afternoon. How long did the competition take?
3. Zakaria took 45 minutes to travel from home to school. If he left home at 6:30 in the morning, what time did he arrive at school in 24 hours format?
4. A football match was played for 1 hour 30 minutes. If the match started at 6:00 in the evening, what time did it end?
5. A pupil arrived at school at 6:15 in the morning. If the pupil was late for 15 minutes, what time was the pupil supposed to arrive?
6. Zuhura is the timekeeper at school. She rings the bell every 40 minutes. If she rang the bell at 8:40 in the morning, what time would the ring the bell again?
7. An examination started at 9:15 in the morning and ended at 12:15 in the afternoon. How long did the examination take?
8. A 40 minutes mathematics lesson ended at 11:20 in the morning. At what time did the lesson start?
9. The bus left Tanga Central Station at 06:00 and arrived in Dodoma at 17:45. How long did the bus travel from Tanga to Dodoma?
10. Vanesa arrived at school at 6:15 in the morning. She was 1 hour and 45 minutes early. What time was she supposed to arrive at school?
11. The plane traveled from Dar es Salaam at 00:45 and arrived in Mwanza at 01:45. How long did the flight take?

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Activity 1 — Learning time measurements through reliable online resources

Description

Explore various examples using reliable programs such as Khan Academy to learn more about converting measurements of time and solving word problems related to the concept of time.

Summary

1. Time is written in either the 12 hours or 24 hours format.
2. If the total number of minutes after adding exceeds 60, convert the minutes into hours and minutes.
3. If the minutes to be subtracted are insufficient, take 1 hour from the hours and convert it into minutes. Add the minutes and then subtract the required minutes.

Revision Exercise

1. Convert the following time measurements:
   (a) 7 days to hours (b) 168 hours to days
   (c) 12 years to months (d) 1 week to minutes
2. A chicken hatches a chick after incubating eggs for 21 days. What is this duration in hours?

3.

hours	minutes
7	35
+ 4	20

4.

hours	minutes
4	45
+ 3	45

5.

hours	minutes
9	25
− 3	45

6.

hours	minutes
6	50
+ 2	35

7.

hours	minutes
8	55
− 2	35

8.

hours	minutes
5	35
− 1	22

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9. A pupil spent 1 hour and 23 minutes cleaning the environment. She spent 1 hour and 36 minutes on studying a mathematics book. What time did she spend on these activities?
10. Rukia spent 1 hour and 38 minutes to complete a mathematics examination while Daniel spent 1 hour 55 minutes to complete the same examination. Find the difference between the time they spent on the examination.
11. The bus started the journey from Sumbawanga at 1220 and arrived in Tabora at 1855. How long did the bus take to travel from Sumbawanga to Tabora?
12. The sports competition started at 8:25 in the morning. If the competition lasted for 4 hours and 53 minutes, what time did the competition end?
13. A pupil started an examination at 8:30 in the morning and finished at 9:45 in the morning. How long did the pupil spend on the examination?
14. A Mathematics lesson starts at 0840 and ends at 0920. How long does the lesson last?
15. A film starts at 2:45 in the afternoon and lasts for 1 hour and 30 minutes. What time does the film end?
16. Aisha began cooking at 5:20 in the evening. If she spent 40 minutes for cooking, what time did she finish?
17. School lessons start at 8:00 in the morning and end at 3:30 in the afternoon. If the break time is 45 minutes, how long do the lessons last excluding the break time?
18. A football match started at 4:00 in the evening and lasted for 90 minutes. If the break time was 15 minutes, what time did the match end?

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Chapter Ten

Tanzanian Currency

Introduction

In this chapter, you will learn the multiplication and division of Tanzanian currency in shillings and cents. You will also learn to solve word problems related to Tanzanian currency. The competencies developed enable you to buy and sell commodities, make currency exchange and many other applications in real life.



Think

Life without the use of money

Multiplication of Tanzanian currency

The Tanzania currency can be multiplied by whole numbers.

Multiplications of Tanzanian currency in shillings

The Tanzanian currency in shillings can be multiplied by whole numbers.

Example 1

shs $750 \times 4 =$

Solution

shs $750 \times 4 =$ shs 3000

Therefore, shs $750 \times 4 =$ shs 3000.

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Example 2

shs 5500
× 20

Solution

shs 5500
× 20
000
+ 11000
shs 110000

Therefore, the answer is shs 110,000.

Exercise 1

1. shs 950 × 7
2. shs 75 × 13
3. shs 560 × 12
4. shs 105 × 85
5. shs 200 × 25
6. shs 960 × 14
7. shs 160 × 6 =
8. shs 960 × 36 =
9. shs 635 × 16 =
10. shs 87 × 15 =
11. shs 340 × 15 =
12. shs 950 × 9 =
13. shs 107 × 30 =
14. shs 750 × 45 =
15. shs 895 × 56 =
16. shs 1000 × 7 =
17. shs 1500 × 50 =
18. shs 10000 × 40 =
19. shs 3250 × 10 =
20. shs 8995 × 20 =

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Multiplication of currency in shillings and cents

The Tanzanian currency in shillings and cents can be multiplied by whole numbers. The cents are multiplied first followed by shillings. The relationship between shillings and cents must be used during the conversions.

Note that, cents are represented by two digits and shs 1 = 100 cents.

Example 1

shs	cts
50	05
×	6

Steps	Solution
1. Multiply 5 cents by 6: 5cts × 6 = 30cts Write 30 in the cents column.	shs 50 cts 05 × 6 — 30
2. Multiply 50 shillings by 6: shs 50 × 6 = shs 300. Write 300 in the shillings column.	shs 50 cts 05 × 6 — 300 30
Therefore, the answer is shs 300 30 cts or shs 300.30.	

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Example 2

shs	cts
250	75
×	10

Steps	Solution
1. Multiply 75 cents by 10: 75 cts × 10 = 750 cts. Convert 750 cents into 7 shillings and 50 cents. Write 50 in the cents column and add 7 shillings to the product of shillings.	shs cts 250 75 × 10 50
2. Multiply shilling 250 by 10: shs 250 × 10 = shs 2500. Add shillings: shs 7 + shs 2500 = shs 2507 . Write 2507 in the shillings column.	shs cts 250 75 × 10 2507 50
Therefore, the answer is shs 2,507 50 cts or shs 2,507.50.	

Example 3

shs	cts
5230	90
×	29

Solution

shs	cts
5230	90
×	29
151696	10

Therefore, the answer is shs 151696.10.

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Exercise 2

1.

shs	cts
500	20
×	4

2.

shs	cts
3115	30
×	5

3.

shs	cts
600	30
×	30

4.

shs	cts
10000	00
×	28

5.

shs	cts
4010	90
×	2

6.

shs	cts
21720	60
×	20

7.

shs	cts
700	00
×	9

8.

shs	cts
35320	70
×	12

9.

shs	cts
1524	36
×	14

Word problems involving multiplication of Tanzanian currency

Word problems on multiplication of Tanzanian currency involve the application of the currency in real-life.

Example 1

A packet of tea leaves is sold at 500 shillings. Find the cost of buying 12 packets of tea leaves of the same type.

Solution

Cost of one packet = 500

Number of packets = 12

Multiply the cost of one packet by the number of packets

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shs 500
× 12
1000
+ 500
shs 6000

Therefore, the cost of buying 12 packets is 6,000 shillings.

Example 2

An exercise book is sold at 800 shillings. What is the cost of 15 exercise books of the same type?

Solution

Cost of one exercise book = shs 800

Number of exercise books = 15

Multiply the cost by the number of exercise books.

shs 800
× 15
4000
+ 800
shs 12000

Therefore, 15 exercise books will be sold at 12,000 shillings.

Example 3

One litre of milk is sold at 1,850 shillings and 50 cents. How much will 17 litres of milk cost?

Solution

Cost of 1 litre = shs 1850 50 cts

Number of litres = 17

Multiply the cost by the number of litres.

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shs	cts
1850	50
×	17
31458	50

Therefore, 17 litres of milk will be sold at 31,458 shillings and 50 cents.

Exercise 3

1. A pen is sold at 250 shillings. How much will 13 pens of the same type cost?
2. A vendor sold a total of 40 plates. If each plate was sold at 650 shillings, what was the total cost for the plates?
3. If the price of one avocado is 850 shillings, how much money is needed to buy ten avocados?
4. A bunch of tomatoes is sold for 720 shillings. What is the cost of 50 bunches of tomatoes?
5. A businessperson bought 50 loaves of bread at 950 shillings each. How much money was spent for buying the bread?
6. A certain primary school pays an electric bill of 36,500 shillings per month. How much does the school pay for a year?
7. A masonry is paid 30,550 shillings and 70 cents for one day's work. How much money does the masonry earn after working for two weeks.
8. Irene earns a profit of 3,889 shillings and 75 cents for selling a pair of shoes. What will be the total profit if she sells 22 pairs?
9. A tray of eggs costs 16,520 shillings and 25 cents. What is the total amount of money required to pay for 40 trays?

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Division of Tanzanian currency

The Tanzanian currency can be divided by numbers. This is done in the same way as dividing any other numbers

Division of Tanzanian currency in shillings

Currency in shillings can be divided by whole numbers.

Example 1

shs 75 ÷ 3 =

Solution

Divide shs 75 by 3: 75 ÷ 3 =shs 25

Therefore, shs 75 ÷ 3 = shs 25.

Example 2

shs 2850 ÷ 38=

Solution

$$
\begin{array}{r}
75 \\
38\overline{)2850} \\
-266 \\
\hline
190 \\
-190 \\
\hline
0
\end{array}
$$

Therefore, the answer is shs 75.

Example 3

shs 9912 ÷ 118=

Solution

$$
\begin{array}{r}
\text{shs} \\
118\overline{)9912} \\
\hline
84
\end{array}
$$

Therefore, the answer is shs 84.

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Exercise 4

1. shs 52 ÷ 4 =
2. shs 247 ÷ 13 =
3. shs 987 ÷ 21 =
4. shs 7260 ÷ 55 =
5. shs 14964 ÷ 87 =
6. shs 146860 ÷ 28 =
7. shs
   14)13440
8. shs
   45)33750
9. shs
   64)40640
10. shs
    57)468084

Division of currency in shillings and cents

The Tanzania currency in shillings and cents can be divided by whole numbers. The shillings are divided first followed by division of cents.

Note that, the relationship between shillings and cents must be considered whenever conversion is needed.

Example 1

shs cts
5)10 50

Steps	Solution
1. Divide shillings 10 by 5: shs 10 ÷ 5 = shs 2. Write in the shillings place.	shs cts 5)10 50 2
2. Divide cents 50 by 5: 50 ÷ 5 = cts 10. Write 10 in the cents place.	shs cts 5)10 50 2 10
Therefore, the answer is shs 2 and 10 cts.	

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Example 2

shs	cts
45) 205312	50

Steps	
1. Divide shillings: shs 205312 ÷ 45 = shs 4562 with remainder shs 22. Write 4562 in the shillings column. Convert shs 22 into cents. Thus, shs 22 = 2200 cts. Add cents: 2200 cts + 50 cts = 2250 cts.	shs cts 45) 205312 50 4562
2. Divide cents: 2250 cts ÷ 45 = 50 cts. Write 50 in the cents column.	shs cts 45) 205290 50 4562 50
Therefore, the answer is shs 4,562 50 cts.	

Example 3

shs	cts
9) 19000	80

Solution

shs	cts
9) 19000	80
2111	20

Therefore, the answer is shs 2111 20 cts.

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Example 4

shs 80.75 ÷ 5 =

Solution

	shs	cts
5)	80	75
	16	15

Therefore, the answer is shs 16.15.

Exercise 5

1.

	shs	cts
15)	46740	60

2.

	shs	cts
3)	45363	00

3.

	shs	cts
12)	24289	20

4.

	shs	cts
15)	45442	50

5.

	shs	cts
8)	3200	40

6.

	shs	cts
7)	91000	70

7. shs 136005.40 ÷ 5 =

8.

	shs	cts
13)	391560	00

9.

	shs	cts
4)	465600	60

10.

	shs	cts
10)	364500	00

11. shs 645000.00 ÷ 8 =

12.

	shs	cts
116)	8526	00

13.

	shs	cts
24)	100000	80

14.

	shs	cts
12)	1740	60

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15. shs cts
24)133248 00

16. shs cts
16)144962 40

17. shs cts
25)75510 00

18. shs cts
12)133248 60

19. shs cts
15)653591 00

20. shs 111.20 ÷ 20 =

21. shs 97500.80 ÷ 40 =

Word problems involving the division of Tanzanian currency

Word problems involving the division of Tanzanian currency help to realize the application of division in real life.

Example 1

A pupil bought 17 exercise books for 14,790 shillings. How much did the pupil pay for each exercise book?

Solution

Cost of 17 exercise books = shs 14790

Number of exercise book=17

Divide the cost by the number of exercise books.

$$
\begin{array}{r}
870 \\
17\overline{)14790} \\
-136 \\
\hline
119 \\
-119 \\
\hline
0
\end{array}
$$

Therefore, the pupil bought each exercise book at 870 shillings.

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Example 2

A poultry farmer sold 38 eggs and earned a total of 17,126 shillings and 60 cents. What was the price of each egg?

Solution

Total sales = shs 17126 cts 60

Number of eggs = 38

Divide the sales by the number of eggs.

	shs	cts
38)	17126	60
	450	70

Therefore, the farmer sold each egg at 450 shillings and 70 cents.

Example 3

Eliakimu was paid a total of 68,763 shillings and 80 cents after working for one week. Calculate his daily wage.

Solution

Total payment = shs 68763 80 cts

Number of days = 7

Divide the payments by number of days.

Payments per day = shs 68763.80 ÷ 7 = 9823.4

= 9823.4

Therefore, Eliakim's daily wage is 9,823 shillings and 40 cents.

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Exercise 6

1. A parent gave a total of 333,500 shillings to 10 children. If this amount of money was shared equally, how much did each child get?
2. A businessperson sold 452 oranges for a total of 146,900 shillings. If the oranges were sold at a uniform price, what was the selling price of each orange?
3. Jamila bought 19 dresses for her business, spending a total of 463,144 shillings. If each dress had the same price, calculate the cost of a single dress.
4. Amir sold 285 mangoes and earned a total of 119,913 shillings and 75 cents. If each mango was sold at the same price, what was the price of each mango?
5. A group of 18 pupils shared 174,499 shillings and 20 cents equally. How much money did each pupil receive?
6. A total of 52 passengers contributed 973,476 shillings and 40 cents for their tickets. If each passenger paid the same fare, what was the cost for one ticket?
7. A teacher bought 87 mathematics books for 842,812 shillings and 50 cents. If each book had the same price, what was the cost of each book?
8. A total of 224 seedlings were sold for 615,148 shillings and 80 cents. Calculate the price of a single seedling if each was sold at the same price.

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Activity — Learning about Tanzanian financial transactions through online resources

Description

Use various reliable programs to learn more about how to make transactions that involve the multiplication and division of Tanzanian currency.

Summary

1. During operations of Tanzanian currency, the cents are represented by two digits.
2. When dividing Tanzanian currency; start with shillings, then cents.
3. When multiplying the Tanzanian currency; start with cents, then shillings

Revision Exercise

1.

shs	
764	
×	6

2.

shs	cts
5675	39
×	8

3.

shs	cts
190	90
×	9

4.

shs	cts
45330	80
×	28

5.

shs	3500
×	9

6.

shs	cts
39850	77
×	79

7.

shs	cts
198	96
×	69

8.

shs	cts
26) 67678	52

9.

shs	cts
6) 45360	96

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10. shs 12400 ÷ 4 =

11. shs 75500 ÷ 20 =

12.
	shs	cts
18)	44284	14

13.
	shs	cts
19)	95460	37

14.
	shs	cts
13)	7653	62

15.
	shs	cts
14)	19007	66

16. One exercise book is sold at 500 shillings and 30 cents. What is the cost of 8 exercise books?

17. A ball is sold at 8,000 shillings and 25 cents. What will be the total cost to buy 4 balls?

18. A total of 1,200 shillings and 75 cents was spent on buying 3 pens that were sold at the same price. What was the selling price of each pen?

19. A pupil bought 9 exercise books for 1,809.45 shillings. What was the price of each exercise book if they were all bought at the same price?

20. An employee receives a monthly salary of 775,355 shillings and 23 cents. What is the total annual salary for this employee?

21. A pack has 12 pencils. If each pencil costs 450 shillings and 25 cents, what is the total cost for 8 packs of these pencils?

22. A farmer sold 98 sacks of maize at 98,500 shillings and 45 cents each. What was the total sales obtained by the farmer?

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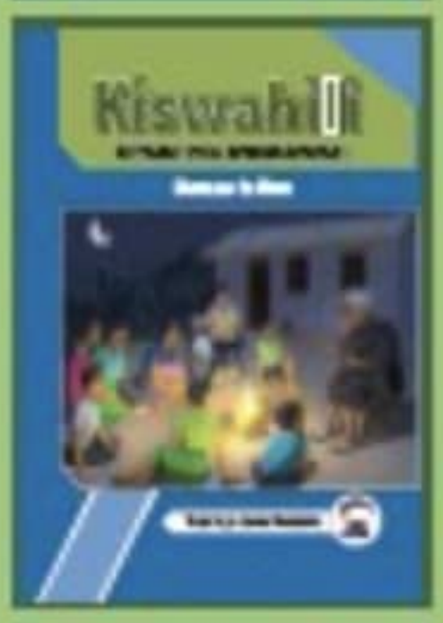
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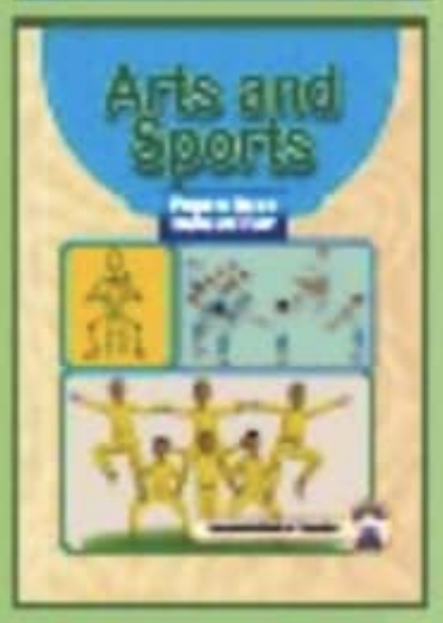


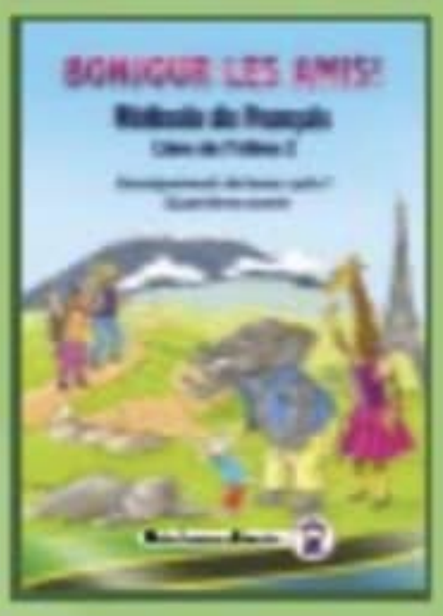
















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